


VIRTUAL!

# Applying Automotive EDR Data to Traffic Crash Reconstruction

# Instructor

Ruth Consulting

# Rick Ruth



- Ford Motor Company Product Development Engineering - Airbag Design and Field Performance  1973-2006
- Ford's lead field user of EDR's 97-06, Ford EDR policy committee, Ford Rep to SAE & EDR Committee, Ford Rep to Bosch CDR
- 2007-present, Ruth Consulting LLC – EDR in Recon all day every day, occasional airbag performance cases
- Teach EDR tech, EDR Analyst 1 & 2 classes for IPTM (Institute of Police Technology and Management, part of UNF) and EDR for SAE.  Presents regularly at national and regional conferences on reconstruction, on EDR's.
- Does research  on EDR accuracy – 19 publications
- Helps prosecutors and cops nationwide in EDR cases – mostly for free (charge for written reports and testimony).
- BSEE Michigan tech, MBA U of Mich, registered PE

# Students In May 2023 Class

| | First Name | Last Name | Employer | Job Title |
|---|---|---|---|---|
| 1 | Alexander | Szepelak | Aperture LLC | Forensic Expert |
| 2 | Alexandra | Boggs | Focus Forensics | Transportation Safety Engineer |
| 3 | Blake | Newton | Collision & Injury Dynamics Inc | Staff Engineer |
| 4 | Dylan | Copestick | DOT/NHTSA | |
| 5 | Ella | Webb | Aperture Llc | Forensic Expert |
| 6 | Evan | McConnell | Rimkus Consulting Group Inc | Senior Consultant |
| 7 | Gabriela | Kwiatkowski | Aperture LLC | Forensic Engineer |
| 8 | Geoff | Desmoulin | | Principal and Sr Biomechanist |
| 9 | Jacob | Ovaska | Explico Engineering Co | Accident Reconstructionist |
| 10 | James | Norris | Parham Engineering Consultant | Accident Reconstructionist |

# Students In May 2023 Class

| | First Name | Last Name | Employer | Job Title |
|---|---|---|---|---|
| 11 | Ko Ming | LAW | KM Consulting | Consultant |
| 12 | Luis | Nolasco | Explico Engineering Co | Scientist |
| 13 | Matthew | Hancock | Focus Forensics | Forensic Consultant |
| 14 | Mattie | Proffitt | Aperture LLC | Forensic Expert |
| 15 | Nathan | Lombardo | SEA Ltd | Mechanical Engineer |
| 16 | Nerina | Salihagic | | Accident Reconstructionist |
| 17 | Rheana | Forristall | | Accident Reconstructionist |
| 18 | Ryan | Porter | Collision & Injury Dynamics Inc | Staff Engineer |
| 19 | Tricia | Morey | Swope Reconstruction | Accident Reconstruction Consultant |
| 20 | Troy | Buhr | | |
| 21 | Yechan | Kim | | Accident Reconstructionist |

# Other Housekeeping

- Roughly 10 min breaks every hour, 30 min lunch
- You learn best if you give your full attention to the class. Multi-tasking is very tempting in the virtual world.
- I'll do my best to keep you engaged
- SAE has an end of class learning assessment to get credit for the class.  You will get an email when the class is over inviting you to take it.

# MODULE 1: US EDR STATUS
# EDR OVERVIEW

# Learning Objectives: Overview

1. Get “big picture” view of EDR’s
2. Basic history of EDR and govt regulation
3. Understand tools to get data
4. Understand typical data gathered and how it has evolved over time
5. Note while minimum data is now standardized, amount of data varies by manufacturer, model, and model year

# Some Brief History

1. Early airbag sensors were mechanical, any electronics only ran diagnostics.
2. For 1994 Model Year GM introduced the first electronic accelerometer sensor to measure crash severity, and used it to make airbag deployment decisions..
3. To know if the system performed properly, GM needed to record the inputs and outputs of the system.

# Early decisions in EDR's

1. The airbag usually deploys in the first 10-20 milliseconds in a typical 30 mph barrier crash.
2. The average crash pulse lasts 100 to 150 milliseconds.
3. GM measured acceleration 800 times per second.
4. Early control modules had only a very small quantity of EEPROM memory built into the microprocessor.

# Early Decisions

1. There was not enough memory to get all the acceleration data.
2. GM decided to calculate the change in velocity from the raw acceleration data and record that once every 10ms.
3. At that time the sensor measured only acceleration in the frontal (longitudinal) direction.
4. GM recorded in airbag deployments, but also decided to record in non-deployments

# Early EDR History

1. In 1999 GM attached the airbag control module to the vehicle CAN bus communication system so it could receive information from other modules in the car.
2. GM phased in recording vehicle speed, throttle, brake on/off, and RPM.
3. The purpose of recording was still for GM to know if its airbag system worked correctly.
4. The tool to read this data were proprietary to GM.

# Early History Continued

1. GM asked the supplier of its “Tech 2” dealer diagnostic tool, Vetronix, to make a tool to read crash data for sale to the public.
2. The Vetronix Crash Data Retrieval System was born in 2000.
3. Bosch Diagnostics bought Vetronix in the mid 2000’s.

# History Marches On

1. Ford began phasing in electronic sensors in 1997 and recorded limited info.
2. Ford contracted with Vetronix/Bosch to read its limited EDR's in the early 2000's.
3. Chrysler added electronic sensors but did not record anything useful until 2005MY.
4. In 2007 they also chose Bosch CDR to allow their EDR's to be read.
5. Toyota began recording in 2001, had proprietary read out tool, switched to Bosch CDR in 2010

# US Government Involvement

1. The US government was reluctant to require EDR's – "big brother is watching".
2. In 2006 they adopted a regulation that would eventually standardize minimum content of **voluntarily** installed EDR's
3. It also CHANGED THE **PURPOSE** to be for ***effective crash reconstruction***.
4. The regulation required that data could be accessed by a publicly available tool.
5. Part 563 became effective September 2012.

# Part 563 EDR Regulation




Conceived in 2002
Adopted in 2006
Effective 9-1-2012
THANKS. NHTSA!!!

(vs. China 2021, Europe 2022)

**TABLE I – DATA ELEMENTS REQUIRED FOR ALL VEHICLES EQUIPPED WITH AN EDR**

| Data Element | Recording Interval/Time[1] (Relative to time zero) | Data Sample Rate Samples per Second |
|---|---|---|
| Delta-V, longitudinal | 0 to 250 ms | 100 |
| Maximum delta-V, longitudinal | 0-300 ms | n.a. |
| Time, maximum delta-V | 0-300 ms | n.a. |
| Speed, vehicle indicated | -5.0 to 0 sec | 2 |
| Engine throttle, % full (or accelerator pedal, % full) | -5.0 to 0 sec | 2 |
| Service brake, on/off | -5.0 to 0 sec | 2 |
| Ignition cycle, crash | -1.0 sec | n.a. |
| Ignition cycle, download | At time of download | n.a. |
| Safety belt status, driver | -1.0 sec | n.a. |
| Frontal air bag warning lamp, on/off | -1.0 sec | n.a. |
| Frontal air bag deployment, time to deploy, in the case of a single stage air bag, or time to first stage deployment, in the case of a multi-stage air bag, driver | Event | n.a. |
| Frontal air bag deployment, time to deploy, in the case of a single stage air bag, or time to first stage deployment, in the case of a multi-stage air bag, right front passenger | Event | n.a. |
| Multi-event, number of events (1,2) | Event | n.a |
| Time from event 1 to 2 | As needed | n.a. |
| Complete file recorded (yes, no) | Following other data | n.a. |

## TABLE II - DATA ELEMENTS REQUIRED FOR VEHICLES UNDER SPECIFIED CONDITIONS

| Data Element Name | Condition for Requirement | Recording Interval / Time[1] (Relative to time zero) | Data Sample Rate (Per Second) |
|---|---|---|---|
| Lateral acceleration | If recorded[2] | 0-250 ms | 500 |
| Longitudinal acceleration | If recorded | 0-250 ms | 500 |
| Normal acceleration | If recorded | 0-250 ms | 500 |
| Delta-V, lateral | If recorded | 0-250 ms | 100 |
| Maximum delta-V, lateral | If recorded | 0-300 ms | n.a. |
| Time maximum delta-V, lateral | If recorded | 0-300 ms | n.a. |
| Time for maximum delta-V, resultant | If recorded | 0-300 ms | n.a. |
| Engine rpm | If recorded | -5.0 to 0 sec | 2 |
| Vehicle roll angle | If recorded | -1.0 up to 5.0 sec[3] | 10 |
| ABS activity (engaged, non-engaged) | If recorded | -5.0 to 0 sec | 2 |
| Stability control (on, off, engaged) | If recorded | -5.0 to 0 sec | 2 |
| Steering input | If recorded | -5.0 to 0 sec | 2 |
| Safety belt status, right front passenger (buckled, not buckled) | If recorded | -1.0 sec | n.a. |
| Frontal air bag suppression switch status, right front | If recorded | -1.0 sec | n.a. |

## Table III - Reported Data Element Format

| Data element | Minimum range | Accuracy[1] | Resolution |
|---|---|---|---|
| [1] Accuracy requirement only applies within the range of the physical sensor. For vehicles manufactured after September 1, 2014, if measurements captured by a sensor exceed the design range of the sensor, the reported element must indicate when the measurement first exceeded the design range of the sensor. | | | |
| Longitudinal delta-V | −100 km/h to + 100 km/h | ±10% | 1 km/h. |
| Lateral delta-V | −100 km/h to + 100 km/h | ±10% | 1 km/h. |
| Maximum delta-V, longitudinal | −100 km/h to + 100 km/h | ±10% | 1 km/h. |
| Maximum delta-V, lateral | −100 km/h to + 100 km/h | ±10% | 1 km/h. |
| Time, maximum delta-V, longitudinal | 0-300 ms, or 0−End of Event Time plus 30 ms, whichever is shorter | ±3 ms | 2.5 ms. |
| Time, maximum delta-V, lateral | 0-300 ms, or 0−End of Event Time plus 30 ms, whichever is shorter | ±3 ms | 2.5 ms. |
| Time, maximum delta-V, resultant | 0-300 ms, or 0−End of Event Time plus 30 ms, whichever is shorter | ±3 ms | 2.5 ms. |
| Vehicle Roll Angle | −1080 deg to + 1080 deg | ±10% | 10 deg. |
| Speed, vehicle indicated | 0 km/h to 200 km/h | ±1 km/h | 1 km/h. |
| Engine throttle, percent full (accelerator pedal percent full) | 0 to 100% | ±5% | 1%. |
| Engine rpm | 0 to 10,000 rpm | ±100 rpm | 100 rpm. |
| Service brake | On or Off | N/A | On or Off. |
| ABS activity | On or Off | N/A | On or Off. |
| Stability control | On, Off, or Engaged | N/A | On, Off, or Engaged. |
| Steering input | −250 deg CW to + 250 deg CCW | ±5% | ±1%. |

# Regulatory Update

1. NHTSA issued NPRM in Dec 2012 to **require** 563-EDR's by Sept 2014 (way past!)
2. NPRM proposed EDR as FMVSS 405, OEM's would have to self certify EDR's. Strong OEM objections were filed.
3. Officially withdrawn Feb 2019 https://www.federalregister.gov/documents/2019/02/08/2019-01651/federal-motor-vehicle-safety-standards-event-data-recorders

4. Mfrs were NOT expected to remove EDR's – But one Land Rover model may have – now has Bosch freeze frame recorder.

# Regulatory Update

- December 4, 2015 Section 24303 of the Fixing America's Surface Transportation Act (FAST Act), Pub. L. No. 119-14 required NHTSA to conduct a study "to determine the amount of time event data recorders installed in passenger motor vehicles should capture and record"
- May 2017 Event Data Recorders Duration Study (EDR Duration Study) completed by Virginia Tech (poorly)
- September 2018 Report to Congress summarizing pre-crash time interval results in determining crash causation
- NHTSA issued NPRM 6-22-2022 Docket NHTSA-2022-0021

# Part 563 NPRM 6-22-2022

1. Proposed rule increases precrash recording time to 20 sec @10X/sec, effective Sept 1st at least one year from final reg publication.
2. Industry strongly disputed need for 20 seconds, noted significant effects on cost, package size, and development time.
3. Insurance, NTSB, others like more free data
4. No Final Rule yet as of 4/14/2023

# Introduction - Main Messages

1. EDR's improve recon **accuracy** & get to **causation** – use with **all other** available scene evidence –EDR is also PRIMARY EVIDENCE.
2. EDR strength is speed **PRIOR** TO *BRAKING* OR *LOSS OF CONTROL, &* DRIVER BEHAVIOR on gas and brake pedal. Modern ABS brakes TOOK AWAY your tire marks, only EDR can "give you back" speed prior to braking. Newer data elements give us EVEN MORE INSIGHT into precrash behavior.
3. Event Data Recording capability and data accessibility ***varies widely*** by **manufacturer, model *and* model year.** Coverage keeps growing with MORE AND BETTER DATA available with each new model released.

# Main Messages

4. NHTSA Part 563 EDR rule Sept 2012 was a game changer – required **Minimum** Data in US vehicles equipped with EDR
5. EDR not required but 99% of US 2020 vehicles have one.
6. EDR's will NEVER replace human beings in figuring out what really happened.
7. US Driver Privacy Act Dec 2015 requires owner permission or a warrant to access.
8. AEB (auto braking) + other driver assist tech= fewer crashes + more data
9. If not already bought, budget for CDR 900 interface $3800 and $2500 new cables

# Main Messages

More and More Data every year!

GM ASCM's – more data than SDM

2020 GM SDM50 – 8 sec precrash, strg, whl spds

2017+ Ford AB12 and RC8, speed @0.2 sec

2017+ Toyota 17 & 19 EDR – X and Y DV sych'd

Toyota VCS data includes PICTURES ~~CDR~~

Most Subaru & Mitsubishi now covered by CDR

GM FCM storing PICTURES CDR 21.2+

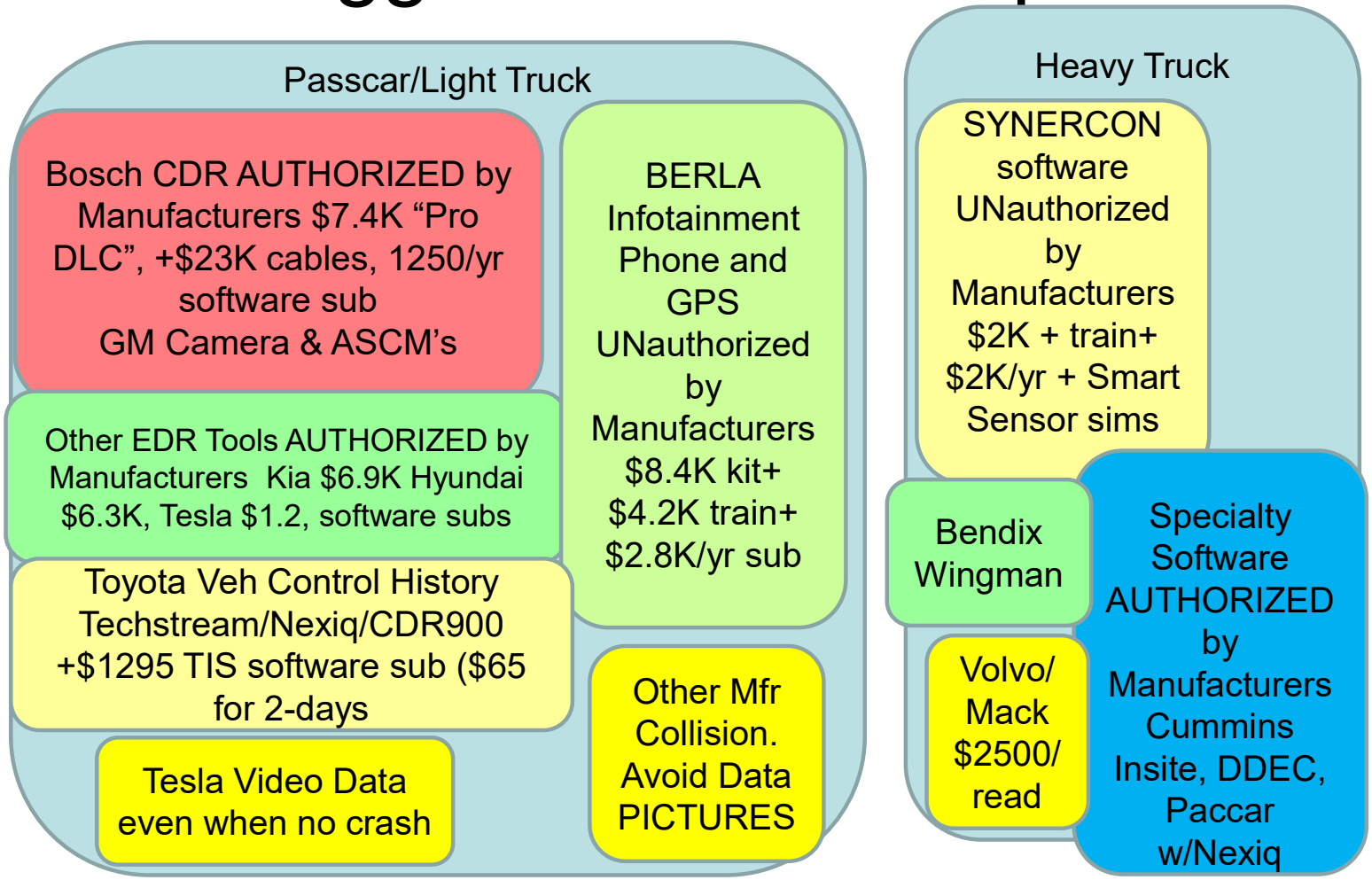
Pedestrian Protection events Toyota 2018+,others

Lots of anomalies to keep us on our toes

# Keeping EDR's in Perspective

1. There are about 300 million vehicles registered in the US. 63+% have an EDR with data accessible by the Bosch CDR system
2. 85+% of 2023's are supported by Bosch.
3. Another 15% have EDR accessible by other tools (10+% Kia/Hyundai, 3+% Tesla). **99+%** have EDR.
4. With *advanced* analysis techniques, we can get ***speed at impact*** for both vehicles even if only one has a 563 EDR.

# The bigger vehicle data picture



# US Lt. Veh EDR & Tool Used

| Manufacturer | 2022 CY per Automotive News | Market Share | EDR First Known Use | Tool? |
|---|---|---|---|---|
| GM | 2,258,283 | 16.3% | 1994 | Bosch CDR |
| Toyota | 2,108,455 | 15.2% | 2001 | Bosch CDR |
| Ford | 1,850,925 | 13.4% | 1997 | Bosch CDR 2001+ |
| Stellantis | 1,553,485 | 11.2% | 2005 | Bosch CDR |
| Hyundai/Kia | 1,474,224 | 10.7% | 2010 | GIT tool co. |
| Honda | 983,507 | 7.1% | 2012 | Bosch CDR 2012+ |
| Nissan | 729,350 | 5.3% | 2006 | Bosch CDR 2013+ |
| VW/Audi | 565,119 | 4.1% | 2014 | Bosch CDR |
| Subaru | 556,581 | 4.0% | 2012 | Bosch CDR now |
| Tesla | 491,000 | 3.5% | 2012 | Tesla Tool |
| BMW | 363,612 | 2.6% | 2013 | Bosch CDR |
| Mercedes-Benz | 350,949 | 2.5% | 2014 | Bosch CDR |
| Mazda | 294,908 | 2.1% | 2011 | Bosch CDR 2011+ |
| Geely (Volvo) | 102,038 | 0.7% | 2011 | Bosch CDR 2011+ |
| Mitsubishi | 85,810 | 0.6% | 2006 | Bosch CDR now |
| Tata (Jag, LR) | 65,602 | 0.5% | early | Send to Supplier |
| Other | 7,944 | 0.1% | | |
| | | | | N/A |
| Industry Total | 13,841,792 | 100.0% | ######## | Bosch CDR 85% |
| updated April 2023 | | | | Non CDR 15% |

Includes ASCM/ FCM

Techstream for VCH

Porsche 2020 Taycan

# 2023 EDR Recording Capability

**EDR Major Data Elements by Manufacturer** (All Mfrs with EDR have the required 15 data elements, all are not listed)

Information is taken from Bosch CDR sample files, Ruth cases, and 2012 MY practice projected into 2019 model year

| | Part 563 Table 1 | | | | | Part 563 Table 2 | | | | | | Beyond 563 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mfr | Event Complete | Key Cycles | Long. ΔV | Speed/Brake/ Throttle or Accel | Multi Event | Lateral ΔV | RPM | Steering Angle | ACM Acceleration | ABS on/off | Roll Angle | ESC Long Accel | ESC Lat Accel | Yaw Rate | Tire Pressure | Panic Brake Assist | Wheel Speeds | Roll Rate | Brake Pressure | OTHER |
| GM | X | X | X | 5@0.5 TA | X | X | X | SDM50 | , Z 300@2 | X | | 50 | 50 | 50 | LO | | 50 | 1 @ 10ms | 50 | 50 GPS Location |
| Ford | X | X | X | 5@0.5 A | X | X | X | 5@0.1 | - | X | | 5@0.1 | | 5@0.1 | LO | | - | 5@0.1 | ** | 2017 AB12 precrash 0.2 sec |
| Chrysler | X | X | X | 5@0.1 TA | X | X | X | 5@0.1 | - | X | | - | | 5@0.1 | PSI | X | X | | | 2018 ACC set speed 2018, Avoidance |
| Toyota | X | X | X | 5@0.5 TA | X | Side | X | 5@0.5 | - | | Peak&atTRG | 5 | in roll | 5@0.5 | - | - | - | In Roll | 5@0.5 | Ped 2018, Veh history rpt |
| Honda | X | X | X | 5@0.5 TA | X | X | X | 5@0.5 | X, Y, Z 250@10 | X | -1+5@0.1 | - | | - | - | - | - | | | AEmBrk. Lanekeep |
| Nissan | X | X | X | 5@0.5 A | X | X | X | 5@0.5 | X, Y 250@10 | - | - | - | | - | - | - | ? | | | |
| Mazda | X | X | X | 5@0.5 T | X | X | X | | X, Y, Z 250@10 | - | | | | | | | - | | | |
| Volvo | X | X | X | 5@0.5T | X | 2016 XC90 | - | 2016 XC90 | X,Y,Z 250@10 | 2016 XC90 | -1+0.3@0.1 | | | | | | | | | |
| BMW | X | X | X | 5@0.5A | X | | X | 5@0.5 | X, Y 300@10ms | X | - | | | | | | | | | |
| Mercedes | X | X | X | 5@0.5A | X | 2017 E class adds | | | | | | | | | | | | | | |
| VW | X | X | X | 5@0.5A | X | X | | | | | | | | | | | | | | Time of Crash |
| Hyundai/Kia | X | X | X | 5@0.5 | X | X | X | 5@0.5 | X,Y,Z 250@10 | X | -1+5@0.1 | - | | - | - | - | - | | 19 Forte | |
| Subaru | X | X | X | 5@0.5A | X | X | X | 5@0.5 | | X | | | | | | | | -1 to+1.5 | | Eyesight |
| Mitsubishi | X | X | X | 5@0.5A | X | X | X | 5@0.5 | X, Y 250@10 | | -1+5@0.1 | | | | | | | | | |

# New users need **both** CAN+ and CDR900



MSRP: $7300

plus shipping & handling

Add to Cart ›

## Bosch CDR Pro Tool Kit

DLC ONLY

The CDR Pro Tool Kit is the heart and soul of the Bosch CDR Tool. This package contains all of the hardware required to perform a DLC/OBD retrieval of EDR data in Bosch supported vehicles. It includes both the CDR 900 and CANplus vehicle interface modules. This package also comes with a 1-year CDR software subscription. NOTE: this package does NOT contain any hardware to perform a "direct-to-module (D2M)" retrieval of EDR data. Please see the accessory cable/adapter packages below for D2M options.

- CDR 900 vehicle interface module
- CDR 900 power cable
- CDR 900/CDR CAN*plus* power supply cable
- CDR 900/CDR CAN*plus* CDR cigarette power adapter
- CDR 900 DLC/OBD cable
- CDR 900 wireless dongles (2)
- CDR 900 extension cable
- CDR 900 legacy cable adapter
- CDR 900 USB cable
- CDR CAN*plus* vehicle interface module
- CDR CAN*plus* DLC/OBD cable
- CDR CAN*plus* GM DLC/OBD cable
- CDR CAN*plus* DLC/OBD adapters (3)
- CDR CAN*plus* PCM adapter
- CDR CAN*plus* USB to Serial port cable
- Deluxe repowering rig
- 1-year CDR software subscription
- Custom carrying case
- Coupon for **FREE online class**: How to Use the Bosch CDR Tool

# Which Interface Module by Manufacturer

| Manufacturer | CAN+ Only | CAN+ OR CDR900 | CDR900 only |
|---|---|---|---|
| GM | All Pre-2016, Some 2016+ | Some 2016+ | ASCM, FCM |
| Ford | All | | |
| Chrysler | All Pre 2016, Some 2016+ | Some | |
| Toyota | | All as of 23.0.2 | |
| Honda | | All as of 23.0.2 | |
| Nissan | All as of 23.0.2 | | |
| Mazda | All as of 23.0.2 | | |
| Volvo | | Most | Some New |
| BMW | | All as of 23.0.2 | |
| Mercedes | | All as of 23.0.2 | |
| VW/Audi | | All as of 23.0.2* | |
| Subaru | | | All as of 23.0.2 |
| Mitsubishi | | | All as of 23.0.2** |

* except early 2000's raider which is a Dodge, CAN+ only

** except late 2000's Routan which is a Chrysler Minivan, CAN+ only

# Help file identifies interface needed

| Mkt | Year/Make | Model | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|---|---|
| 13 | 2020 Chevrolet | Aveo (note 6) | ACM | Data | CANplus | F00K108287 | F00K108454 | Center tunnel |
| 4 | 2020 Chevrolet | Blazer (note 6) | ACM | Data | CANplus or CDR 900 | CANplus - F00K108287 CDR 900 - click here | CANplus - F00K108454 CDR 900 - click here (use listed cable above) | Center tunnel |
| | | | ASCM (Note1) | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 833 | In the rear cargo storage area, under the left rear trim panel, at the front of the wheel well |

CAN+ only

Either one

CDR900only

# HARDWARE CHECKLIST 23.1

New 3-digit Part ID

| Part Number | Description | Released |
|---|---|---|
| 1699200778 | Cable, ID835, CDR Subaru ACM 1 — 2019+ Forester/Acsent | 19.0 |
| 1699200779 | Cable, ID836, CDR Subaru ACM 2 — 2015+ Legacy/Outback | 19.0 |
| 1699200783 | Cable, ID837, CDR Mitsubishi ACM 1 — Eclipse Cross | 19.0 |
| 1699200784 | Cable, ID838, CDR Mitsubishi ACM 2 — Outlander | 19.0 |
| 1699200785 | Cable, ID839, CDR Mitsubishi ACM 3 — Many Older models | 19.0 |
| 1699200786 | Cable, ID840, CDR Mitsubishi ACM 4 — Japan Only models | 19.0 |
| 1699200776 | Cable, ID833, CDR GM ASCM 1 — Advanced Safety | 19.1 |
| 1699200777 | Cable, ID834, CDR GM ASCM 2 — Advanced Safety | 19.1 |
| 1699200797 | Cable, ID842, CDR VW Group ACM 4 — 2020 Golf, Neo | 19.1 |
| 1699200800 | Cable, ID843, CDR Mazda ACM — 2019+ Mazda 3 | 19.1 |
| 1699200838 | Cable, ID845, CDR GM ACM — SDM-50 2020 Corvette | 19.1 |
| 1699200837 | Cable, ID844, CDR Subaru ACM 3 — 2020 Legacy/Outback | 19.4 |
| 1699200839 | Cable, ID846, CDR Honda ACM — 2021+ Honda Gen 4 | 19.4 |
| 1699200787 | Cable, ID841, CDR Daimler ACM (requires CDR 500 — 2021 S-class | 19.6 |
| 1699200882 | Cable, ID847, CDR PSA ACM 1 — 2021 Puegot in China | 21.1 |
| 1699200883 | Cable, ID848, CDR PSA ACM 2 — 2021 Puegot in China | 21.1 |
| 1699200983 | Cable, ID850, CDR VWG ACM 5 — 2021 Golf + China models | 21.1 |
| 1699201032 | Cable, ID853, CDR Mitsu ACM 5 — Outside US/Canada | 21.1 |
| 1699503814 | Cable, ID858, CDR Maserati / Ferrari ACM | 21.4 |
| 1699503636 | Cable, ID859, CDR Lucid ACM | 21.4 |
| 1699200985 | Cable, ID852, CDR GM ACM | 23.0 |
| 1699503637 | Cable, ID854, CDR GM ASCM3 | 23.0 |

Nothing New in 23.1

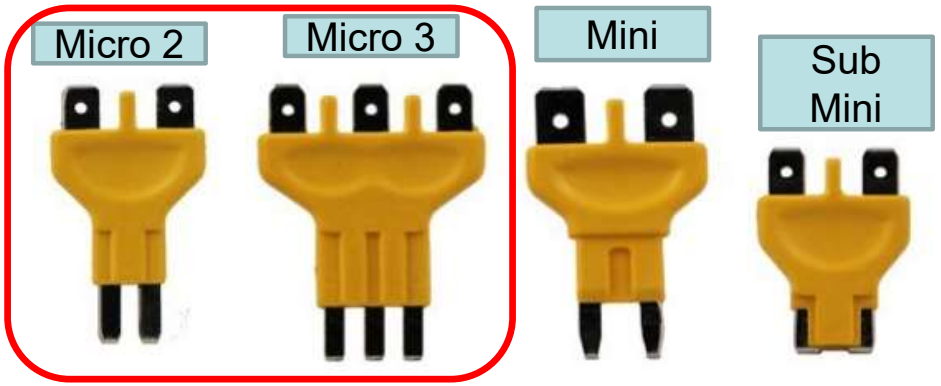
Coming 2023: 855 GM ASCM and 856 GM FCM

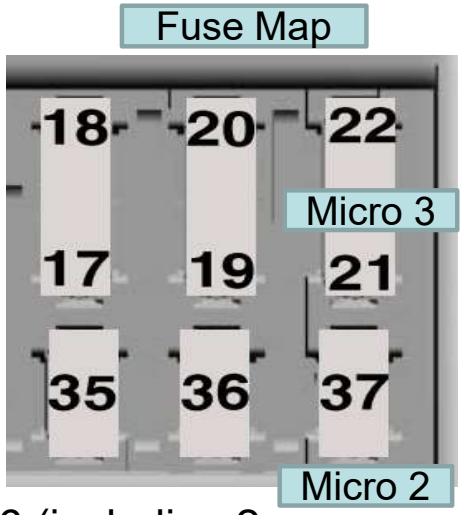
# Bosch CDR and Windows 11

- Windows 11 became supported in CDR 21.3

# New Back Powering Gear Needed

1. Manufacturers (GM+) are replacing mini and micro fuses with Micro 2 and Micro 3 fuses to pack more fuses into smaller spaces. The complete back power kit must now have several different yellow clips to fit all the different vehicles on the road. GM started it, Ford followed, more to come. Mini fits into sub mini, don't need sub mini.




2. Buy General Technologies Corp GTC063 set of 6 (including 2 bigger ones you don't really need) at the site below for $19.95

https://www.amazon.com/General-Technologies-Corp-GTC063-Connector/dp/B0787JKMTH?ref_=bl_dp_s_web_9003703011

# EMERGING TRENDS

1. Worry re: cyber security, (hackers trying to take over vehicle control systems).
2. Many are installing “gate-way” modules or features to protect ECU’s from hacking.
3. Back power ACM fuse AND cyber security (gateway) module fuse. DTM still an alt.
4. Recently needed on 2017 RAM 3500, 2018 Toyota, all VW, 2020 Ford Explorer
5. Ford “extended power” fuse or “Battery Electronic Control Module” for ACM & “smart data link” gateway– no RCM or SRS

# Recent CDR Software Updates

- 19.5 Aug 13, 2020 + 21's,
- 19.6 Dec 1 China releases + more
- 21.0 Jan 21, 2021 China releases + more
- 21.1 May 14 Many Mitsubishi
- 21.2 Aug 3 GM Fwd Camera Module
- 21.3 Oct. 28 21 Volvo, 22 Ford, Audi. WIN11
- 21.4 Jan 5, 2022 - +GM ASCM, 22 FCA,
- 21.5 April 8– Ferrari, more '22, mitsu
- 23.0 October 7– Many 2023's
- 23.1 March 31, 2023 – Euro support, 2023/4's

# Connectivity Issues with 21.4/21.5

- Several users have reported connectivity issues when using software level 21.4 or 21.5.  They have rolled back to 19.4 and the gotten data successfully.
- If you have comms issues, try a software rollback.
- The comms issues are more prevalent with CAN+, if you have a choice use CDR900, use it first.
- The 23.0 release is expected to correct these issues, but there has not been enough time to verify that yet.

# CDR Budgeting

| | Own everything | Middling Approach | A Minimal Approach |
|---|---|---|---|
| **Initial Expenses** | | | |
| Initial Technician Training | $200/$595 | $200/$595 | $200/$595 |
| Hardware (Cabels Etc...) | $31050 Premium Pro Kit) | $19,650 Deluxe Pro Kit) | $7,650 Pro Kit) |
| Software | Included | Included | Included |
| Popular Cables | Included | 50 Popular Cables/Adapters Included | ~$6000 (Choose 20 Popular Cables/Adapters - Estimate) |
| Analyst Training (Travel Extra) | $995 | $995 | $995 |
| **Sub Total** | $32645 | $21645 | $15245 |
| | | | |
| **Estimated Annual Recurring Expenses** | | | |
| Software Updates | $1500 | $1500 | $1500 |
| New Cables | $ 5*$285=1425 | | |
| Unexpected Cables | | $ 2*$285=570 ot have - Shipped overnight) | $440 ( 2*$285=570 e - Shipped overnight) |
| Analyst Training (Travel Extra, or Online = $0) $1000 Every Other Year | $500 | $500 | $500 |
| **Sub Total** | 3425 | 2575 | 2575 |
| * Prices Subject to Change | | | |

# New EDR Tool?

1. Collision Sciences based in Toronto has developed a DLC only tool and Bluetooth software that runs from your phone to read EDR data from vehicles with functioning electrical systems.
2. Business model is still evolving, but entry price for the hardware is very low compared to Bosch
3. Raw data is uploaded to cloud, free teaser preview
4. User pays modest fee to get report which includes value added analysis such as fraud risk assessment or risk of injury (vs. Bosch reports strictly data with no analysis.)
5. Developers are talented engineers - figuring out the decoder – and working on getting decoding data directly from manufacturers.
6. Works on Kia/Hyundai + most CDR supported vehicles

# CDR/EDR Training

1. Bosch Training Provider Page lists IPTM (Rick Ruth+), Crash Data Specialists (Brad Muir+), CSI (Rusty Haight), Crash Data Group and Northwestern (was CSI, now + Muir/Russel).

   SAE (Rick R) not listed, teaches analysis only.

2. There is no “certification” – your qualifications depend on your training and experience

3. IPTM ONLINE technician training $595 and ONLINE analyst training $995. Live training too post pandemic

4. SAE Analyst EDR was Virtual, live 4/22, virtual again!

5. Online tech basic from Crash Data Group $200

6. Advanced EDR Technician class by Kent Boots

# NHTSA CISS DATABASE UPDATE

1. https://crashviewer.nhtsa.dot.gov/CISS/SearchIndex
2. 11,358 cases available from 2016-2020 calendar years – last updated Jan 2022 for 2020 CY
3. Searchable by manufacturer, model year, model, EDR data available, driver assist technology available, etc
4. Find similar cases, check polarity, see typical EDR report in 5,913 of those cases
5. Link to VIN decoder, Canadian Specs, Vin check digit calc'tr Welcome to VIN Decoding :: provided by vPIC (dot.gov)

# European Reg Impact on US EDR in 23.1

- OEMs with Euro presence (VW/Audi & Toyota so far) will provide a place for user to enter Vehicle Type, Variant, and Version into CDR before saving file.  Can enter later.
- Also EU GDPR privacy reg PROHIBITS storing full VIN
- Last 6 VIN characters replaced by ****** prior to saving
- Expect to see more manufacturers implementing this as next phase of Euro Reg gets closer

# Details by Manufacturer

NOTE the big picture OVERVIEW presentation will end here, but SAE students are encouraged to review the remainder of the presentation for a quick “what’s new in EDR’s”

# Honda 23.0 CDR Coverage

| Model | Location | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ACURA | | Last 3 digits of Bosch CDR cable number | | | | | Gen 2 | | Gen 3 | | | Gen 4 | | |
| RL | Under dashboard cente | | 547 | | | | | | | | | | | |
| TL | Under dashboard cente | | 547 | | | | | | | | | | | |
| TSX | Under dashboard cente | | 547 | | | | | | | | | | | |
| ZDX | Under dashboard cente | | 547 | | | | | | | | | | | |
| MDX | Under dashboard cente | | 547 | | 387/789 | | | 387/810 | | | | ? | 846 | |
| NSX | Under dashboard cente | | | | | | | 387/789 | | | | | | |
| RDX | Under dashboard cente | | | 547 | | | 387/789 | | 387/825 | | | | | |
| ILX | Under dashboard cente | | | 547 | | | 387/789 | | | | | | | |
| RLX | Under dashboard cente | | | | 387/789 | | | | | | | | | |
| TLX | Under dashboard cente | | | | | 387/789 | | | | 387/810 | | 846 | | |
| INTEGRA | | | | | | | | | | | | | | 846 |
| HONDA | | | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | |
| ACCORD | Under dashbord center | | | 387/789 | | | 387/810 | | 387/825 | | | | | |
| CIVIC 2 DR | Under dashbord center | | 547 | | | | 387/810 | | | | | | 846 | |
| CIVIC 4 DR | Under dashbord center | | 547 | 387/789 | | | 387/810 | | | | | | 846 | |
| CLARITY | | | | | | | | | | | | 387/810 | | |
| CROSSTOUR | Under dashbord center | | 547 | | | | | | | | | | | |
| CR-V | Under dashbord center | | 547 | | | | 387/789 | 387/810 | | | | | 846 | |
| CR-Z | Under dashbord center | | 547 | | | | | | | | | | | |
| FCX Clarity | Under dashbord center | | | | | | | 387/810 | | | | | | |
| FIT | Under dashbord center | EV | 547 | | | 387/789 | | 387/810 | | | | | 846 | |
| HR-V | | | | | | | 387/789 | | 387/810 | | | | 846 | |
| INSIGHT | Under dashbord center | | 547 | | | | | | | 387/810 | | | | |
| ODYSSEY | Under dashbord center | | 547 | | 387/789 | | | 387/810 | | | | | | |
| PASSPORT | Under dashbord center | | | | | | | | | 387/810 | | | | |
| PILOT | Under dashbord center | | 547 | | | | 387/789 | 387/810 | | | | | | 846 |
| RIDGELINE | Under dashbord center | | 547 | | | | | 387/810 | | | | | | |

2011- NO time series EDR But may have Delta V single value

Honda only assists law enforce -ment voluntar ily

# Honda EDR Evolution

1. One set of DL's thru 2015
   a) Roll rate polarity varies by supplier – polarity added under graph
   b) Accel data instantaneous
2. 2016-17 Gen 2 adds "Honda sensing" & Cruise. Updated Data Lims.
3. 2018 – 3rd Gen to include "ig switch status"
   a) Airbag deploys even if ign off until car stops
   b) Markets outside US/Japan do not get precrash data
4. 2021 – 4th Gen –'21-'22(please send files!)

# Honda Gen 2 EDR

- If equipped may have semi autonomous vehicle data elements

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3**

| Time Stamp (sec) | PCM Derived Accelerator Pedal Position, % full | Forward Collision Warning (Not Warning/ Warning) | Collision Mitigation Braking System (Not Engaged/ Engaged) | Collision Mitigation Braking System, Forward Collision Warning (On/Off) | Lane Departure Warning (Not Warning/ Warning) | Road Departure Mitigation (Not Engaged/ Engaged) | Road Departure Mitigation, Lane Departure Warning (On/Off) |
|---|---|---|---|---|---|---|---|
| -5.0 | 16 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -4.5 | 8 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -4.0 | 11 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -3.5 | 13 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -3.0 | 18 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -2.5 | 18 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -2.0 | 17 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -1.5 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |
| -1.0 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |
| -0.5 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |
| 0.0 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |

# Honda Gen 2 Semi Auton  cont'd

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 3 of 3**

| Time Stamp (sec) | Adaptive Cruise Control (Not Engaged/ Engaged) | Adaptive Cruise Control (On/Off) | Lane Keeping Assist (Not Engaged/ Engaged) | Lane Keeping Assist (On/Off) | Cruise Control (Not Engaged/ Engaged) | Cruise Control (On/Off) |
|---|---|---|---|---|---|---|
| -5.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -4.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -4.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -3.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -3.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -2.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -2.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -1.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -1.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -0.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| 0.0 | Not engaged | On | Not engaged | On | Not Engaged | On |

# *Reminders: UNIQUE* TO HONDA

1. Most manufactures record the speed calculated in the powertrain control module as it is being sent **TO** the speedometer on the CAN bus.
2. Honda records the speed being displayed **ON** the speedometer, after the PCM calculated speed has been dampened or “smoothed”.
3. The last data point labeled 0.0 is sampled at AE. -0.5 is -0.01 to -0.50. While sampled at AE, speedometer processing delays what is being reported, so it lags the true speed at impact during hard braking.

# Chevrolet SDM Phase In 21_5 87%

SDM50

X DV

+ SPEED

+Y DV

563

3 events

GREEN = 300 ms Longitudinal Delta V | Red = Precrash Data & 150ms Longitudinal Delta V | Blue = Precrash Data & 300ms XY Delta V | 3 Events | Part 563 Intent | SDM50 | GM

| Model | 1994 | 1995 | 1996 | 1997 | 1998 | - | 1999 | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Astro | | | 2888 | 2888 | 2888 | - | 2888 | 2829 | 2829 | 2829 | 2829 | 2829 | 2829 | | | | | | | | | | | | | | | | | | |
| Avalanche | | | | | | | | | | 2829 | 3003 | 3003 | 3003 | 3003 | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | * | | | | | | | | | |
| Aveo | | | | | | | | | | | | | | | | | 3320 | 3320 | 3320 | | 454 | * | | | | 454 | | | | | |
| Beat (India) | | | | | | | | | | | | | | | | | | | | | | 454 | 454 | | | | | | | | |
| Blazer | | | | | 2888 | - | 2888 | 2829 | 2829 | 2829 | 2829 | 2829 | 2829* | * | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 |
| Bolt EV | | | | | | | | | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 |
| Camaro | | | 2888 | 2888 | 2888 | - | 2829 | 2829 | 2829 | 2829 | | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Caprice | 3002 | 3002 | 3002 | | | | | | | | | | | | | | | | 452 | 454 | police | police | police | police | police | police | | | | | |
| Captiva | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | * | Sport | Sport | | | | | | | |
| City Express | | | | | | | | | | | | | | | | | | | | | | | 780 | 780 | 780 | 780 | 780 | | | | |
| Colorado | | | | | | | | | | | | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 852 |
| Corvette | | | | 2888 | 2888 | - | 2829 | 2829 | 2829 | 2829 | 2829 | 2829 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 3003 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 845 | 845 | 845 | 845 |
| Cruze | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | | | |
| Equinox | | | | | | | | | | | | | 3004 | 3004 | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Express | | | 2888 | 2888 | 2888 | - | 2888 | 2888 | 2829 | 2829 | 3003 | 3003 | 3003 | 3003 | 3003 | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Impala | | 3002 | 3002 | | | - | | 2829 | 2829 | 2829 | 2829 | 2829 | 2829 | 3320 | 3320 | 3320 | 3320 | 3320 | 3320 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | | | | |
| Kodiak | | | | | | | | | | | 2829 | 2829 | 2829 | 2829 | 2829 | 2829 | 2829 | | | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| Low Cab Fwd | | | | | | | | | | | | | | | | | | | | | | | | | | | 454 | | | | |
| Malibu | | | | 2888 | 2888 | - | 2888 | 2829 | 2829 | 2829 | 2829 | 3321*** | 3321 mix | 3321 | 3321 | 3321 | 3321 | 3321 | 3321 | 3321 | 3321 (2012 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Matiz | | | | | | | | | | | | | | | | | | | | 454 | | | | | | | | | | | |
| Orlando | | | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | | | | | | | | | |
| Silverado | | | | 2888 | 2888 | - | 2888 | MIX | 2829 | 2829 | 3003 | 3003 | 3003 | 3003 | 3320 ROS*mix | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 852 |
| Sonic | | | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | | | |
| Spark | 1994 | 1995 | 1996 | 1997 | 1998 | - | 1999 | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | |
| SSR | | | | | | | | | | | 3003 | 3003 | 3003 | 3003 | | | | | | | | | SS | 454 | 454 | 454 | | | | | |
| Suburban | | | | 2888 | 2888 | - | 2888 | MIX | 2829 | 2829 | 3003 | 3003 | 3003 | 3003 | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 845 | 845 | 852 |
| Tahoe | | | | 2888 | 2888 | - | 2888 | MIX | 2829 | 2829 | 3003 | 3003 | 3003 | 3003 | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 845 | 845 | 852 |
| Tracker | | | DLC | DLC | DLC | - | DLC | DLC | DLC | DLC | DLC | DLC | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 | |
| TrailBlazer | | | | | | | | | | 2829 | 2829 | 2829 | 3293 | 3293 | 3293 | 3293 | 3293 | | | | | | | | | | | | 454 | 454 | 454 |
| Traverse | | | | | | | | | | | | | | | | | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Trax | | | | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | |
| Volt | | | | | | | | | | | | | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | | | | |
| | 14% | 48% | 76% | 94% | 100% | | 20% | 74% | 86% | 96% | 98% | 100% | 14% | 43% | 71% | 79% | 82% | 91% | 92% | 22% | 96% | | 100% | | 100% | | | | | |
| PHASE IN | Delta V | Delta V | Delta V | Delta V | Delta V | | Pre-Crash | Pre-crash | Pre-crash | Pre-Crash | Pre-Crash | Pre-Crash | Pre + XY | Pre + XY | Pre + XY | Pre + XY | Pre + XY | Pre + XY | Pre + XY | 563 | 563 | 563 | 563 | | | | | | | | |

Freeze Frame Data

** = 150ms of Longitudinal Delta V instead of 300ms

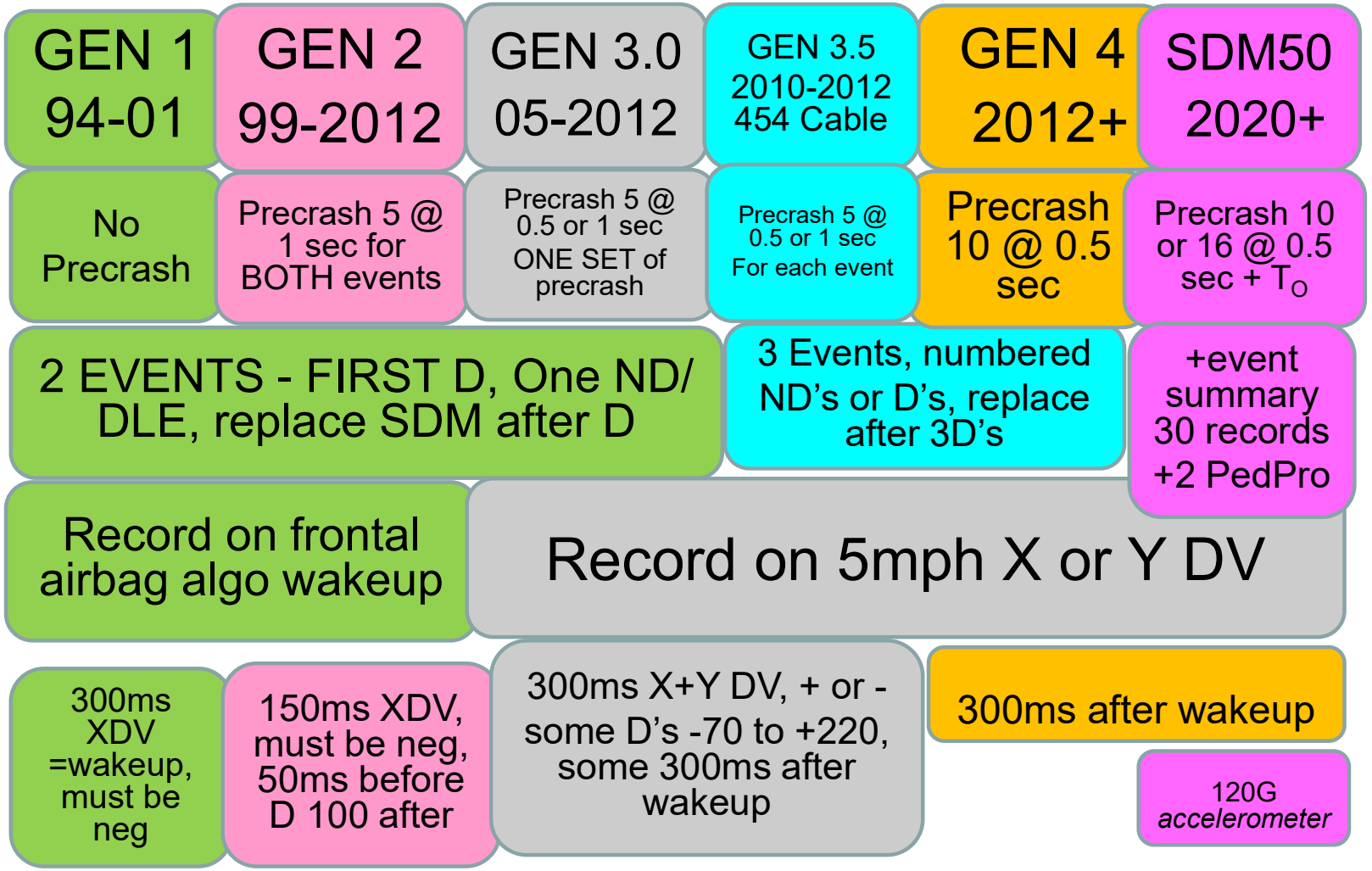
GM full chart printed in color in your book
and is on student reference disk

# Chevrolet Recent Changes 87% 563

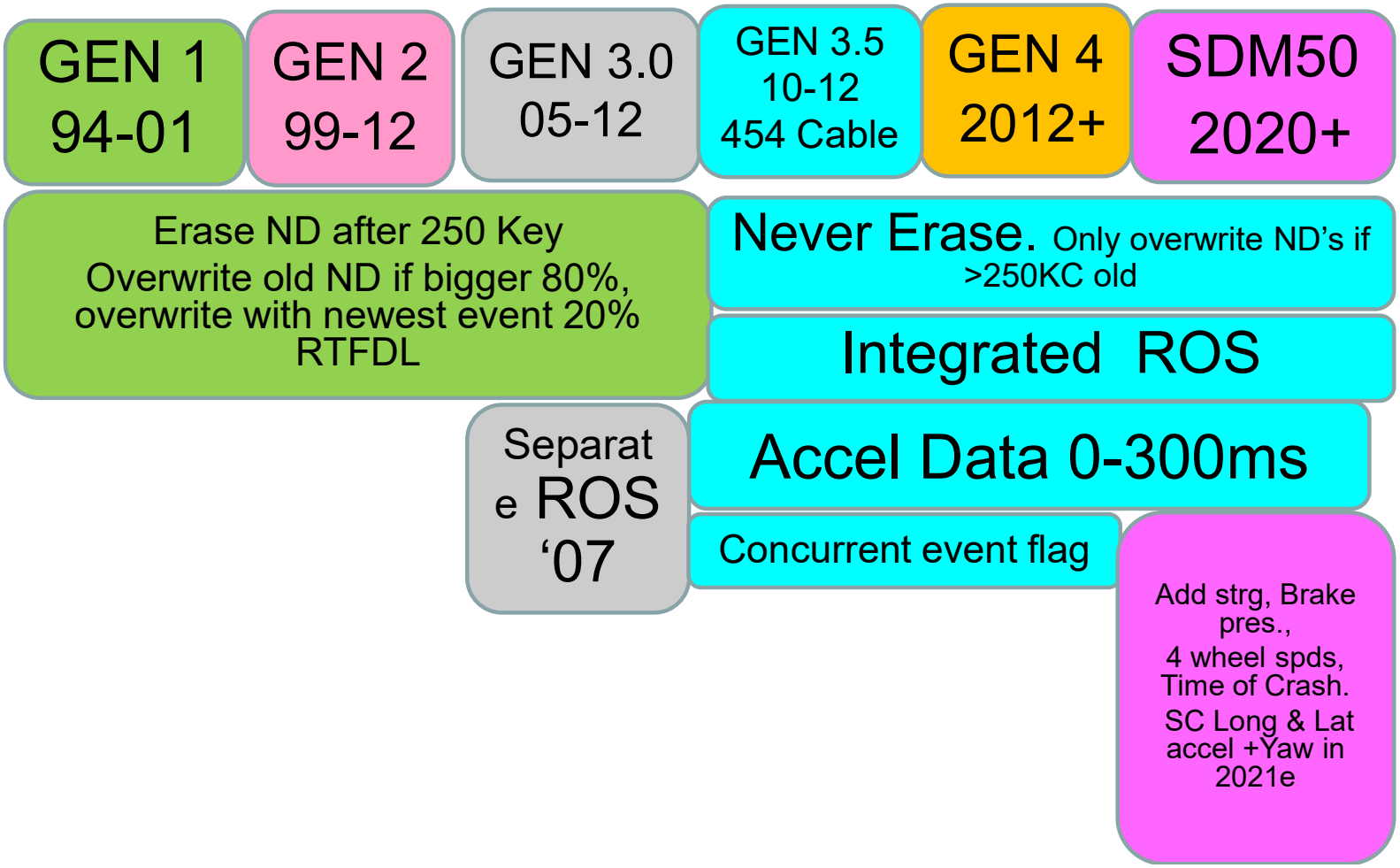
| | Part 563 Intent | | | | | | | | SDM50 | GM | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | **2012** | **2013** | **2014** | **2015** | **2016** | **2017** | **2018** | **2019** | **2020** | **2021** | **2022** | **2023** |
| Astro | | | | | | | | | | | | |
| Avalanche | 454 | 454 | * | | | | | | | | | |
| Aveo | | 454 | * | | | | 454 | | | | | |
| Beat (India) | | | 454 | 454 | | | | | | | | |
| Blazer | | | | | | | | 454 | 454 | 454 | 454 | 454 |
| Bolt EV | | | | | | | 454 | 454 | 454 | 454 | 454 | 454 |
| Camaro | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Caprice | **454** | police | police | police | police | police | police | | | | | |
| Captiva | 454 | 454 | * | **Sport** | **Sport** | | | | | | | |
| City Express | | | | **780** | **780** | **780** | **780** | **780** | | | | |
| Colorado | 3003 | | | **454** | **454** | **454** | **454** | **454** | 454 | 454 | 454 | **852** |
| Corvette | 3003 | 3003 | 454 | 454 | 454 | 454 | 454 | **454** | **845** | **845** | **845** | **845** |
| Cruze | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **454** | | | |
| Equinox | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | 454 | 454 | 454 | 454 |
| Express | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | **454** | **454** | 454 | 454 | 454 |
| Impala | **454** | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **454** | | | |
| Kodiak | **2012** | **2013** | **2014** | **2015** | **2016** | **2017** | **2018** | **2019** | **2020** | **2021** | **2022** | **2023** |
| Low Cab Fwd | | | | | | | | 454 | | | | |
| Malibu | 3321 | 3321 (20 | 2 454 | 454 | 454 | 454 | 454 | 454 | **454** | 454 | 454 | 454 |
| Matiz | 454 | | | | | | | | | | | |
| Orlando | 454 | 454 | 454 | | | | | | | | | |
| Silverado | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **454** | **454** | **852** |
| Sonic | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | | | |
| Spark | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **454** | **454** | |
| SSR | | | | **SS** **454** | **454** | **454** | | | | | | |
| Suburban | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **845** | **845** | **852** |
| Tahoe | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **845** | **845** | **852** |
| Tracker | | | | | | | 454 | 454 | **454** | **454** | **454** | |
| TrailBlazer | | | | | | | | | | **454** | **454** | **454** |
| Traverse | 454 | 454 | 454 | **454** | **454** | **454** | **454** | 454 | **454** | **454** | **454** | **454** |
| Trax | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | **454** | **454** | **454** | |
| Volt | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | | | | |

No Precrash in 2013-5 Express/Savanna

# GM EDR Evolution



# GM EDR Evolution #2



# 2020 Corvette SDM50 Data

**Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 1 of 6**

| Time (sec) | Accelerator Pedal, % Full (%) | ABS Activity | Brake Pedal Position (%) | Driver Applied Brake Pedal Pressure (kPa) | Driver Applied Brake Pedal Pressure Detected | Brake Pedal Moderate Travel Achieved |
|---|---|---|---|---|---|---|
| -8.0 | 0 | Off | 0 | 0 | False | False |

**Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 2 of 6**

| Time (sec) | Cruise and Speed Limiter Switch Status | Cruise Secondary Switch Status | Cruise Control Active | Adaptive Cruise Control Active Authenticated | Transmission Estimated Gear | Transmission Shift Lever Position | Reduced Engine Power Mode Indicator On |
|---|---|---|---|---|---|---|---|
| -8.0 | No Activation | No Activation | Inactive | Inactive | Ninth | Forward Range A | Off |

**Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 3 of 6**

| Time (sec) | Engine RPM (RPM) | Engine Torque (foot-pounds) | Engine Throttle, % Full (%) | Speed, Vehicle Indicated (MPH [km/h]) | SC accel + Yaw 2021 | Right Front Wheel Angular Velocity (RPM) | Left Front Wheel Angular Velocity (RPM) |
|---|---|---|---|---|---|---|---|
| -8.0 | 1,301 | 50.5 | 16 | 44.1 [71] | | 595 | 598 |

**Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 4 of 6**

| Time (sec) | Right Rear Wheel Angular Velocity (RPM) | Left Rear Wheel Angular Velocity (RPM) | Steering Wheel Angle (deg) | Propulsion System Active | Run/Start Input | Ignition Prolongation Time (sec) | Secondary Collision Prolongation Timer (sec) |
|---|---|---|---|---|---|---|---|
| -8.0 | 599 | 598 | 0 | True | High | 5.0 | 0.0 |

**Pre-Crash Data -8.0 to 0.0 sec (Record 1) - Table 5 of 6**

| Time (sec) | Antilock Brake System Failed | Brake Pedal Override Flag | Traction Control System Present | Traction Control System Failed | Traction Control System Enabled | Traction Control System Active | Traction Control System Driver Intent |
|---|---|---|---|---|---|---|---|
| -8.0 | False | False | True | False | True | Traction Control Off | Enabled |

# More 2020 Corvette SDM50

- 3 AACN records in Eng Translation Rpt
- GPS Lat/Long at crash
- Turn Signal Switch Status
- Second and third row seat belt status
- Up to 30 Event Summary records give clear order of events
- Outside Air Temp, Brake Overrides
- Traction Control activate
- Auxilliary Crash Sensor Data

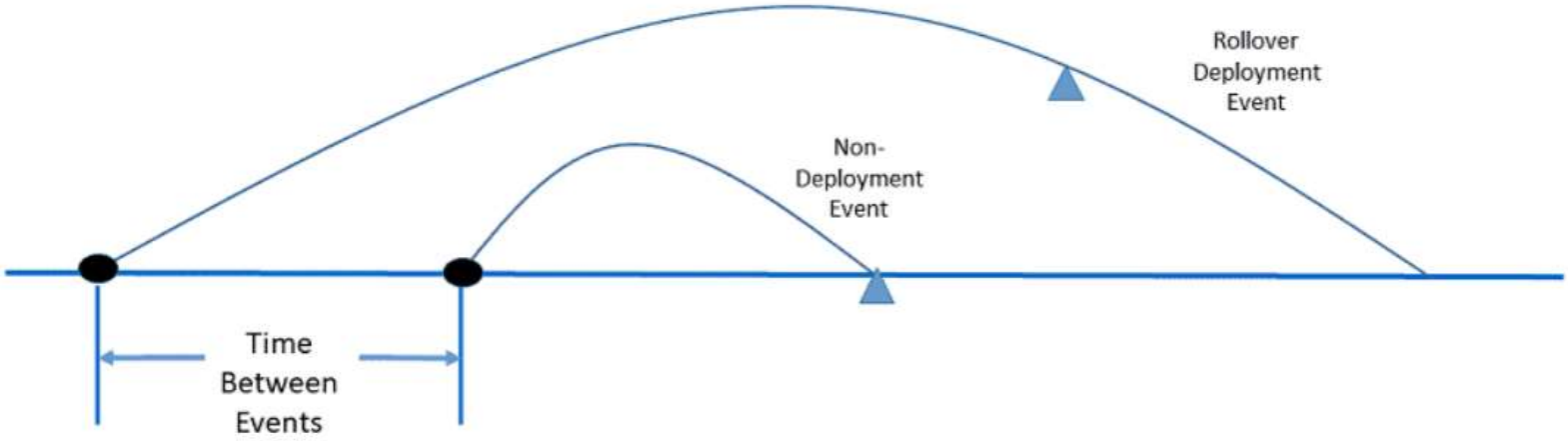
EVERYTHING GM COULD IMAGINE

# GM Gen 3.5 & 4 Features

1. 2010 454 cable SDM's do not require replacement after a D (until 3 D's).
2. Events & deployments all numbered vs. labeled D/ND.
3. Events may not be in order algorithms started, they are in order they qualified for recording (see graphic next page)
4. **Align precrash data for all events to see relationship**
5. Black plastic SDM housings
6. Accelerometer max values listed in DL's

# GM Concurrent Event Flag

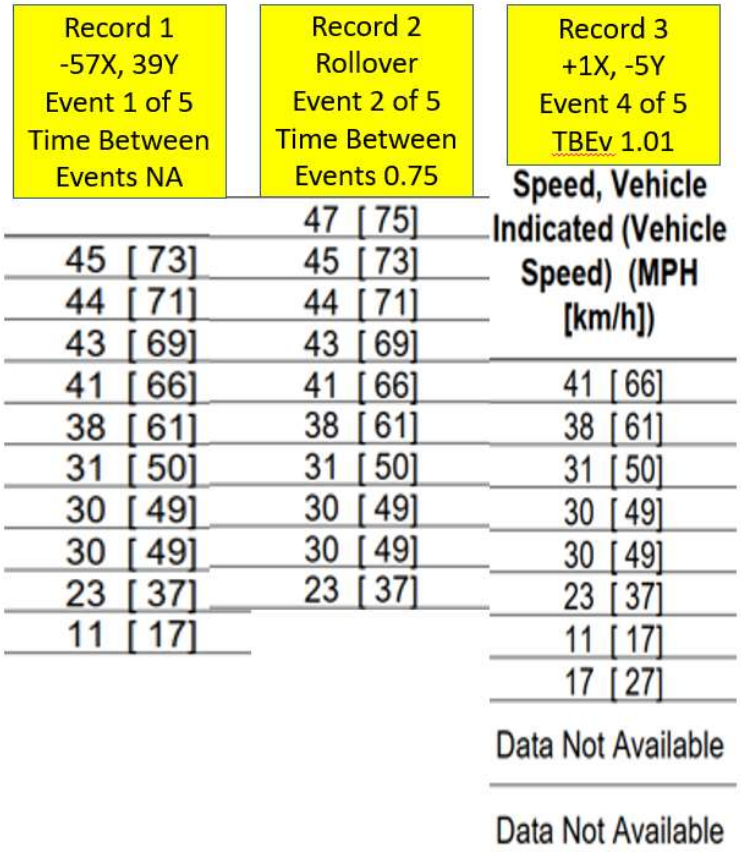
-A Concurrent Event is when two events are happening nearly simultaneously. The "Concurrent Event Flag Set" parameter will indicate "Yes" if one event begins, but before that event is qualified, another event begins and is qualified.
A Non-Deployment event typically becomes qualified if that event exceeds the 5 MPH (8 km/h) delta V recording threshold and the event has concluded. A deployment event (FSR or Rollover) becomes qualified when a deployment has been commanded for that event.
Example of a Concurrent Event:
A Rollover event begins. Before the Rollover event is qualified, a Non-Deployment event begins and is qualified. Sometime after the Non-Deployment event is qualified, the Rollover event is qualified. The Non-Deployment event will be recorded in the first open record even though the Rollover event enabled before the Non-Deployment event. The Rollover event will be recorded in the next open record. The "Concurrent Event Flag Set" parameter will indicate "Yes" for the Rollover event. The "Time Between Events" parameter will indicate the time from the start of the Rollover event to the start of the Non-Deployment event.




| Event Recor #1 | Event Record #2 |
|---|---|
| Event Record Type = Non – Deployment | Event Record Type = Rollover |
| Concurrent Event Flag = No | Concurrent Event Flag = Yes |
| Time Between Events = NA | Time Between Events = XX seconds |

# When Concurrent Event Flag is Yes, Overlay Precrash Data from all events

- The event number gives when the event QUALIFIED for recording, not when I started
- Time between events is time between consecutive wake-ups, NOT always time between qualifying events
- Recap Delta V or, if rollover, event type over precrash data to assist understanding



# Noteworthy GM news

1. Conti SDM-30 data incomplete file not accurately interpreted by CDR – not fixable. Must send to Conti for readout and interp. (data lims detail how)
2. GM note 6 (no key needed) vehicles initially had 2 fuses, SDM bat and SDM ignition. Newer GM vehicles may have only ONE SDM fuse, normally bat, but police packages may only power it with ignition

1. Foreign coverage – China since 99 (but only Gov't/NPS can use CDR), Australia/Korea/Thailand since 07

# GM Active Safety Control Module

1. First installed in 2013 Cadillac ATS
2. Only with GM option UGN or UKL
3. Initially released in CDR 17.9 to get raw data file only for future interpretation.
4. Interpretation capability added in 19.1
5. REQUIRES CDR 900 interface!
6. Some vehicles have TWO ASCM modules.

# GM ASCM CDR 21.5 Coverage

| Model | Location | Model Year | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 digits Bosch CDR cable ID shown | | | | | | | | | | | | |
| **BUICK** | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| ENCLAVE | cargo left front of wheel well | | | | | | 833 1B | | | | | |
| ENVISION | rear seat left trim by wheel wl | | | | 833 1A | 833 1B | | | | 854 3-LC | | |
| LACROSSE | trunk left | | | | | 833 1B | | | | | | |
| REGAL | trunk left | | | | | | 833 1B | | | | | |
| **CADILLAC** | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| ATS | trunk left | 833 1A | | | | | 833 1B | | | | | |
| CT4 | Trunk upper shelf | | | | | | | | 854 3-LC | 855 3-HC | | |
| CT5 | Trunk upper shelf | | | | | | | | 854 3-LC | 855 3-HC | | |
| CT6 | Primary right, secondary left | | | | 833 1A | | 833 1B, 834 2 | | | | | |
| CTS SEDAN | trunk left | | 833 1A | | | | 833 1B | | | | | |
| CTS | trunk left | | | | 833 1A | | 833 1B | | | | | |
| ESCALADE | under left qtr window trim | | | 833 1A | | | 833 1B | | | 854/855 LC/HC | | |
| LYRIC | | | | | | | | | | | | 833 1B |
| SRX | under left rear seat | 833 1A | | | | | | | | | | |
| XT4 | cargo left front of wheel well | | | | | | | 833 1B | | | | |
| XT5 | cargo left front of wheel well | | | | | 833 1B | | | | | | |
| XT6 | | | | | | | | | 833 1B | | 833 1B/834 2 | |
| XTS | trunk left | 833 1A | | | | | 833 1B | | | | | |
| **CHEVROLET** | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| BOLT | trunk left | | | | | | | | | | 834 2 | |
| BLAZER | trunk left | | | | | | | 833 1B | | | | |
| MALIBU | trunk left | | | | 833 1A | 833 1B | | | | | | |
| SILVERADO | | | | | | | | | | | | 855 |
| SUBURBAN | | | | | | | | | | | 854 3-LC | |
| TAHOE | | | | | | | | | | | 854 3-LC | |
| TRAVERSE | trunk right | | | | | | 833 1B | | | | | |
| VOLT | trunk left | | | | | 833 1B | | | | | | |
| **GMC** | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| ACADIA | cargo left front of wheel well | | | | | 833 1B | | | | | | |
| HUMMER EV | | | | | | | | | | | 855 3-HC | |
| SIERRA | | | | | | | | | | | | 854 3-LC |
| YUKON | | | | | | | | | | 854 3-LC | | 854/855 |

# GM Active Safety Control Module

| Mkt | Year/Make | Model | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|---|---|
| 4 | 2018 Cadillac | ATS (note 6) | ACM | Data | CANplus | CANplus - F00K108287 | CANplus - F00K108454 | Center tunnel |
| | | | ASCM (Note1) | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 833 | In the luggage compartment, left side, near rear, on outboard side of brace, behind trim |
| 6 | 2018 Cadillac | ATS-L (note 6) | ACM | Data | CANplus | CANplus - F00K108287 | CANplus - F00K108454 | Center tunnel |
| 4,6 | 2018 Cadillac | CT6 (note 6) | ACM | Data | CANplus or CDR 900 | CANplus - F00K108287 CDR 900 - click here | CANplus - F00K108454 CDR 900 - click here (use listed cable above) | Center tunnel |
| | | | ASCM (Note2) how to image | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 834 | In the luggage compartment, right center, mounted to the right side in the battery well. The secondary ASCM is on the left. |
| | | | ASCM (Note1) | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 833 | In the luggage compartment, right center, mounted to the right side in the battery well |

# GM Active Safety Module




1. Must be downloaded separately from ACM
2. Icon will appear after VIN input for any vehicle that MIGHT have Super Cruise
3. Module /data will only be present if option “UKL” (Super Cruise) or “UGN” (Adv Driver Assist) has been purchased for the subject vehicle (most DO NOT have it).

## GM ASCM EDR – **New Icon**

Effective with V 19.6 of the software, the ASCM “RADAR” Icon has been replaced by new Icon



OLD




NEW

- Best guess is that there is more upcoming automated sensing data, from other mfrs, that have more than just frontal radar.

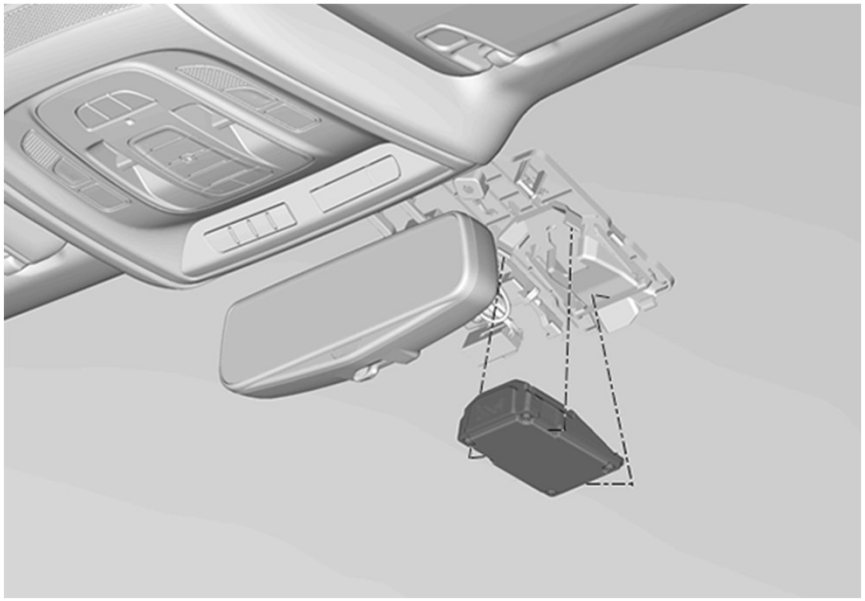
# GM Active Safety Module

1. Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 4 seconds before the end of an auto braking event or 4 seconds before an airbag deployment. ONE EVENT ONLY
2. Data is recorded in non-volatile memory at the next proper power-down of the module. Data may **not** be stored if the battery power is lost before the power-down.

# GM Auto Emergency Braking

1. AEB will prevent many crashes and reduce the severity of crashes still happening
2. If there is still a crash, the AEB system may store up to 3 camera images in FCM*:
   a) 4 sec before event (*Fwd Camera Module)
   b) At event
   c) 4 sec after event
3. Some accessible with CDR 21.2 – others must be requested with proper legal authority from GM

# Front Camera Module (FCM)

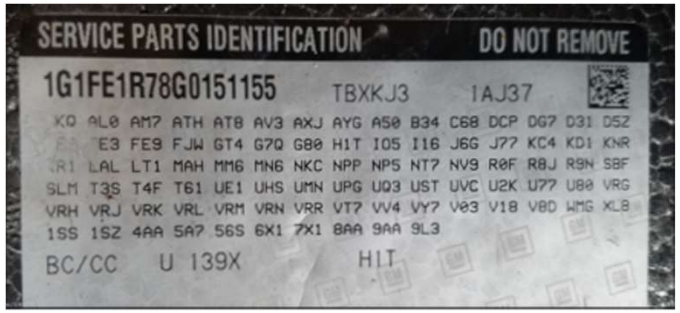


FCMs are mounted on the windshield near the rear view mirror.

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# GM FCM EDR

- GM released the ability to read data from Front Camera Modules(FCM) with V 21.2 of the CDR Software (Date: 8/2021)
- The data consists of photos taken before, during, and after the crash.
- Not all GM cars are equipped with FCMs
- Some cars have FCMs but are not supported at this time
- We can identify which cars have an FCM by
  - Looking for presence of the camera module near the rear view mirror
  - Verifying the (RPO) UEU and UGN in the vehicles Service Parts Identification sticker, or through the QR code on the FMVSS Sticker (see photos)






# FCM access released Aug 2021 in GREEN

| Model | Location | Model Year | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 digits Bosch CDR cable ID shown | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| **BUICK** | | | | | | | | | | | |
| ENCLAVE | cargo left front of wheel well | | | | | | ASCM | | | | |
| ENCORE | | | | | | | | | FCM | | |
| ENVISION | rear seat left trim by wheel wl | | | | ASCM | | | FCM | | | |
| LACROSSE | trunk left | | | | | ASCM | | | | | |
| REGAL | trunk left | | | | | | ASCM | | | | |
| **CADILLAC** | | | | | | | | | | | |
| **ATS** | | | | | | | | FCM | | | |
| CT4 | | | | | | | | | | | |
| CT6 w/o UKL | Trunk right | | | | | ASCM | without supercruise | | | | |
| CT6 w/UKL | Primary right, secondary left | | | | 834 | | Supercruise Option Code UKL | | | | |
| CTS SEDAN | trunk left | | ASCM | | | | | | | | |
| CTS | trunk left | | | | ASCM | | | FCM | | | |
| ESCALADE | under left qtr window trim | | | ASCM | | | | | | | |
| SRX | under left rear seat | ASCM | | | | | | | | | |
| XT4 | cargo left front of wheel well | | | | | | | ASCM | | | |
| XT5 | cargo left front of wheel well | | | | | ASCM | | | | | |
| XT6 | | | | | | | | | | ASCM | |
| XTS | | | | | | | | FCM | | | |
| **CHEVROLET** | | | | | | | | | | | |
| BLAZER | trunk left | | | | | | | ASCM | | | |
| CRUZE | | | | | | | | FCM | | | |
| EQUINOX | | | | | | | | FCM | | | |
| MALIBU | trunk left | | | | ASCM | | | | | | |
| | | | | | | | | | | | |
| SILVERADO | | | | | | | | FCM | | | |
| SPARK | | | | | | | | FCM | | | |
| TRAVERSE | trunk right | | | | | | ASCM | | | | |
| VOLT | trunk left | | | | | ASCM | | FCM | | | |
| **GMC** | | | | | | | | | | | |
| ACADIA | cargo left front of wheel well | | | | | ASCM | | | | | |
| SIERRA | | | | | | | | FCM | | | |
| TERRAIN | | | | | | | | FCM | | | |

# GM FCM EDR

- The Lookup file will identify which GM vehicles may have an

| Mkt | Year/Make | Model | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|---|---|
| 4,6 | 2019 Cadillac | XTS (note 6) | ACM | Data | CANplus | CANplus - F00K108287 | CANplus - F00K108454 | Center tunnel, forward |
| | | | ASCM (Note1) | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 833 | In the luggage compartment, under the left side trim panel |
| | | | FCM (Note1) how to image | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 856 | Center - top interior of windshield |

ACM Data may be present

ASCM Data may be present

FCM Data may be present



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# GM FCM EDR

- The Lookup file will identify which GM vehicles may have an

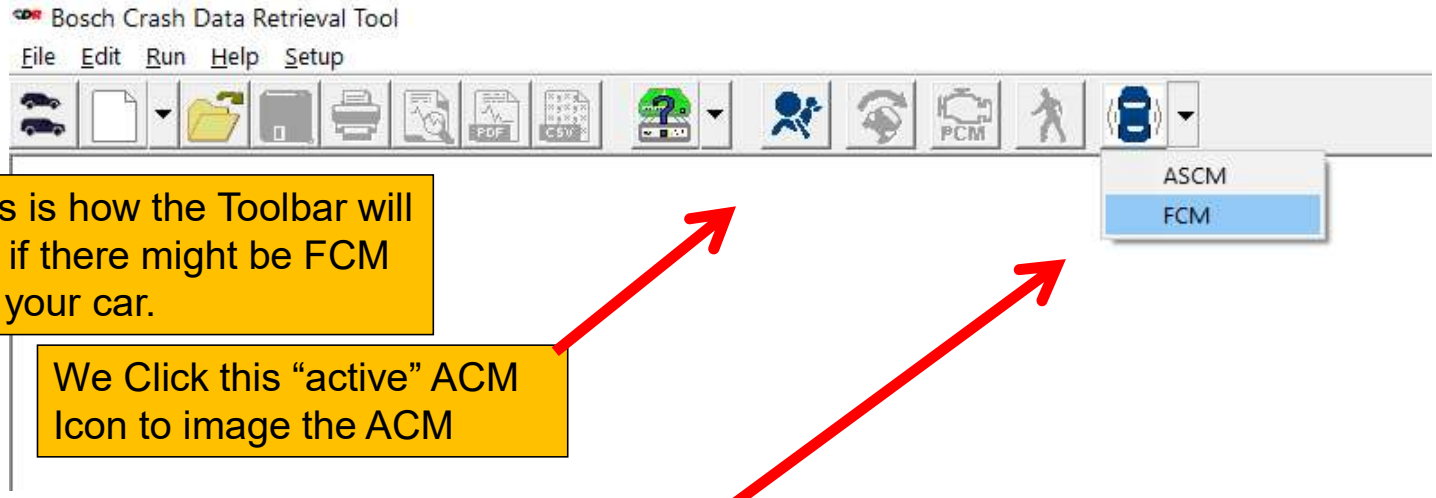
| Mkt | Year/Make | Model | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|---|---|
| 4,6 | 2019 Cadillac | XTS (note 6) | ACM | Data | CANplus | CANplus - F00K108287 | CANplus - F00K108454 | Center tunnel, forward |
| | | | ASCM (Note1) | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 833 | In the luggage compartment, under the left side trim panel |
| | | | FCM (Note1) how to image | Data | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 856 | Center - top interior of windshield |

Note we get hints on how to DL the FCM – How to image



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# GM ASCM EDR – Bosch CDR 900




And this is how the Toolbar will appear if there might be FCM data in your car.

We Click this “active” ACM Icon to image the ACM

Then once the ACM Download is complete and saved, we click this “active” Dropdown Icon to image the FCM and then save that file as well. Now we have 3 things to image!!

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# GM ASCM EDR – Bosch CDR 900




We repeat the idea that we discussed in the DLC training: If you fail to download and save the data from all available modules you are driving away and leaving unread data in the car.

And, just because you can read the ACM via the DLC does no guarantee that you have communication with the FCM, ASCM, FCM, ROS, etc .



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# GM FCM EDR

- There are several network pathways in modern vehicles
- Damage may allow DLC communication to one module and prevent comm to another module.
- Do the research to see if your car is equipped with all indicated modules (ACM, ASCM, FCM) and read as necessary (DLC or DTM or maybe one of each)
- At this time FCMs are only readable on 2019MY and later GM's
- If you previously read the SDM and ASCM on an FCM now-supported vehicle, you may have to return to read the data from this new module type
- The FCM raw data was NOT saved in any ASCM files you may have read.

# GM FCM EDR

- When a CDR file contains image data, the user can export the data using options from the CDR software:




We Click this “Icon” to export FCM images

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# GM FCM EDR

- The Front Camera Module (FCM) can record two events. Recording is triggered by “events of interest”. The following are possible “events of interest” that trigger data recording listed in the priority order:
- Airbag Deployment Event,
- Pretensioner Deployment Event,
- Non-Deployment Event,
- Pedestrian Braking,
- Collision Imminent Breaking (CIB)
- Pedestrian Warning,
- Front Collision Alert,
- Lane Departure Warning.



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# GM FCM EDR

- The new event is stored only if it is a higher priority than the priority of one of the currently stored events. The new event overwrites the lowest priority event that is already stored.
- If all previously stored events are of equal priority (and are lower priority than the new event), the oldest stored event is the one being overwritten.
- The image data is not rendered directly in the CDR program .



- It must be exported to a file in the images' native format.

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# GM FCM EDR

- Photos are black and white *.pgm files (**P**ortable **G**ray **M**ap)
- You will need a *.pgm file viewer – many free *.pgm apps are available.
  (Try www.coolutils.com/online/PGM-to-JPG)
- Users can regenerate the images collected by the FCM as many times as they wish recreating the file as it existed when the data was collected from the vehicle.
- Yet again we see that having the Raw file IS the evidence – As an analyst, *if sharing with another,* you want the raw FCM cdrx file, not someone else's downloaded *.pgm images.
- The data in the FCM CDRx file remains untouched and cannot be changed.
- The images can be downloaded and generated from the Raw file as many times as necessary.






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# GM FCM EDR

- If the exported image file appears to be changed by another user, you can re-export the image directly from the CDR file it was created; therefore, removing any doubt that the data source was changed after the file was generated.
- One more issue… The FCM image will be upside down.





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# GM FCM EDR

- The solution is to “Invert” or Flip the photo. DO NOT “rotate” it.



- Original FCM “upside down” photo



- A successfully “inverted” or “flipped” photo

(if you convert pgm to jpg, grab the photo at the top and drag the top thru the bottom of the photo to invert it)

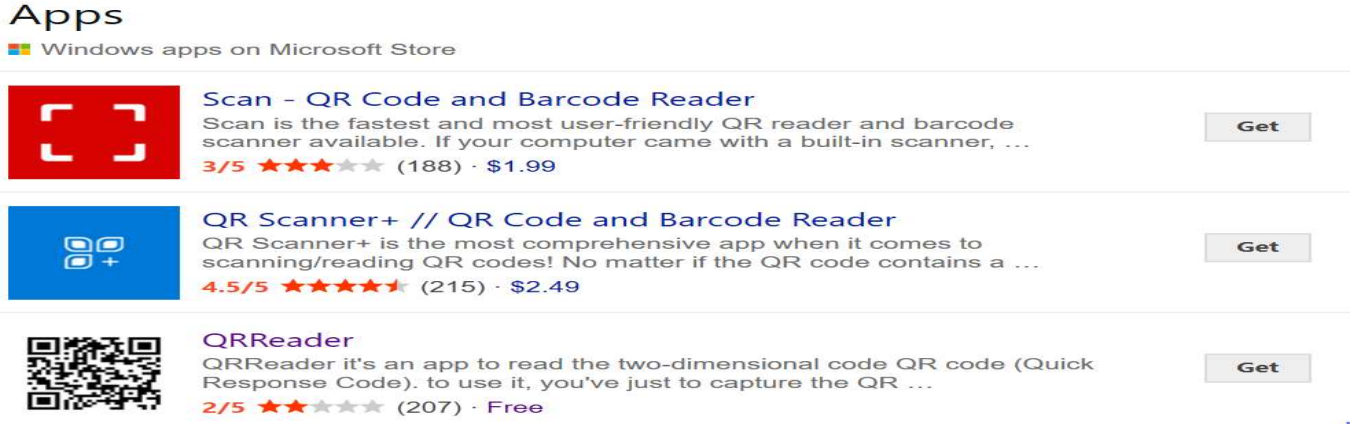


- A confusing, inaccurate “rotated” photo.

(Note car appears to be driving on *wrong side of road*)

# GM RPO Sticker Change

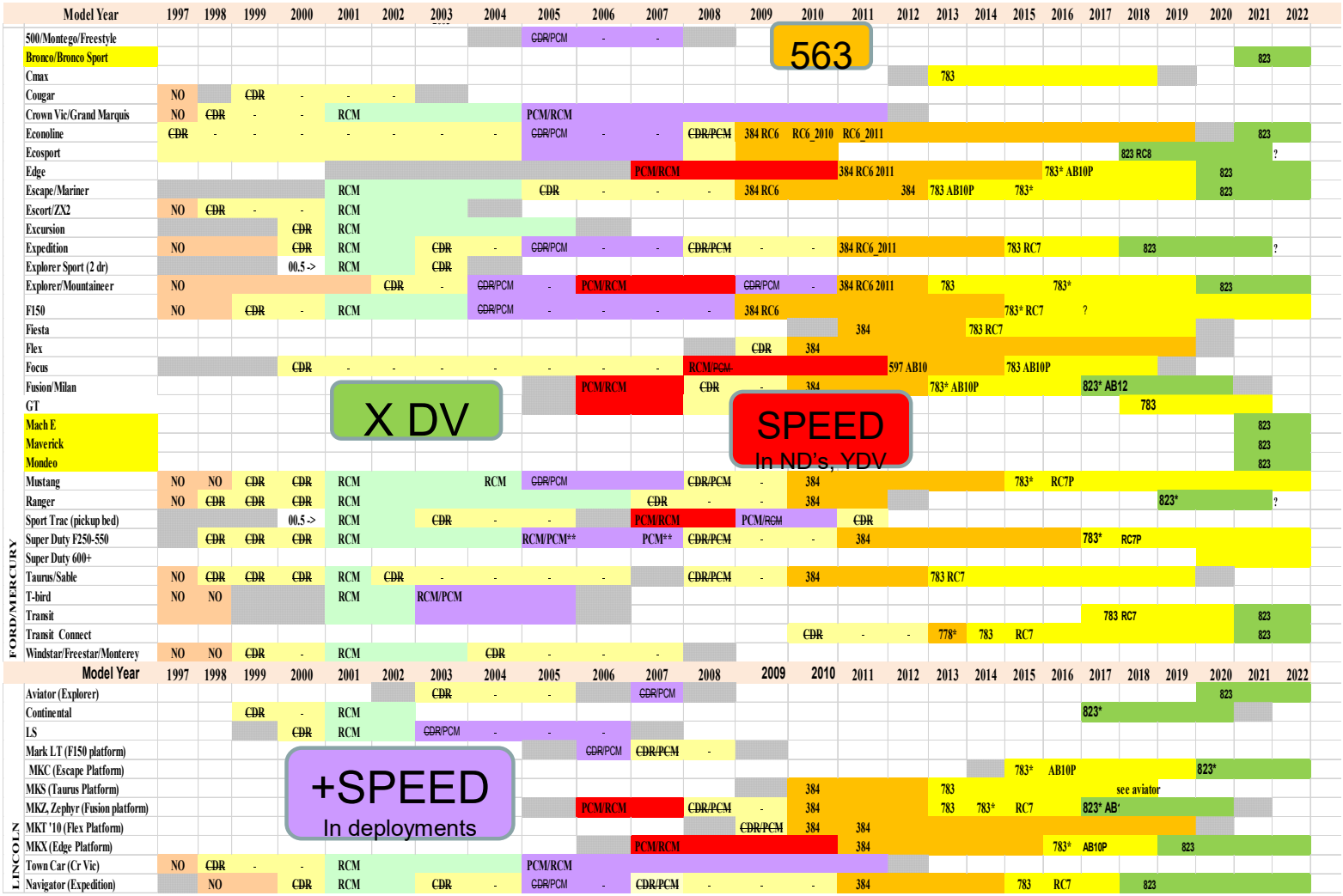
1. GM formerly had a sticker with 3 digit RPO (Regular Production Option) codes printed in the glove box or trunk
2. 2018-ish switched to a “QR code”
3. Google search QR code reader and install one on your phone.  Some free, some cost



# ETR Access Removed

- CDR 21.0 removes the ability to see and save the Engineering Translation Report as a PDF file.
- Presumably a policy decision
- If ETR is important to you (speed in Gen3 or later SDM says invalid), reinstall an older version of CDR software to access it

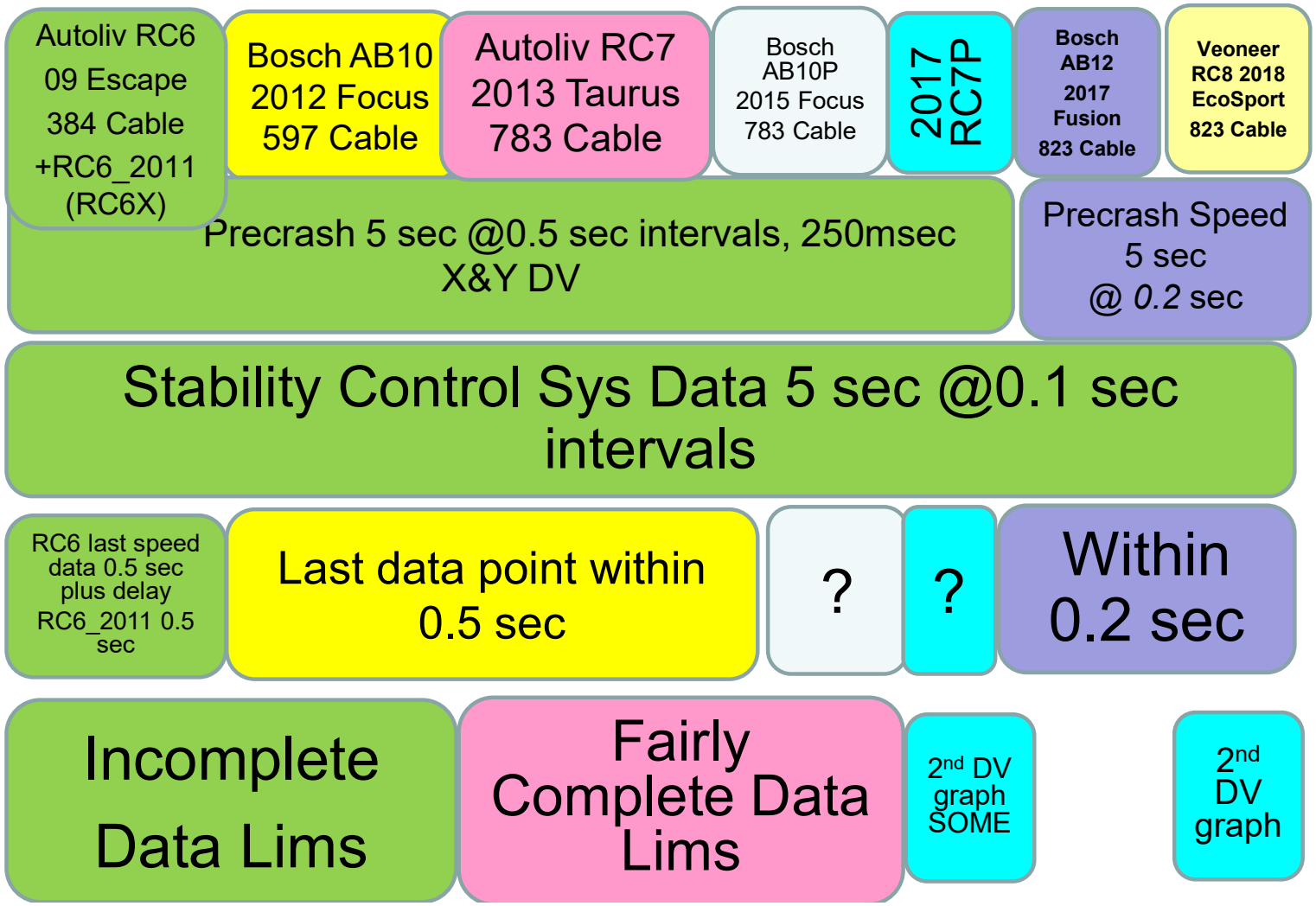
# Ford CDR 21.3 Coverage 69%



# Ford w/Stability Control Data 21.5

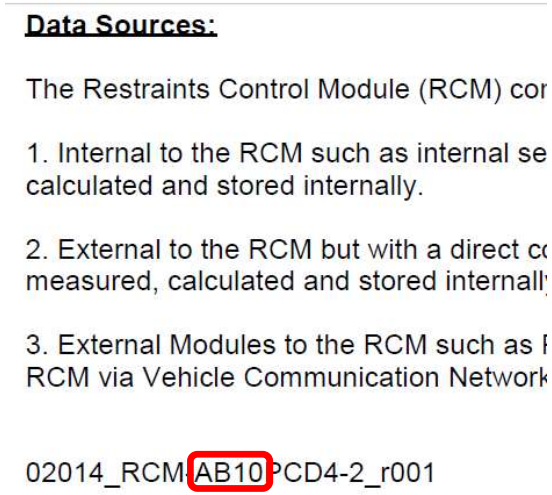
| | Model Year | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FORD/MERCURY (Merc Discontinued 2011) | 500/Montego/Freestyle | | | | | | | | | | | | | | |
| | Bronco/Bronco Sport | | | | | | | | | | | | | 823 | |
| | Cmax | | | | | 783 | | | | | | | | | |
| | Crown Vic/Grand Marquis | | | | | | | | | | | | | | |
| | Econoline | 384 | | | | | | | | | | | | 823 | |
| | Ecosport | | | | | | | | | | 823 RC8 | | | | |
| | Edge | | | 384 | | | | | 783* AB10P | | | | 823 | | |
| | Escape/Mariner | 384 | | | 384 | 783 AB10P | | 783* | | | | | 823 | | |
| | Expedition | - | - | 384 | | | | 783 RC7 | | | 823 | | | | |
| | Explorer/Mountaineer | ~~CDR~~/PCM | - | 384 | | 783 | | | 783* | | | | 823 | | |
| | F150 | 384 | | | | | | 783* RC7 | | ? | | | | | |
| | F250-F550 Super Duty | - | - | 384 | | | | | | 783* | RC7P | | | | |
| | F600-750 Medium Duty | | | | | | | | | | | | | 783 | |
| | Fiesta | | | 384 | | | 783 RC7 | | | | | | | | |
| | Flex | ~~CDR~~ | 384 | | | | | | | | | | | | |
| | Focus | | | | 597 AB10 | | | 783 AB10P | | | | | | | |
| | Fusion/Milan | - | 384 | | | 783* AB10P | | | | 823* AB12 | | | | PHEV | PHEV |
| | GT | | | | | | | | | | 783 | | | | |
| | Mach E | | | | | | | | | | | | | 823 | |
| | Maverick | | | | | | | | | | | | | 823 | |
| | Mondeo | | | | | | | | | | | | | 823 | |
| | Mustang | - | 384 | | | | | 783* | RC7P | | | | | | |
| | Ranger | - | 384 | | | | | | | | | 823* | | | ? |
| | Sport Trac (pickup bed) | PCM/~~RCM~~ | | ~~CDR~~ | | | | | | | | | | | |
| | Taurus/Sable | - | 384 | | | 783 RC7 | | | | | | | | | |
| | Transit T150-T350 | | | | | | | 783 RC7 | | | | | | 823 | |
| | Transit Connect | | ~~CDR~~ | - | - | 778* | 783 | RC7 | | | | | | 823 | |
| | Windstar/Freestar/Monterey | | | | | | | | | | | | | | |
| | | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| LINCOLN | Aviator (Explorer) | | | | | | | | | | | | 823 | | |
| | Continental | | | | | | | | | 823* | | | | ? | |
| | MKC/Corsair (Escape Platform) | | | | | | | 783* | AB10P | | | | 823* | | |
| | MKS (Taurus Platform) | | 384 | | | 783 | | | | | see aviator | | | | |
| | MKZ, Zephyr (Fusion platform) | - | 384 | | | 783 | 783* | RC7 | | 823* AB' | | | | | |
| | MKT '10 (Flex Platform) | ~~CDR/PCM~~ | 384 | 384 | | | | | | | | | | | |
| | MKX/Nautilus (Edge Platform) | | | 384 | | | | | 783* | AB10P | | 823 | | | |
| | Town Car (Cr Vic) | | | | | | | | | | | | | | |
| | Navigator (Expedition) | - | - | 384 | | | | 783 | RC7 | | 823 | | | | |

# Ford Part 563 EDR Subfamilies



# Identifying Ford EDR families

1. At the end of the data limitations section, just below the last paragraph, on the left, Bosch puts what version of data limitations is being printed. For Ford, most data limitations by family.
2. No difference between DL
   a) 2009 RC6 and
   b) 2011 RC6X.
3. Open cdrx copy in notepad, see RC6_2011



# RC8 Precrash Data Look

**Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record)**

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Speed, Vehicle Indicated, Quality Factor | Accelerator Pedal, % Full | Accelerator Pedal, % Full, Quality Factor | Service Brake, On/Off | Service brake, on/off Quality Factor | Engine RPM |
|---|---|---|---|---|---|---|---|
| - 5.0 | 59.8 [96] | OK | 18 | OK | OFF(Driver_Not_Braking) | OK | 2,174 |

**Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 1 of 2**

| Time (sec) | ABS Activity (Engaged, Non-Engaged) | Stability Control Active | Traction Control via Engine | Traction Control via Brakes | Stability Control Lateral Acceleration (g) | Stability Control Longitudinal Acceleration (g) |
|---|---|---|---|---|---|---|
| - 5.0 | Off/non-engaged | Off/non-engaged (No) | Off/non-engaged (Inactive) | Off/non-engaged (Inactive) | -0.01 | 0.02 |

**Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 2 of 2**

| Time (sec) | Stability Control Yaw Rate (deg/sec) | Stability Control Roll Rate (deg/sec) | Steering Wheel Angle (Pinion) (deg) | Steering Wheel Angle (Pinion) Quality Factor |
|---|---|---|---|---|
| - 5.0 | 0.06 | -1.53 | 0.0 | OK |
| - 4.9 | -0.13 | -1.18 | 0.0 | OK |

# Other RC8 features

- Acceleration data added
- Second Delta V  graph centered around deployment +/- 125ms (in SOME RC7P)
- Unique decoder for occupant size

**Pre-Crash Data -5 to 0 sec [1 sample/sec] (First Record)**

| Time (sec) | Occupant Size Classification, Front Passenger (Child size Yes/No [Hex value]) | Extended Power Status |
|---|---|---|
| - 5.0 | No [$03] | No Fault |
| - 4.0 | No [$03] | No Fault |

# Ford Occupant Size Classification

**Pre-Crash Data -1 sec (First Record)**

| | |
|---|---|
| Ignition cycle, crash | 359 |
| Frontal air bag warning lamp, on/off | OFF |
| Occupant size classification, front passenger (Child size Yes/No [Hex value]) | No [$08] |
| Safety belt status, driver | Driver Buckled |

- The Hex Value provides additional information than just child yes/no
- The Decoder is different for each supplier and major module type
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

| RC6/7 | AB10 | AB12 | RC8 | Classification |
|---|---|---|---|---|
| $01 | $00 | $00 | $00 | Empty |
| $02 | $01 | $05 | $02 | Child or Occupied Disable |
| $04 | $05 | $04 | $01 | Indeterminate (gray zone between child and adult) |
| $08 | $02 | $01 | $03 | Occupied Enable Small Adult (typical 110 lb 5th female) |
| $10 | $03 | $02 | $04 | Occupied Enable Medium (typical 172 lb 50th male) |
| $20 | $04 | $03 | $05 | Occupied Enable Large Adult (typical 223 lb 95th male) |

# RC8 Lower Recording Threshold?

- Data Limitations State 5mph DV over 150ms threshold for recording
- But events with less than 5mph are seen

**Deployment Data (First Record)**

| | |
|---|---|
| Maximum Delta-V, Longitudinal (MPH [km/h]) | -2.02 [-3.25] |
| Time, Maximum Delta-V Longitudinal (msec) | 300.0 |

- Remember, nothing deployed unless a device is listed with a deploy time
- This is an non-deployment

# Unusual Ford Event Names

- Fuel Cutoff Level 1 or 2 (RC7P, RC8, AB12)
- Unlocked Event
- Locked Frontal, Side, or Rollover  Event
- Rollover Threshold Exceeded (AB12)
- Side Acceleration Threshold Exceeded (AB12)
- Shutdown Request Threshold Exceeded Level 1,2
- Event Record 1 or 2

Multiple events (front, side, rollover) may be listed in the events recovered but there is only one event in memory – they are “all in one”.

Some events like Fuel Cutoff 1 may NOT store data.

# 2017 MKZ AB12

**Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 1 of 2**

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Speed, Vehicle Indicated, Quality Factor | Accelerator Pedal, % Full | Accelerator Pedal, % Full, Quality Factor | Service Brake, On/Off | Service brake, on/off Quality Factor | Engine RPM | Cruise Control Driver Accelerator Pedal Override |
|---|---|---|---|---|---|---|---|---|
| - 5.0 | 99.2 [160] | OK | 0 | OK | Off | OK | 2,816 | Cruise_Reg_Not_Overridden |
| - 4.8 | 98.7 [159] | OK | 0 | OK | Off | OK | 2,800 | Cruise_Reg_Not_Overridden |
| - 4.6 | 98.3 [158] | OK | 0 | OK | Off | OK | 2,786 | Cruise_Reg_Not_Overridden |
| - 4.4 | 98.2 [158] | | 0 | OK | On | OK | 2,770 | Cruise_Reg_Not_Overridden |
| - 4.2 | .8 [157] | OK | 0 | OK | On | OK | 2,758 | Cruise_Reg_Not_Overridden |

Every 0.2 Sec!

**Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 2**

| Time (sec) | Brake Pre-Charge Request | Brake Assist Sensitivity Level | Brake Deceleration Request (m/s^2) | Large Driver Steering or Accel Pedal Input | Collision Mitigation System Fault | Collis Mitiga Syste Enabled |
|---|---|---|---|---|---|---|
| - 5.0 | N/A | N/A | 236 | N/A | N/A | N/A |
| - 4.8 | N/A | N/A | 236 | N/A | N/A | N/A |

Collision Avoidance

# 2017 MKZ AB12 add'l data

## Pre-Crash Data -5 to 0 sec [2 samples/sec] (First Record)

| Time (sec) | Driver Gear Selection (Auto Trans) | Wheel Torque Requested (Nm) | Total Arbitrated Brake Torque (Nm) | Ignition Status | Speed Control Status |
|---|---|---|---|---|---|
| - 5.0 | Drive | 532 | 0 | Run | Off |
| - 4.5 | Drive | 120 | 0 | Run | Off |
| - 4.0 | Drive | -20 | 0 | Run | Off |

## Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 1 of 2

| Time (sec) | ABS Activity (Engaged, Non-Engaged) | Brake Powertrain Torque Request 1 | Brake Powertrain Torque Request 2 | Traction Control via Brakes | Stability Control Lateral Acceleration (g) | Stability Control Longitudinal Acceleration (g) |
|---|---|---|---|---|---|---|
| - 5.0 | Non-engaged | No | Inactive | No | -0.02 | 0.09 |

## Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record) - Table 2 of 2

Had this before

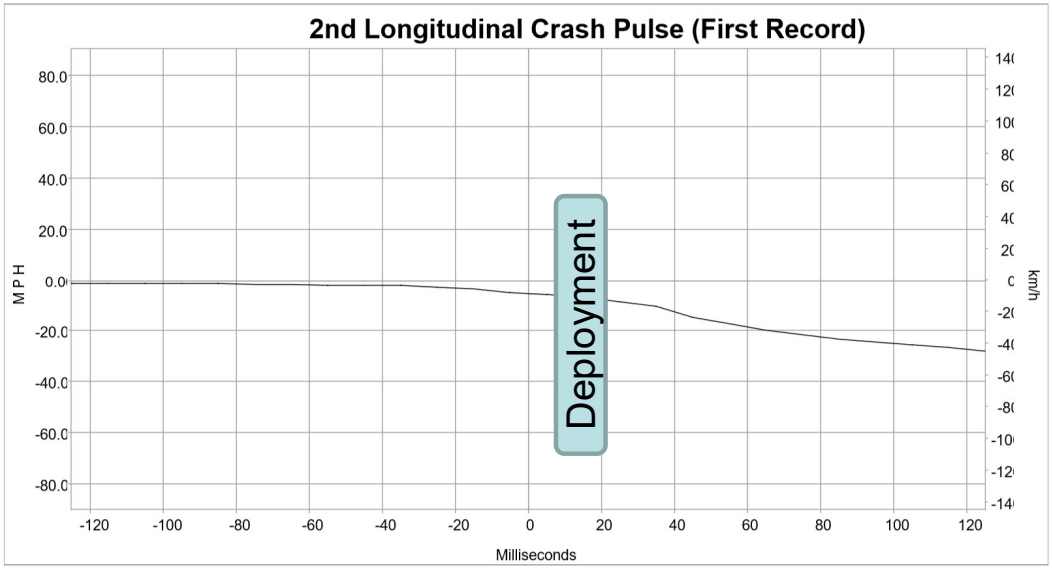
| Time (sec) | Stability Control Yaw Rate (deg/sec) | Stability Control Roll Rate (deg/sec) | Steering Wheel Angle (deg) | Steering Wheel Angle Quality Factor |
|---|---|---|---|---|
| - 5.0 | 0.06 | 0.24 | -0.9 | OK |
| - 4.9 | -0.08 | 0.00 | -0.6 | OK |
| - 4.8 | 0.45 | 0.86 | 0.0 | OK |

# Ford AB12 Anomaly

- File found with “Event Record 2” with no “Event Record 1”. Reported to Bosch.
- Check polarities of stability control system data. Ford Steering, Yaw, and Lat Accel 2009+ should all be positive right, negative left, and same sign – ~~but in some Lat Accel is reversed and is **not** noted in data limitations~~ (CHECKED - FIXED IN 19.5.3)

# New RC7P Feature- 2017 F250

- Ford has added a second graph to capture the deployment decision +/-125ms. This is important in hit-curb-then-hit-tree cases where the deployment was previously off the end of the mandated 250ms after wakeup graph



# Ford Global Real Timer

Added for AB12/ RC7P/RC8

Timer starts when car is built, is hooked to battery, runs continuously

Month of build is on door jamb sticker (exact day/time is not)

Can be used to tie your recording to your crash

Great for knowing a more precise time between events when not within 5 seconds

**Pre-Crash Data -1 sec (First Record)**

| | |
|---|---|
| Ignition cycle, Crash | 1,782 |
| Frontal Air Bag Warning Lamp, On/Off | Off |
| Safety Belt Status, Driver | Belted |
| Seat Track Position Switch, Foremost, Status, Driver | Rearward |
| Seat Track Position Switch, Foremost, Status, Front Passenger | Rearward |
| Safety Belt Status, Front Passenger | Unbelted |
| Brake Telltale | Off |
| ABS Telltale | Off |
| ESC/TC Telltale | Off |
| ESC/TC Off Telltale | Default Mode |
| Powertrain Wrench Telltale | Off |
| Powertrain Malfunction Indicator Lamp (MIL) Telltale | Unchanged Off |
| Global Real Time (seconds) | 47,951,087 |

# Polarity on Stability Control 09+

1. RC6 Data Limitations do not give polarity for stability control data. For lateral it is positive left or CCW. Opposite of right hand rule.
2. New RC7 and AB12 DL's show the stability control lateral polarity as stated above, all 3 backwards of right hand rule

# 09+ Ford Stability Control Data

**Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)**

| Times (sec) | Stability Control Lateral Acceleration (g) | Stability Control Longitudinal Acceleration (g) | Stability Control Yaw Rate (deg/sec) |
|---|---|---|---|
| - 1.4 | 0.086 | -0.045 | 0.12 |
| - 1.3 | | -0.031 | 1.5 |
| - 1.2 | | -0.647 | -1.5 |
| - 1.1 | | -0.75 | -2.5 |
| - 1.0 | | -0.709 | -1.5 |
| - 0.9 | | -0.763 | -1.5 |
| - 0.8 | | -0.826 | -1.5 |
| - 0.7 | | -0.792 | -0.75 |
| - 0.6 | | -0.895 | 1.25 |
| - 0.5 | 0.014 | -0.83 | 1.87 |
| - 0.4 | 0.03 | -0.835 | 0.75 |
| - 0.3 | 0.022 | -0.607 | 0.37 |
| - 0.2 | 0.107 | -0.653 | 0.75 |
| - 0.1 | 0.05 | -0.823 | -3.37 |
| 0.0 | -0.004 | -0.811 | -3.75 |

Reaction

Integrate lat accel - get velocity – integrate again – get displacement

Drag Factor!

Integrate yaw – get **approach angle change**!! Take tangent of cumulative angle, multiply by speed, get **sideways movement** – how far out of lane vehicle went!!

# Autoliv RC6 Anomalies (09-10)

1. There are timing delays in reporting the 0.5 sec interval precrash data in the early Ford Autoliv RC6 modules (09 F150-Escape-Econoline and many 2010). Identify by lag between stability control longitudinal (braking) g's and brake on time and speed lost. Use Stability Control System long. accel for timing of accel release and brake on.
2. Fixed by "RC6X", identifiable only by opening COPY of .cdrx in notepad, observe "RC6_2011" in header (and by agreement between stability control and 0.5 sec interval precrash data).

# 2012-2015 AB10 Anomaly

1. The “Bosch AB10” ACM found in 2012+ Focus and 2015 Escape and other vehicles **produced before Nov. 2015**, which have a **second event**, incorrectly label pre-crash **time** to impact in the second event.
2. Time between events is accurate – use it
3. Second event precrash (both speed and stability control sections) include extra points at irregular intervals labeled as if they occurred at regular intervals.

# AB10 Anomaly cont'd

1. If you can see the first event in the second event's data, it will lead you to think the time between events is longer than it really is.
2. The data in the second event did occur, just not at the time it said it did.
3. The second event last precrash speed is still within 0.5 seconds of the second event.
4. There is no software fix for EDR's produced before Nov. 2015. Fix implemented 11/2015.
5. No mention in DL's yet.

# Bosch AB10P Unique Info

1. ~~Longitudinal Accel Stability Control Info is positive for slowing down, negative for speeding up – *opposite of what data limitations say*.~~ Fixed in latest software
2. ~~Lateral Accel data is positive right, negative left – different than the other Ford modules.~~ Fixed in latest software
3. Steering and Yaw remain positive left and are identified as such in data limitations.

# Ford Red Group Anomalies

- 2006-7 RC5 Lateral Delta V polarity is not stated in the minimal data limitations. Two recent cases with clear facts showed opposite of right hand rule in 21.0.
- RC5 Lat Delta V graph only 60ms from AE
- 2007+ AB9 Lateral Delta V graph only present if SAB deploys, duration -58ms to deployment.
- AB9 Lifetime operating timer discrepancy between event and readout noted

# RC6+ Steering Angle Glitch

1. Ford stores one value for steering relative to original zero position set at factory
2. If vehicle steering is serviced, original zero may be lost and system learns a new one from average of all new driving.
3. EDR may write down ORIGINAL position relative to zero – so a FEW models serviced may appear to have WRONG value for zero.
4. No warning in DL's about this yet

# 2020/21 F250+ no EDR events

- Ford accidentally turned off the EDR in 20/21 F Super Duty (F250, 350, 450, and 550) ACM's built from 2/18/2019 thru 7/10/2021
- 12/3/2021 Ford issued a Customer Satisfaction Program 21P19 advising of this. Vehicle brought to dealers for other service will reflash the ACM to turn the EDR back on.
- You can ask dealer to run VIN and tell you if 21P19 has been performed before you go to download it so ask not to waste your time.

# Backpowering New Fords

1. Airbag system will stays on until car slows to a near stop. Switched battery feed replaces ignition switch feed.
2. Airbag fuse (interior fuse box) Micro 3

| | | |
|---|---|---|
| 19 | 7.5A [7] | Extended power module. |

3. 2020 Explorer moves to engine compt

| | | |
|---|---|---|
| 146 | 15A | Battery electronic control module. ACM Power |

4. ALSO REQUIRES GATEWAY (INTERIOR FUSE BOX)

| | | |
|---|---|---|
| 13 | 7.5A | Steering column control module.<br>Smart datalink connector. GATEWAY<br>Instrument cluster. |
| 7 | 10A | Smart data link connector power. DLC |

# For Ford EDR **not supported** by Bosch CDR
## See "Unsupported Fords" on my website

Many requests can go directly to Ford's suppliers.

Other requests for assistance to Ford legal:

Email to EDRFORD@FORD.COM

(this is an address for a Bowman and Brooke outside counsel that handles these requests for Ford)

Contact Rick Ruth for additional information prior to making request

As of this writing, Veoneer (former Autoliv) charges $2000 to all, Bosch and Conti are no charge to law enforcement but $1500 to private parties.

Veoneer now says it cannot download modules >15 years old

Email to rcmdownloads@Veoneer.com is being returned as undeliverable in last 2 weeks. Use Deborah.Cox@veoneer.com.

**Recent 05 Freestar RCM sent to Takata was returned with Hex Data only and no guide to interpretation (no charge, no value!)**

**ACM suppliers work from home due to COVID, not allowed in lab!**

Available on my website

# Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy (this copy has been updated 4-01-2021, after the original )16 CDR
presentation at the Jan 2016 CDR Summit)



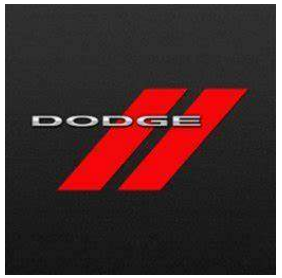
# Stellantis (formerly FCA)

## Includes Chrysler, Jeep, Dodge, Ram, Fiat, Lancia, Renault, Citroen and Peugot Data Analysis































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# Stellantis 21.5 CDR Coverage 64%

| Last 3 digits of Bosch CDR DTM adpater/cable number s | | | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Chrysler | 200 | Console, aft of shifter | | | | | | | 790/516 | | 387/516 | | 387/785 | | | | | | | |
| Chrysler | 300 | Center stack | | 228 3,5 | 228 5 | | | | 387/598 | | | | | | | | | | | |
| Chrysler | Aspen | Console, aft of | | | 226 4 | 226 | | | | | | | | | | | | | | |
| Chrysler | Cirrus (not sold in US) | s | | | | | | 387/516 8 | | | | | | | | | | | | |
| Chrysler | Grand Caravn Voygr | Center st | | | | | | 387/385 8 | | | | | | | | | | 387/785 | | |
| Chrysler | Pacifica | Center stack | | | 228 5 | | | | | | | | | | 387/785 | | | | | |
| Chrysler | PT Cruiser | Console, fwd of | | | 228 5,7 | 228 5 | | | | | | | | | | | | | | |
| Chrysler | Sebring | Console, aft of | | | 226 4 | 226 | | 387/516 8 | | | | | | | | | | | | |
| Chrysler | Town & Country | Center stack | | | | 385 6 | | 387/385 8 | | | 387/785 | | | | | | | | | |
| Dodge | Avenger | Console, aft of | | | | 226 | | 387/516 8 | 387/790/516 | | 387/516 | | | | | | | | | |
| Dodge | Caliber | C | | | 226 4 | 226 | | 387/516 8 | 387/790/516 | | | | | | | | | | | |
| Dodge | Caravan | Center stac | | | | 385 6 | | 387/385 8 | | | 387/785 | | | | | | | | | |
| Dodge | Challenger | Center stack | | | | 228 | | | 387/228 10 | 387/228 | | | 387/598 | | | | | | | |
| Dodge | Charger | Center stack | | 228 3,5 | 228 5 | | | | 387/598 | | | | | | | | | | | |
| Dodge | Dakota/Raider | Center stack | | | 228 5 | | | | 228 | Mex Only | | | | | | | | | | |
| Dodge | Dart | Center Stack | | | | | | | | | 387/785 | | | | | | | | | |
| Dodge | Durango | Console, aft | 226 2,4 | 226 2,4 | 226 2,4 | 226 2 | 226 2 | | 387/228 10 | 387/228 | | 387/598 | | | | | | | | |
| Dodge | Journey | Consol | | | | | 385 | 387/385 8 | | | 387/385 | 387/385 +516PPT | | | | 387/821 | | | | |
| Dodge | Magnu | Center stack | | 228 3,5 | 228 5 | | | | | | | | | | | | | | | |
| Dodge | Nitro | Console, aft of | | | 227 4 | 227 | | 387/516 8 | | | | | | | | | | | | |
| Dodge/Ram | Promaster City | FW of shifter | | | | | | | | | | | 387/802 | + 709 DLC Adapter | | | | | | |
| Dodge/Ram | Promaster (Big Van) | Center Tunnel | | | | | | | | | | 387/829 new 17_4 | | + 709 DLC Adapter | | | | | | |
| Dodge/Ram | Ram Cab Chassis | CF Seat/console | | 226 4 | | | | 387/453 8 | | | 387/785 | | | | | | | | | |
| Dodge/Ram | Ram Pickup (25-3500) | CF Seat/console | | 226 4 | | | 226 9 | 387/453 8 | | | | | | | | | and classic 1500 | | | |
| Dodge/Ram | Ram Pickup (1500) | CF Seat/console | | 226 4 | | 226 | 453 8 | 387/453 8 | 387/453 | | | | | | | | 785 new 1500 | | | |
| Dodge/Ram | Ram Van | Center Stack | | | | | | | | | 387/785 | | | | | | | | | |
| Dodge/SRT | Viper | Center Stac | | | | | | | | | 387/785 | | | | | | | | | |
| | | | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | |
| Jeep | Commander | onsole, aft | | 228 3,5 | 228 5 | | | | | | | | | | | Grand commander china only | | | | |
| Jeep | Compass/Patriot | Center stack | | | 226 4 | 226 | | 387/516 8 | 790/516 | | 387/516 | | 387/ *808* | | 808/819 | 808/819 | 819 | | | |
| Jeep | Grand Cherokee | Console, aft of | | 228 3,5 | 228 5 | | | | 387/228 10 | 387/228+ | 387/228 | 387/598 | | | | | | | | |
| Jeep | Gladiator | Center Console | | | | | | | | | | | | | | | | 821 | | |
| Jeep | Liberty/Cherokee | Console, aft of | | | | 227 | | 387/516 8 | | | ? | 387/785 | | | | | | | | |
| Jeep | Renegade | Center Stack | | | | | | | | | | | | 387/819* | | | | | | |
| Jeep | Wrangler | Console, aft of | | | | | | | | | 387/546 | | | | | | 387/821 | | | |
| Jeep | Wagoneer & Gr Wag | | | | 227 4 | 227 | | 387/546 8 | | | | | | | | | | | | 785 |

Call FCA

Call FCA ?

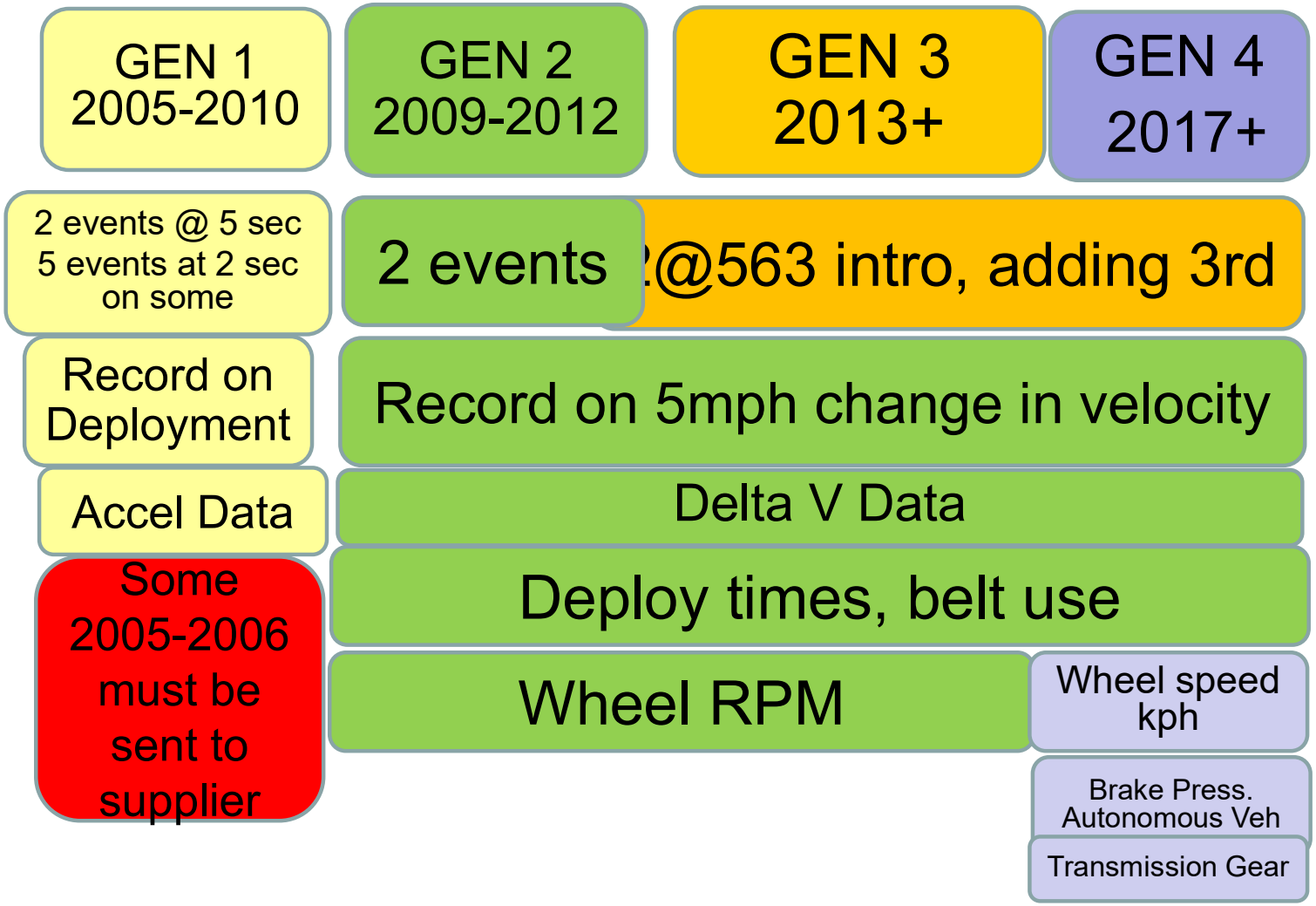
Frontal Deploy

Frontal/Side Deploy

Non Deploys

563 Non Deploys

# FCA EDR Evolution (under dev't)



# Output New Look –

Removed multiple brake elements, added ABS/SC, moved strg to P1, hi res
Two tables at 0.1 sec interval, others tables @ 0.25 sec intervals

**Pre-Crash Data [10 samples/sec] (Most Recent Event - table 1 of 2)**
(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | Pre-Crash Recorder Status | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full | Service Brake | Engine RPM | ABS Activity | Stability Control | Steering Input (deg) |
|---|---|---|---|---|---|---|---|---|
| -5.0 | Complete | 96 [155] | 0.00 | Off | 3,878 | No | On | 1 |
| -4.9 | Complete | 96 [154] | 0.00 | Off | 3,835 | No | On | -1 |
| -4.8 | Complete | 96 [154] | 0.00 | Off | 3,726 | No | On | -5 |
| -4.7 | Complete | 96 [154] | 0.00 | Off | 3,621 | No | On | -10 |
| -4.6 | Complete | 96 [154] | 0.00 | Off | 3,489 | No | On | -18 |
| -4.5 | Complete | 95 [154] | 0.00 | Off | 3,344 | No | On | -21 |
| -4.4 | Complete | 95 [153] | 0.00 | Off | 3,210 | No | On | -20 |
| -4.3 | Complete | 95 [153] | 0.00 | Off | 3,109 | No | On | -19 |
| -4.2 | Complete | 95 [153] | 0.00 | Off | 3,068 | No | On | -19 |
| -4.1 | Complete | 95 [152] | 0.00 | Off | 3,108 | No | On | -18 |
| -4.0 | Complete | 95 [152] | 0.00 | Off | 3,079 | No | On | -18 |

# Wheel RPM replaced by speed

See driven vs non driven, left vs right, ABS working…

**Pre-Crash Data [10 samples/sec] (Most Recent Event - table 2 of 2)**

(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | ABS MIL | Yaw Rate (deg/sec) | Wheel Speed LF (km/h) | Wheel Speed RF (km/h) | Wheel Speed LR (km/h) | Wheel Speed RR (km/h) |
|---|---|---|---|---|---|---|
| -5.0 | Off | 0.40 | 154.37 | 154.52 | 154.55 | 154.52 |
| -4.9 | Off | -0.48 | 154.28 | 154.16 | 154.25 | 154.25 |
| -4.8 | Off | -1.76 | 154.40 | 154.09 | 153.88 | 153.77 |
| -4.7 | Off | -2.80 | 154.28 | 153.95 | 153.88 | 153.67 |
| -4.6 | Off | -4.24 | 154.01 | 153.64 | 153.85 | 153.55 |
| -4.5 | Off | -6.80 | 153.98 | 153.16 | 153.80 | 153.19 |
| -4.4 | Off | -7.12 | 153.80 | 152.88 | 153.70 | 152.80 |
| -4.3 | Off | -5.84 | 153.49 | 152.67 | 153.37 | 152.62 |
| -4.2 | Off | -5.12 | 153.13 | 152.19 | 153.25 | 152.38 |
| -4.1 | Off | -4.64 | 152.80 | 152.04 | 152.91 | 151.91 |
| -4.0 | Off | -4.64 | 152.49 | 151.73 | 152.46 | 151.65 |
| -3.9 | Off | -4.00 | 152.13 | 151.55 | 152.13 | 151.31 |
| -3.8 | Off | -3.68 | 151.83 | 151.25 | 151.95 | 151.13 |

# Current Gear & More @ 0.25 sec

**Pre-Crash Data [4 samples/sec] (Most Recent Event - table 1 of 2)**

(the most recent sampled values are recorded prior to the event)

NEW

Changes at END of shift

| Time Stamp (sec) | Pre-Crash Recorder Status | Raw Manifold Pressure (kPa) | PCM MIL | ETC Lamp | Current Gear | Reverse Gear Status |
|---|---|---|---|---|---|---|
| -5.00 | Complete | 20.00 | Off | Off | D5 | Not_Reverse |
| -4.75 | Complete | 20.00 | Off | Off | D5 | Not_Reverse |
| -4.50 | Complete | 21.60 | Off | Off | D5 | Not_Reverse |
| -4.25 | Complete | 23.20 | Off | Off | D5 | Not_Reverse |
| -4.00 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.75 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.50 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.25 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.00 | Complete | 22.40 | Off | Off | D6 | Not_Reverse |
| -2.75 | Complete | 21.60 | Off | Off | D6 | Not_Reverse |
| -2.50 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |
| -2.25 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |
| -2.00 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |
| -1.75 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |

‹ 10.63 in ‹

# Tire pressure @ 0.25 sec intervals

**Pre-Crash Data [4 samples/sec] (Most Recent Event - table 2 of 2)**

(the most recent sampled values are recorded prior to the event)

NEW

| Time Stamp (sec) | Tire Pressure Position | Tire Pressure Status | Tire Pressure Value (PSI) | Cruise Control Lamp Status | Cruise Control Engaged Status | ACC On/Off | ACC Set Speed (MPH [km/h]) |
|---|---|---|---|---|---|---|---|
| -5.00 | LHF | Normal | 34 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.75 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.50 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.25 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.00 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.75 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.50 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.25 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.00 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -2.75 | RHF | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -2.50 | RHF | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |

# Integrating FCA Yaw Rate

From 08 charger case

| | |
|---|---|
| - 3 deg/sec * 0.1 sec = | -0.3 degrees |
| -11 deg/sec * 0.1 sec = | -1.1 degrees |
| -18 deg/sec * 0.1 sec = | -1.8 degrees |
| -15 deg/sec * 0.1 sec = | -1.5 degrees |
| - 9 deg/sec * 0.1 sec = | -0.9 degrees |
| - 4 deg/sec * 0.1 sec = | -0.4 degrees |
| - 1 deg/sec * 0.1 sec = | -0.1 degrees |
| Total Swerve Right | 6.1 degrees |

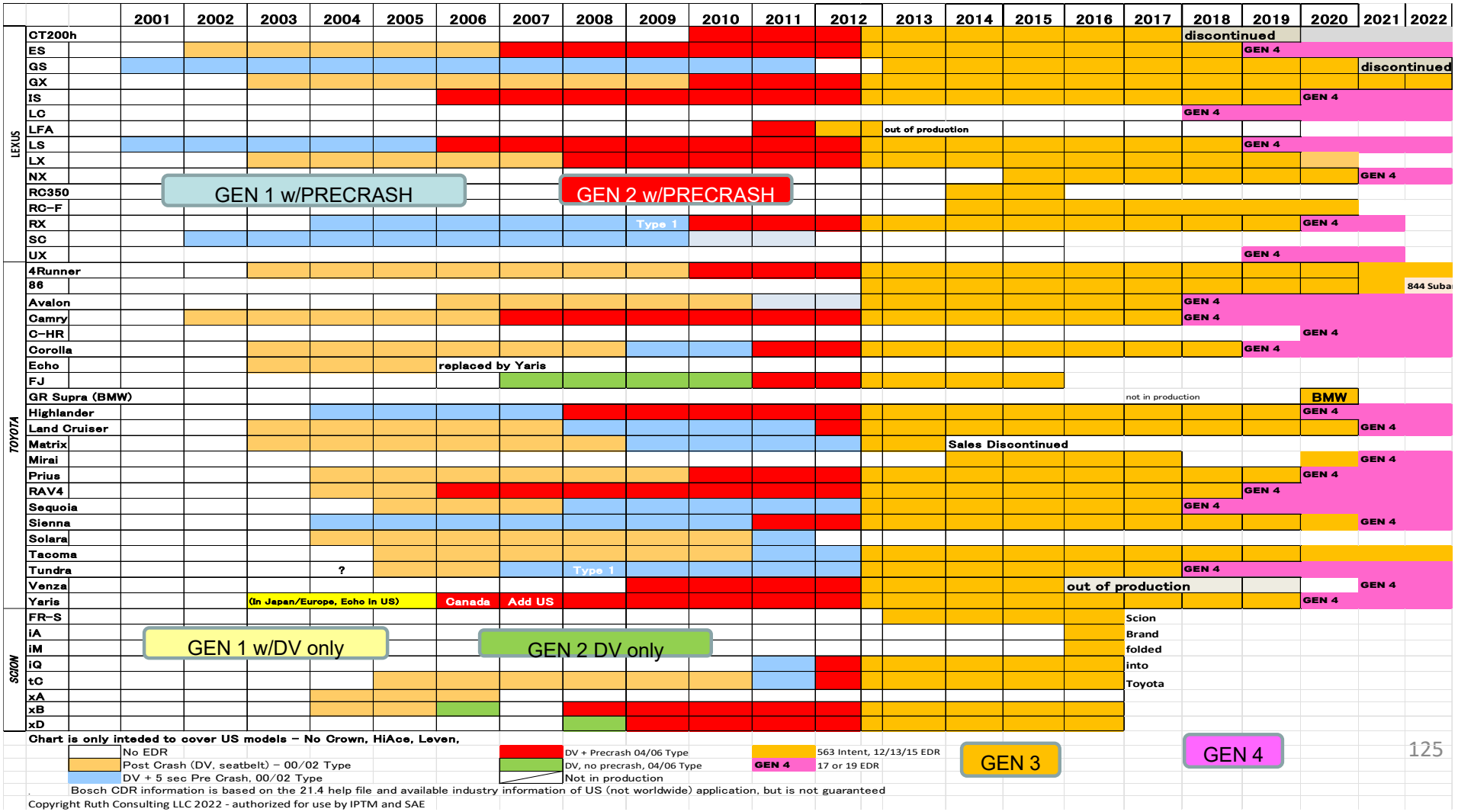
**Note 1: this is how far nose of car is pointing right – *CG* motion may *lag***

***Note 2: Polarity opposite RH rule this model***

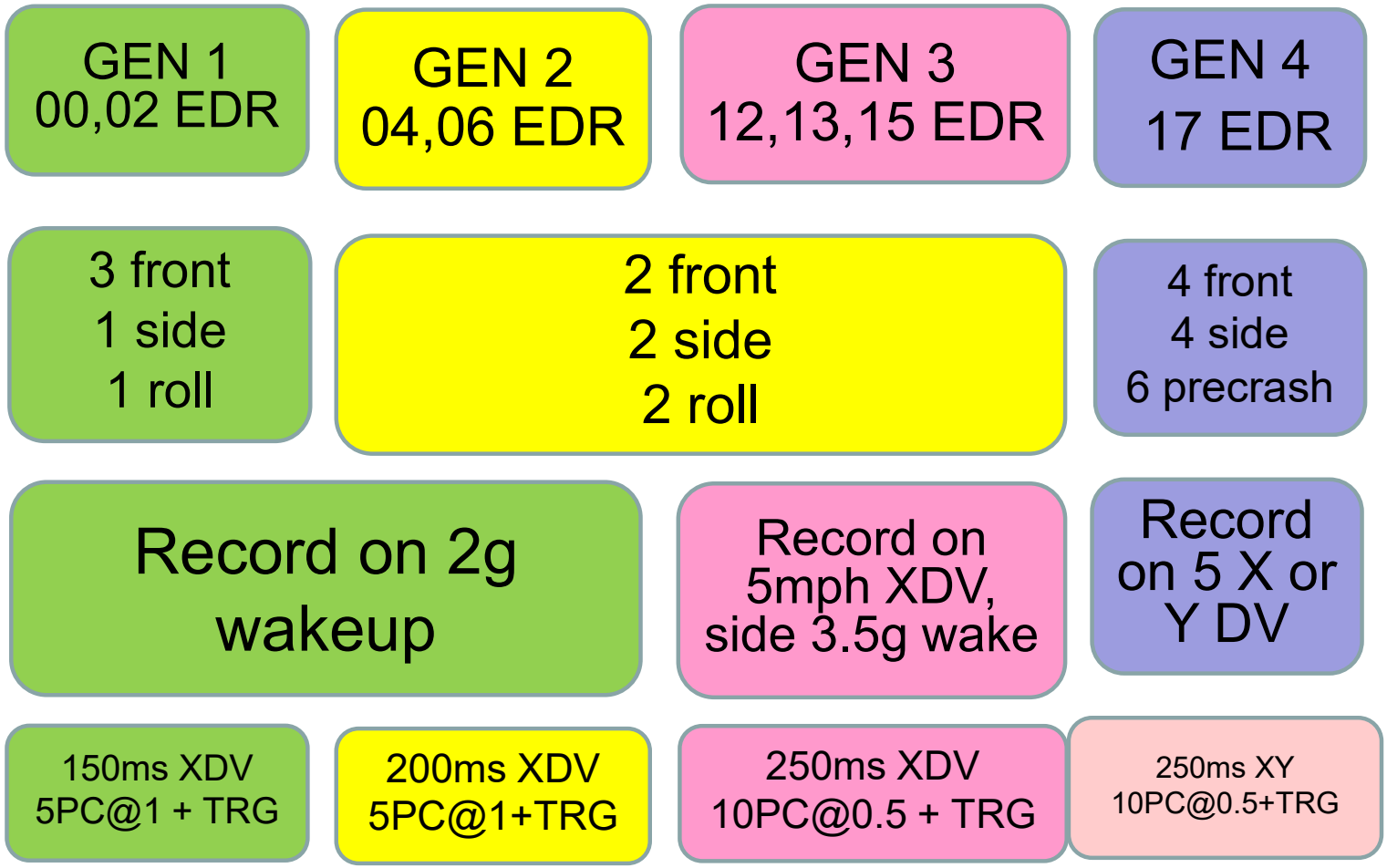
# New Chrysler Info

1. Frontal sensors are 100G at 45 degrees (100*cos 45$^{o}$ = 71g straight ahead).
2. “Sensor Design Range Exceeded” line on graph means clipping above that level
3. Buckle switch NOT normally used to determine if pre-tensioner fires (Exceptions on some non US built vehicles)
4. Fiat Spider = Mazda Miata, threshold <5mph
5. Master Cylinder Pressure on 2018 Charger!
6. Driver assistance data elements coming in
7. 2019+ not reading in 21.4, revert to 19.5 to read

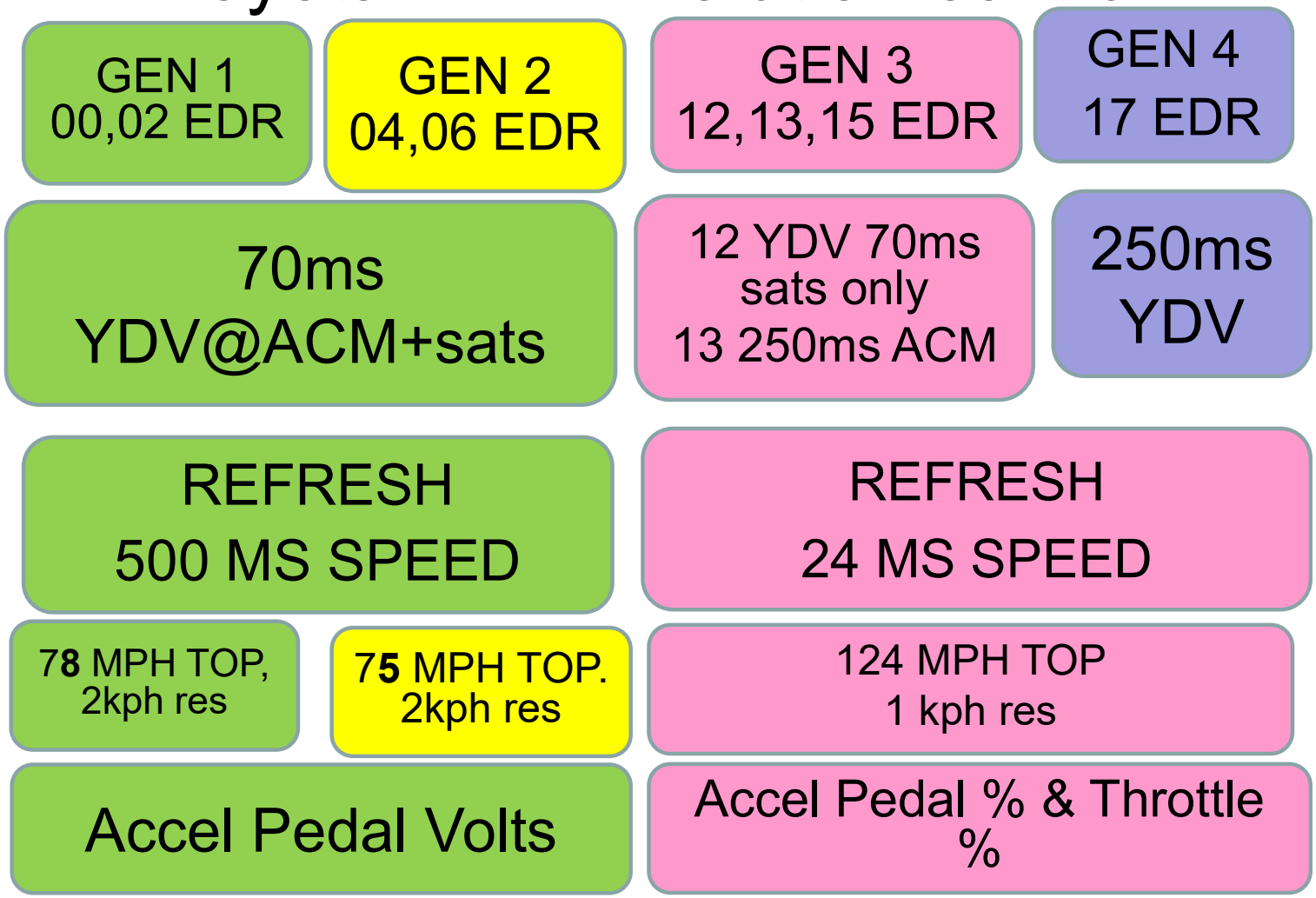
# Toyota US CDR 21.4 Coverage 84%



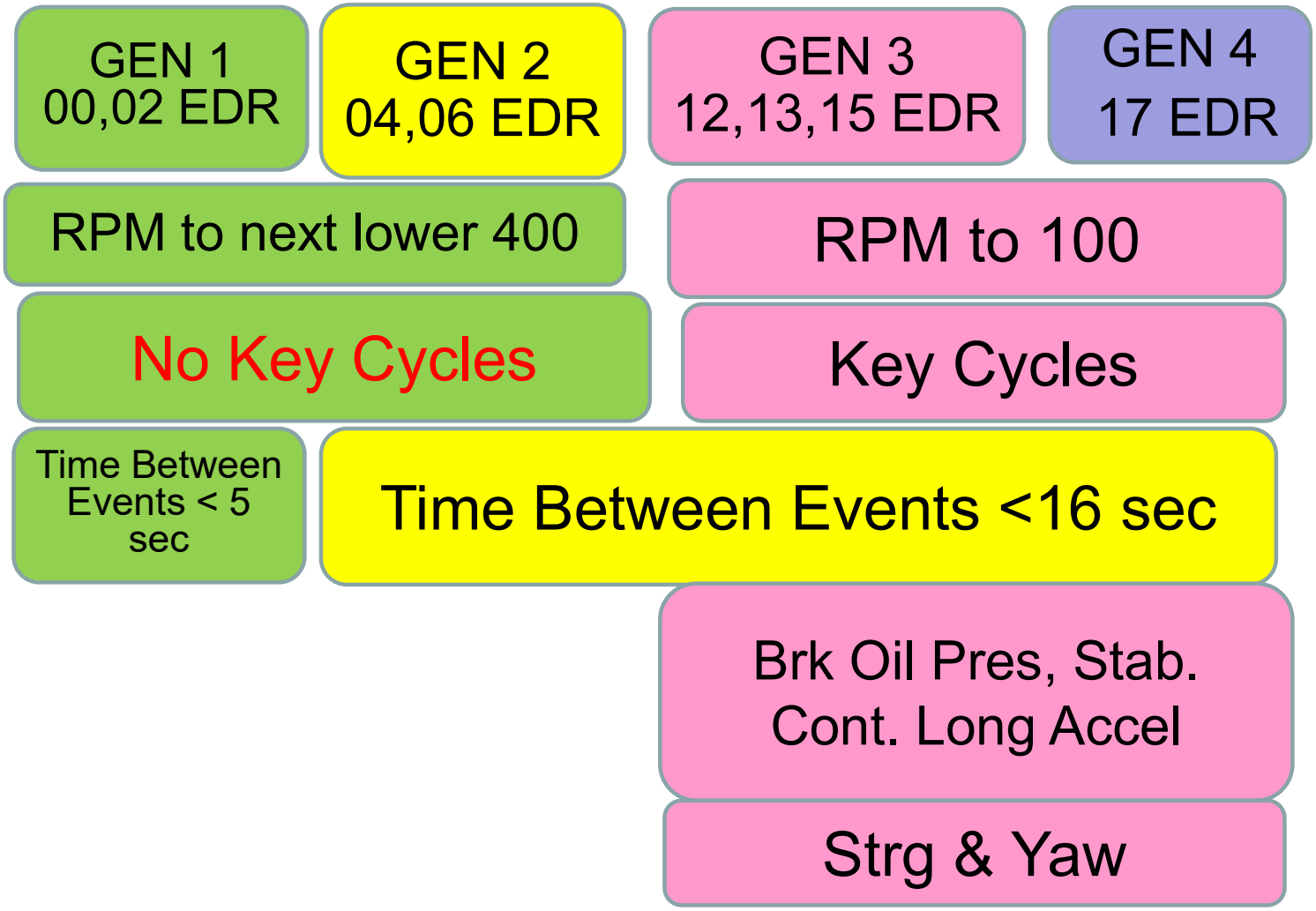
# Toyota EDR Evolution



# Toyota EDR Evolution cont'd



# Toyota EDR Evolution cont'd



# New “17 EDR”

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM added to frontal events for recon
- More precrash registers (6 vs 2 prior)
- More front/rear & side registers (4 vs 2)
- Separate side events still reported, but do NOT have precrash associated with them. Side events have detail of side satellite sensors (generally ***not used*** by Recons).
- Odometer data element added

# 17 EDR Event Record Summary
# more events, detail on recording status

**Event Record Summary at Retrieval**

| Events Recorded | TRG Count | Crash Type | Time (msec) | Pre-Crash Recording Status | Diagnostic Data Recording Status | Occupant Data Recording Status | Crash Info Recording Status | Time Series Recording Status |
|---|---|---|---|---|---|---|---|---|
| Most Recent Event | 10 | Side Crash | 0 | N/A | Complete | N/A | | plete |
| 1st Prior Event | 9 | Frontal/Rear/Si | -898 | Complete | plete | Complete | Complete | Complete |
| 2nd Prior Event | 8 | ide Crash | -925 | N/A | | | N/A | Complete |
| 3rd Prior Event | 6 | Side Crash | N/A | | plete | N/A | N/A | Complete |
| 4th Prior Event | 5 | Rollover | N/A | omplete | Complete | Com | Incomplete | Incomplete |
| 5th Prior Event | 4 | Frontal/Rear/Side Crash | N/A | Complete | Co | Complete | Complete | Complete |
| 6th Prior Event | 3 | Side Crash | N/A | N/A | Complete | N/A | N/A | Complete |
| 7th Prior Event | 2 | Rollover | N/A | Complete | Complete | Complete | Complete | Complete |

Pages NOT numbered like Gen3

#7 missing

Lost count due to 7 missing

No precrash for sides

# Same Data, New Look to Precrash

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4**

| Time (sec) | Vehicle Speed (MPH [km/h]) | Accelerator Pedal, % Full (%) | Percentage of Engine Throttle (%) | Fuel Injection Quantity (mm^3/st) | Engine RPM (RPM) | Motor RPM (RPM) | Service Brake, ON/OFF |
|---|---|---|---|---|---|---|---|
| -5.00 | 29.8 [48] | 49.5 | 24.5 | Invalid | 2,400 | Invalid | OFF |
| -4.50 | 32.3 [52] | 47.0 | 24.5 | Invalid | 2,500 | Invalid | OFF |
| -4.00 | 34.2 [55] | 38.0 | 20.0 | Invalid | 2,700 | Invalid | OFF |
| -3.50 | 35.4 [57] | 32.5 | 16.0 | Invalid | 2,800 | Invalid | OFF |
| -3.00 | 36.7 [59] | 26.5 | 13.5 | Invalid | 2,400 | Invalid | OFF |
| -2.50 | 37.3 [60] | 22.5 | 11.0 | Invalid | 2,400 | Invalid | OFF |
| -2.00 | 37.3 [60] | 19.0 | 8.5 | Invalid | 2,200 | Invalid | OFF |
| -1.50 | 38.5 [62] | 19.0 | 6.0 | Invalid | 1,900 | Invalid | OFF |
| -1.00 | 37.9 [61] | 18.0 | 5.5 | Invalid | 1,900 | Invalid | OFF |
| -0.50 | 37.9 [61] | 18.0 | 5.0 | Invalid | 1,800 | Invalid | OFF |
| TRG(0) | 36.0 [58] | 100.0 | 26.0 | Invalid | 1,700 | Invalid | OFF |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4**

| Time (sec) | ABS Control Status | BOS Control Status | Brake Oil Pressure (Mpa) | Longitudinal Acceleration , VSC Sensor (m/s^2) | Yaw Rate (deg/s) | Steering Input (degrees) | Shift Position |
|---|---|---|---|---|---|---|---|
| -5.00 | OFF | OFF | 0.00 | 1.579 | -0.98 | -4.5 | D |
| -4.50 | OFF | OFF | 0.00 | 1.795 | 0.00 | -1.5 | D |
| -4.00 | OFF | OFF | 0.00 | 1.292 | 0.49 | 0.0 | D |
| -3.50 | OFF | OFF | 0.00 | 0.861 | 0.00 | -1.5 | D |
| -3.00 | OFF | OFF | 0.00 | 0.718 | 0.00 | -1.5 | D |
| -2.50 | OFF | OFF | 0.00 | -0.072 | 0.00 | -1.5 | D |
| -2.00 | OFF | OFF | 0.00 | 0.000 | 0.00 | -1.5 | D |
| -1.50 | OFF | OFF | 0.00 | 0.000 | -0.49 | -1.5 | D |
| -1.00 | OFF | OFF | 0.00 | 0.287 | 0.00 | -1.5 | D |
| -0.50 | OFF | OFF | 0.00 | -0.144 | 0.00 | -1.5 | D |
| TRG(0) | OFF | OFF | 0.48 | -1.507 | 61.00 | -19.5 | D |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4**

| Time (sec) | Sequential Shift Range | Cruise Control Status | VSC Control Status | READY Signal | Drive Mode, Power Train | Drive Mode, Snow | Drive Mode, EV |
|---|---|---|---|---|---|---|---|
| -5.00 | Undetermined | OFF | ON (enable) | Invalid | Normal | OFF | Invalid |
| -4.50 | Undetermined | OFF | ON (enable) | Invalid | Normal | OFF | Invalid |

# Pedestrian Protection Module

## System Status at Pedestrian Event (1st Prior Event, TRG 1)

| | |
|---|---|
| TRG Count (times) | 1 |
| TRG Count not for Pedestrian (times) | SNA |
| Ignition Cycle, Crash (times) | 353 |
| Odometer signal (miles [km]) | 1,195 [1,923] |
| Trip count (times) | 392 |
| Time count (msec) | 1,283,900 |
| Time count input system | Normal |
| Airbag Warning Lamp, On/Off | OFF |
| Pedestrian Warning Lamp, On/Off | OFF |
| Time from Previous TRG (msec) | 32767 or greater |
| Time from Pre-Crash to TRG (msec) | 300 |
| Time between TRG not for Pedestrian and TRG for Pedestrian (msec) | SNA |
| TRG Establishment not for Pedestrian, within 5 sec. | ON |
| Freeze Signal | ON |
| Recording Status, Pre-Crash | Complete |
| Recording Status, Crash Info. | Complete |
| Recording Status, EDR Correlation Info. | Complete |

## Deployment Command Data (1st Prior Event, TRG 1)

| | |
|---|---|
| Front Pop-Up Hood, Time to Deploy (msec) | 3.0 |
| Rear Pop-Up Hood, Time to Deploy (msec) | 3.0 |
| Airbag for Pedestrian Deployment, Time to Deploy (msec) | SNA |

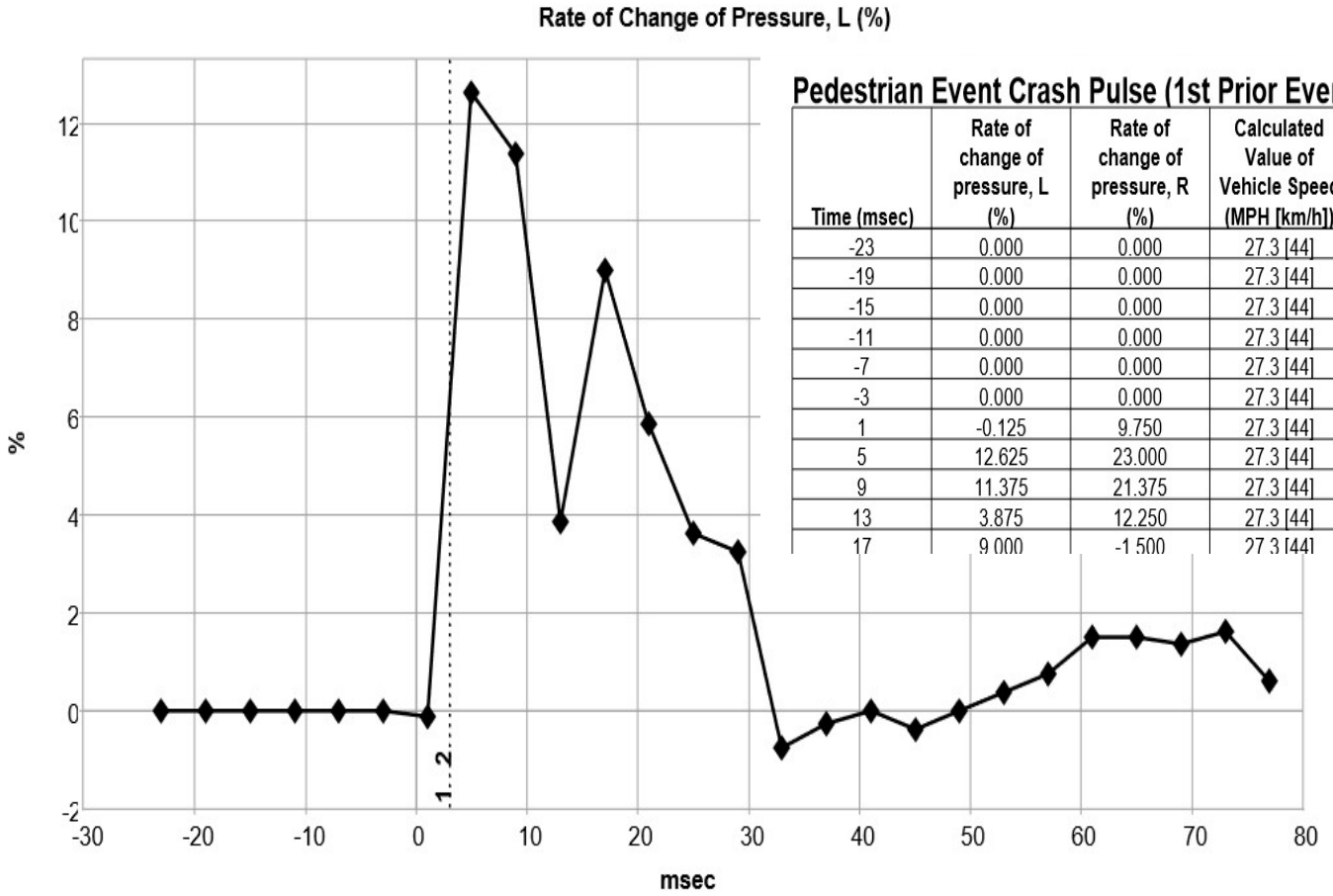
## DTCs Present at Time of Event (1st Prior Event, TRG 1)

| | |
|---|---|
| Recording Status, Diagnostic | Complete |
| Ignition Cycle Since DTC was Set (times) | 0 |
| Warning ON Time Since DTC was Set, for Pedestrian (min) | 0 |
| Diagnostic Trouble Codes | None |

## Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Time from TRG to Next Sample (msec) | 1 |

# Data from L/R bumper pressure sensors on 15 & 17 EDR




**Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)**

| Time (msec) | Rate of change of pressure, L (%) | Rate of change of pressure, R (%) | Calculated Value of Vehicle Speed (MPH [km/h]) |
|---|---|---|---|
| -23 | 0.000 | 0.000 | 27.3 [44] |
| -19 | 0.000 | 0.000 | 27.3 [44] |
| -15 | 0.000 | 0.000 | 27.3 [44] |
| -11 | 0.000 | 0.000 | 27.3 [44] |
| -7 | 0.000 | 0.000 | 27.3 [44] |
| -3 | 0.000 | 0.000 | 27.3 [44] |
| 1 | -0.125 | 9.750 | 27.3 [44] |
| 5 | 12.625 | 23.000 | 27.3 [44] |
| 9 | 11.375 | 21.375 | 27.3 [44] |
| 13 | 3.875 | 12.250 | 27.3 [44] |
| 17 | 9.000 | -1.500 | 27.3 [44] |

# Toyota Help File Organization

## Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Cli list of supported ACM part numbers.

Click here for tips on how to use this *Supported Vehicles* help topic

This Started in Nov 2015

**IMPORTANT:** To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this the help file.

**General Toyota Application Notes:**

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is n indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage not

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

**Vehicle Specific Application Notes:**

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W X Y Z

**4Runner (Toyota)** **2003 and later** **Market: 7**

| Year | EDR Generation | CDR Cable & Adapter | Module | Module Location |
|---|---|---|---|---|
| 2002 - 2008 | Gen 1 (02EDR) | F00K108617 (no adapter) | ACM | Under Center Stack |
| 2009 - 2011 | Gen 2 (06EDR) | F00K108616 & F00K108387 | ACM | Under Center Stack |
| 2012 and later | Gen 3 (12EDR/13EDR) | F00K108616 & F00K108387 | ACM | Under Center Stack |

Help file may be CALENDAR year not model year,
EDR may not yet be present in Model Year of first year listed

# Toyota Vehicle Lookup



# Toyota 12 EDR Drawback

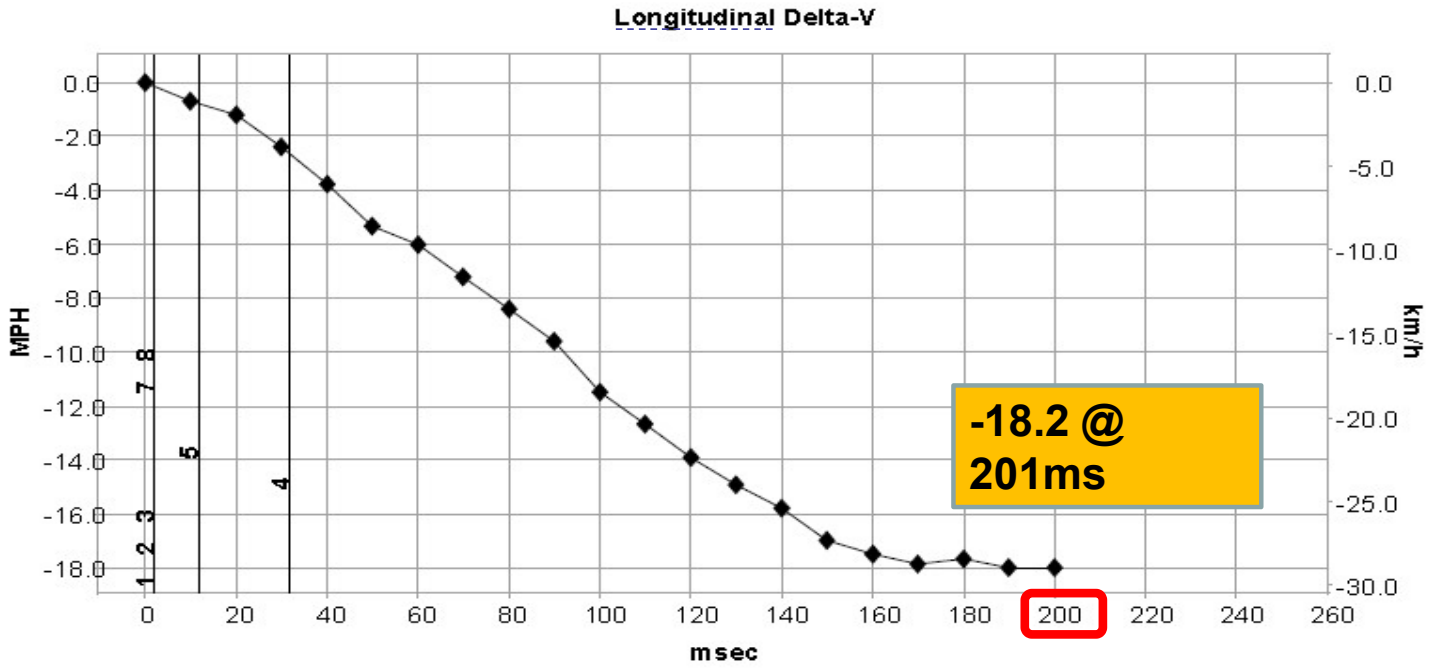
1. Lateral acceleration is only recorded in a side algorithm wakeup.
2. When it is recorded, it is only for 68ms
3. It is only recorded for the side satellites, not for the central control module
4. If trying to calculate a PDOF you must be very aware of the different recording timing and sensor location
5. Fixed in “13EDR” – get 250ms of Y accel

# Delta V analysis – What's the PDOF? The Total?? 12 EDR

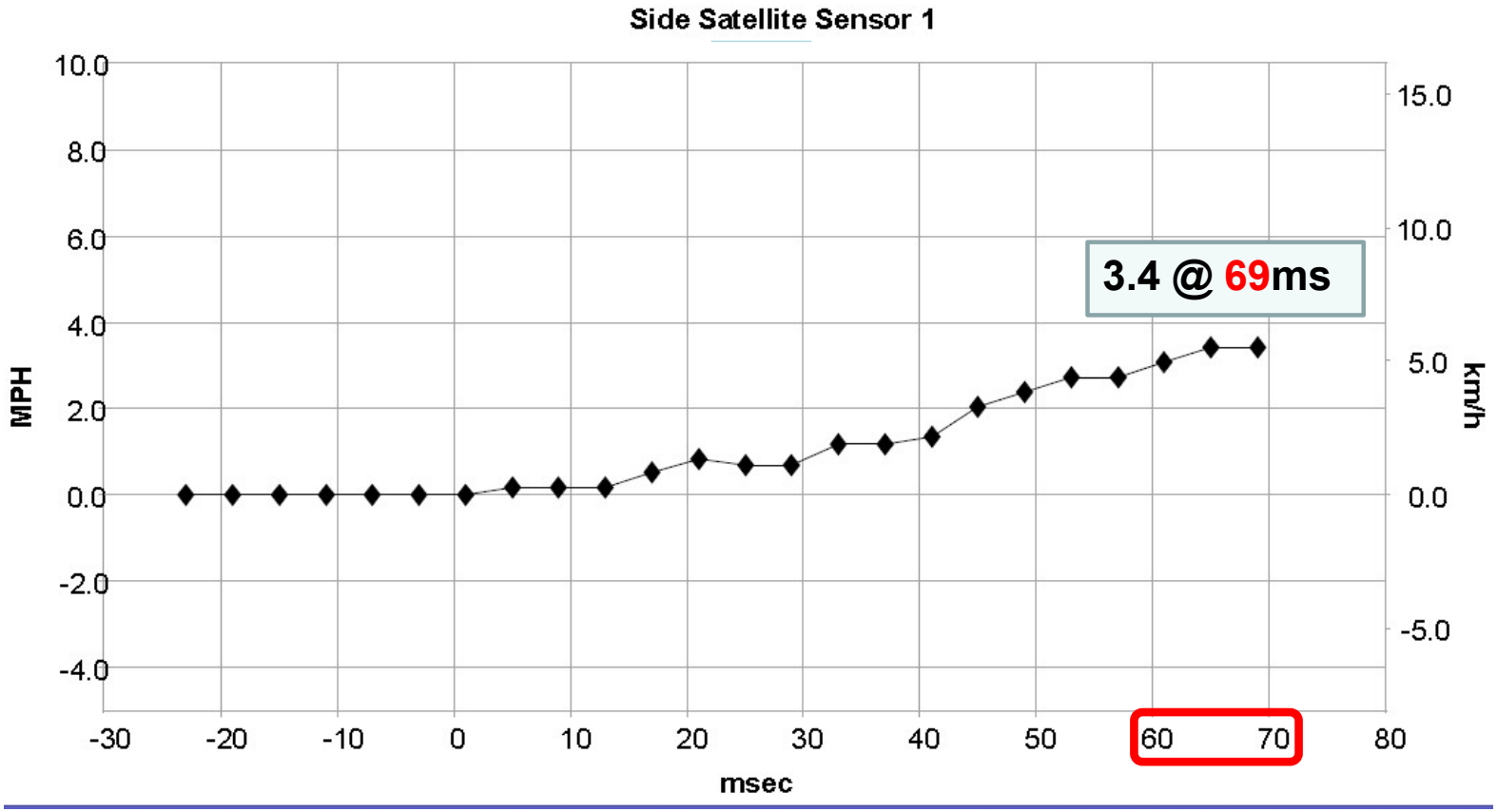
- PDC

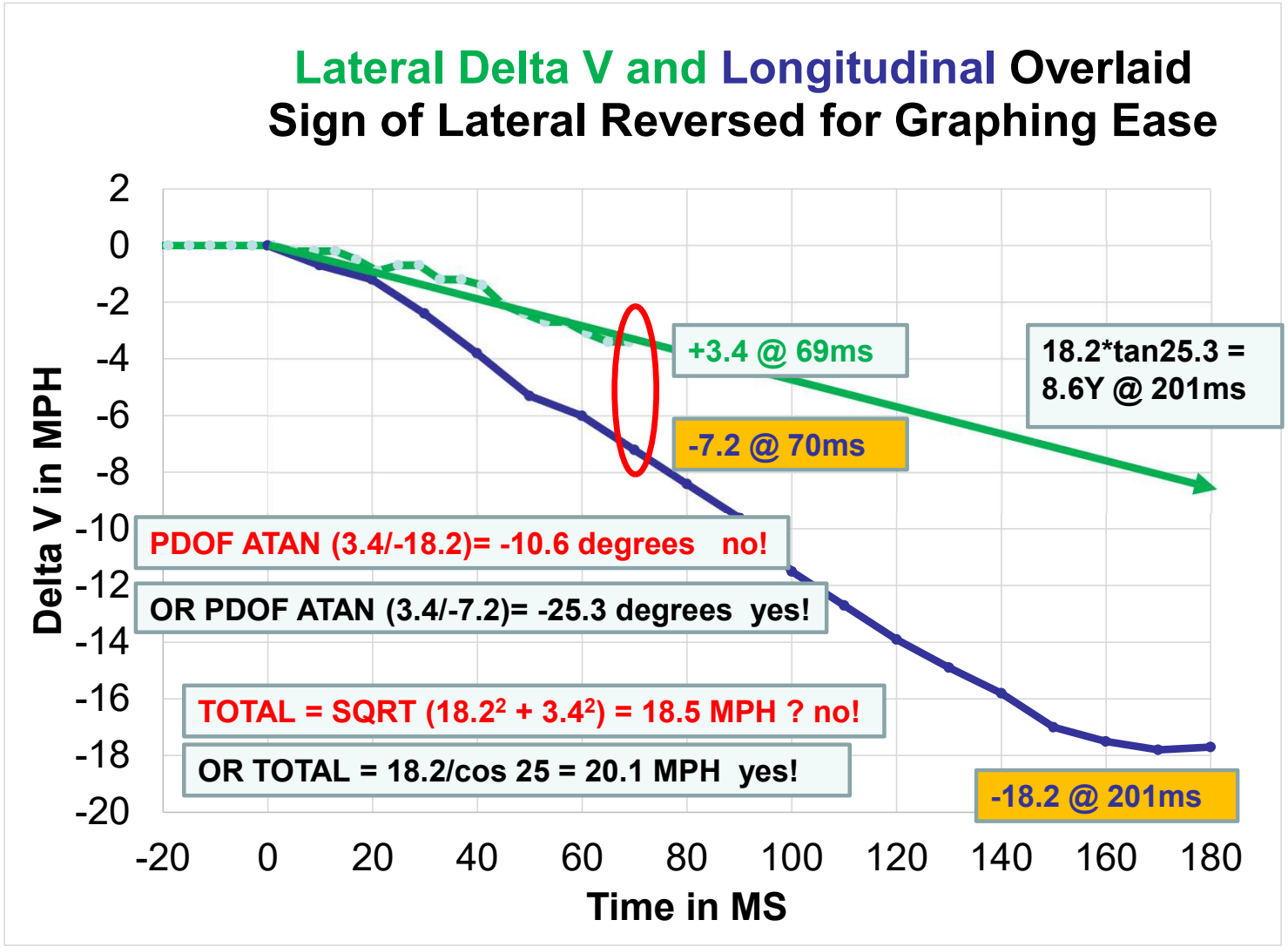
**Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)**

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Time from Time Zero to TRG (msec) | 2.0 |
| Length of Delta-V (msec) | 200 |
| Max. Longitudinal Delta-V (MPH [km/h]) | -18.2 [-29.2] |
| Time, Maximum Delta-V, Longitudinal (msec) | 201.0 |
| Power Supply Status at Max. Delta-V | ON |



# 12 EDR Lateral Delta V -




Lateral Delta V and Longitudinal Overlaid
Sign of Lateral Reversed for Graphing Ease
Delta V in MPH
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-20
0
20
40
60
80
100
120
140
160
180
Time in MS
+3.4 @ 69ms
18.2*tan25.3 =
8.6Y @ 201ms
-7.2 @ 70ms
PDOF ATAN (3.4/-18.2)= -10.6 degrees no!
OR PDOF ATAN (3.4/-7.2)= -25.3 degrees yes!
TOTAL = SQRT (18.2² + 3.4²) = 18.5 MPH ? no!
OR TOTAL = 18.2/cos 25 = 20.1 MPH yes!
-18.2 @ 201ms

# VCH Freeze Frame Data 2013+

- New feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, "**V**ehicle **C**ontrol **H**istory"
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC.
- Accessible only with Toyota Techstream service tool – not part of Bosch CDR system. Annual s/w subscription or 2 day subs available.
- CDR900 or Nexiq works as pass thru device same as Toyota Mongoose hardware
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a history of erratic driver behavior
- Details in SAE 2016-01-1495 Appendix D. New **2019-01-0632** "**Reconstructing Vehicle Dynamics from On-Board Event Data**" **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events).**
- **Photos stored by pre-collision braking system may be accessed with Techstream system!**

## TOYOTA/LEXUS Vehicles with Vehicle Control History Data

| | | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 |
|---|---|---|---|---|---|---|---|---|---|
| LEXUS | CT200h | | | | | | | | |
| | ES | | | | | | | | |
| | GS | | | | | | | | |
| | GX | | | | | | | | |
| | IS | | 250 and 350 | | | | | | |
| | LS | | | | | | | | |
| | LX | | | | | | | | |
| | NX | | | | | | | | |
| | RC | | | | | | | | |
| | RX | | | | | | | | |
| TOYOTA | Avalon | | | | | | | | |
| | Camry | | | | | | | | |
| | Corolla | | | | | | | | |
| | Prius | | | C and V | and Prius | and Prime | | | |
| | Highlander | | | | | | | | |
| | Land Cruiser | | | | | | | | |
| | 4Runner | | | | | | | | |
| | Tacoma | | | | | | | | |
| | Tundra | | | | | | | | |
| | Sequoia | | | | | | | | |
| | RAV4 | | | | | | | | |
| | Sienna | | | | | | | | |
| | Yaris | | | | | | | | |
| SCION | iM | | | | | | | | |

# Toyota/Lexus with Photos stored

Gen 1 TSS/LSS pictures only with auto braking, Gen 2 pictures in more circumsta

| | | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|---|---|
| LEXUS | CT200h | | | | discontinued | | | | | |
| | ES | | LSS | LSS | LSS | ? | ? | ? | 2.5 | |
| | GS | | LSS | LSS | LSS | ? | ? | discontinued | | |
| | GX | | | | | | LSS | LSS | LSS | |
| | IS | | | LSS | LSS | ? | ? | ? | 2.5 | |
| | LC | | | | LSS | LSS | LSS | LSS | LSS | |
| | LS | | | | LSS+ | 2.0 | 2.0 | 2.0 | 2.5 | |
| | LX | | LSS | LSS | LSS | ? | ? | ? | 2.5 | |
| | NX | | | | LSS | ? | ? | ? | 3.0 | |
| | RC-F | | | | LSS | ? | ? | ? | LSS | |
| | RX | | LSS | LSS | LSS | ? | ? | ? | 2.0 | |
| | UX | | | | | | | | 2.0 | |
| TOYOTA | 4Runner | | | | | | P | P | P | |
| | 86 | | | | | | | | Subaru | |
| | Avalon | | P | P | P | P | P | P | 2.5+ | |
| | Camry | | | | P | P | P | 2.5+ | 2.5+ | |
| | C-HR | | | | P | P | P | 2.5 | 2.5 | |
| | Corolla | | | P/C | P | 2.0 | 2.0 | 2.0 | 2.0 | |
| | Echo | | | | | | | | | |
| | FJ | | | | | | | | | |
| | Highlander | | | P | P | P | 2.0 | 2.5+ | 2.5+ | |
| | Land Cruiser | | P | P | P | P | P | P | | |
| | Mirai | | | | | 2.0 | 2.0 | 2.5+ | 2.5+ | |
| | Prius | | P/C | P/C | P | P/C | P | 2.0 | 2.0 | |
| | RAV4 | | P/C | P/C | P | 2.0 | 2.0 | 2.0 | 2.0 | |
| | Sequoia | | | | P | P/C | P | P | P | |
| | Sienna | | | | P | P/C | P | 2.0 | 2.0 | |
| | Supra (BMW) | | | | | | (made by BMW) | | | |
| | Tacoma | | | | P | P | P | P | P | |
| | Tundra | | | | P | P | P | P | 2.5 | |
| | Venza | out of production | | | | | | 2.0 | 2.0 | |
| | Yaris | | | C | C | | | | | |

Chart is only inteded to cover US models. Content MAY depend on trim level or model.

# Toyota TechStream / PCS & VCH Data

For IATAI 2021 Conference

Copy in  SAE EDR student reference materials


**This presentation developed by**

**Brad Muir**

**of**

CRASH DATA SPECIALISTS

WWW.CDR-TRAINERS.COM

**Edited for time and Presented by**

**Rick Ruth**

**of**

Institute of Police Technology and Management

IPTM

FOUNDED 1980

# TSS 1.0

- For Images to be recorded and linked the

  VEHICLE must brake
- A PCS warning buzzer is not sufficient for images to be linked with VCH and recorded
- If vehicle begins to brake and driver responds, then related images should be stored
- CHECK: perform CDR download FIRST. Look for deceleration with no driver input: i.e. no brake switch = on, no brake oil pressure, but speed drops more than idling down

# TSS 2.0

- Adds new Image capture categories:
  - Vehicle behavior related image data: 1 Set of images
  - PCS related image data: 3 Sets of images (Pre-Crash, AEB)
  - LKA related image data: 1 Set of images (Lane Keep)
  - ICS related image data : 3 Sets of images
    (Intelligent Clearance Sonar – short range)
  - Collision related image data : 2 Sets of images

# TSS 2.5 / 2.5+

- 2021 Camry and Highlander and Lexus IS to start
- Camera / Millimetre Wave Radar optimized improve response range and enhanced low light pedestrian and daytime bicycle recognition.
- Pre-Collision System adds new Features

| TSS-C | TSS-P | TSS 2.0 | TSS 2.5+ |
| --- | --- | --- | --- |
| TSS-C features Pre-Collision System, Lane Departure Alert, and Automatic High Beam. | TSS-P enhances TSS-C, and adds Dynamic Radar Cruise Control to its suite of systems. | Builds on TSS-P, and consists of up to six active safety and driver assistance systems. | TSS 2.5+ adds new capabilities to the Pre-Collision and Dynamic Radar Cruise Control Systems, and enhances Lane Tracing Assist. |

# 20 Images from the Toyota FCM @ 0.6 sec



# Sample VCH Summary

## Vehicle Diagnostic Report

2020 Rav4 [redacted]
004320 mile

[redacted]

[redacted]

Printed By: Default User(1)

[redacted]

**Vehicle Control History Report**
(1 of 2)

Current Key Cycle : 00372
Current Key Cycle Elapsed Time : 00000:17:23
Maximum Number of Overwrite of the Recording Area : 30000

Key Cycle : All

| Key Cycle | Elapsed Time | Clock Type | Time and Date | Trigger | Image | ODO (mile) | Number of Overwrite of the Recording Area |
|---|---|---|---|---|---|---|---|
| 00030 | 00000:53:56.4 | - | 2/1/2020 4:24:56 PM | Steering angle speed over certain number was detected (related to LDA/LKA/LTA) | Icon | 51 | 1 |
| 00044 | 00000:19:06.4 | - | 2/4/2020 11:00:03 AM | PCS operation history | | 264 | 1 |
| 00051 | 00000:01:15.3 | - | 2/6/2020 9:30:21 AM | Steering angle speed over certain number was detected (related to LDA/LKA/LTA) | Icon | 315 | 1 |
| 00055 | 00000:24:15.8 | - | 2/7/2020 8:50:59 AM | Steering angle speed over certain number was detected (related to LDA/LKA/LTA) | Icon | 340 | 1 |

# A way to get ped hits??

**Vehicle Control History**

| Key Cycle | All |
|---|---|

| Current Key Cycl |
|---|
| 00372 |

| Key Cycle | Elapsed Time | Clock Type | Time and Date | Trigger |
|---|---|---|---|---|
| 00143 | 00000:47:31.8 | - | 2/24/2020 11:24:28 AM | Steering angle speed over certain number was dete |
| 00144 | 00000:19:09.1 | - | 2/24/2020 7:22:43 PM | PCS operation history |
| 00145 | 00000:07:55.7 | - | 2/24/2020 7:53:02 PM | Sudden braking history |
| 00162 | 00000:43:37.7 | - | 2/28/2020 6:26:13 PM | PCS operation history |
| 00178 | 00000:03:40.9 | - | 3/4/2020 5:09:47 PM | Steering angle speed over certain number was dete |
| 00182 | 00000:24:30.4 | - | 3/6/2020 5:50:27 PM | Sudden braking history |
| 00182 | 00000:24:30.4 | - | 3/6/2020 5:50:27 PM | PCS operation history |
| 00182 | 00000:24:30.5 | - | 3/6/2020 5:50:27 PM | Steering angle speed over certain number was dete |
| 00182 | 00000:24:30.8 | - | 3/6/2020 5:50:27 PM | Accelerator signal and brake signal input simultane |
| 00226 | 00000:00:04.3 | - | 3/27/2020 6:58:48 AM | Accelerator signal and brake signal input simultane |

# Toyota “13 EDR” & “15 EDR”

1. **13 EDR** Adds Lateral ***Accel*** data from ACM in frontal events for 200ms (Fixes biggest drawback of **12 EDR**)

**Longitudinal/Lateral Crash Pulse (Most Recent Event,**

| Time (msec) | Longitudinal Delta-V (MPH [km/h]) | Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2) | F |
|---|---|---|---|
| 0 | 0.0 [0.0] | 0.0 | |
| 10 | 0.3 [0.6] | -1.9 | |
| 20 | 0.5 [0.8] | 0.0 | |

2. ***Calculate*** Lat **Delta V** for each 10ms interval
3. $\mathbf{DV_{10ms}}$ = ***Accel***/9.8*(32.2/1.466)*0.01s
4. Add up intervals to get ***total*** **Delta V**
5. **Template available to calculate it**

# 2018+ Toyota “17EDR”

1. “Front/rear” 5mph X events replaced by “Front/Rear/Side”, events triggered by 5mph X ***or Y*** DV or side deploys. Y DV graph added.
2. 6 precrash memory spaces, 4 DV spaces for each crash type (vs 2 each prior)
3. Separate side events still reported with 70ms side accel in $m/sec^2$, but do NOT have precrash associated with them.

# 17EDR Precrash (new look, same data)

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4**

| Time (sec) | Vehicle Speed (MPH [km/h]) | Accelerator Pedal, % Full (%) | Percentage of Engine Throttle (%) | Fuel Injection Quantity (mm^3/st) | Engine RPM (RPM) | Motor RPM (RPM) | Service Brake, ON/OFF |
|---|---|---|---|---|---|---|---|
| -5.00 | 29.8 [48] | 49.5 | 24.5 | Invalid | 2,400 | Invalid | OFF |
| -4.50 | 32.3 [52] | 47.0 | 24.5 | Invalid | 2,500 | Invalid | OFF |
| -4.00 | 34.2 [55] | 38.0 | 20.0 | Invalid | 2,700 | Invalid | OFF |
| -3.50 | 35.4 [57] | 32.5 | 16.0 | Invalid | 2,800 | Invalid | OFF |
| -3.00 | 36.7 [59] | 26.5 | 13.5 | Invalid | 2,400 | Invalid | OFF |
| -2.50 | 37.3 [60] | 22.5 | 11.0 | Invalid | 2,400 | Invalid | OFF |
| -2.00 | 37.3 [60] | 19.0 | 8.5 | Invalid | 2,200 | Invalid | OFF |
| -1.50 | 38.5 [62] | 19.0 | 6.0 | Invalid | 1,900 | Invalid | OFF |
| -1.00 | 37.9 [61] | 18.0 | 5.5 | Invalid | 1,900 | Invalid | OFF |
| -0.50 | 37.9 [61] | 18.0 | 5.0 | Invalid | 1,800 | Invalid | OFF |
| TRG(0) | 36.0 [58] | 100.0 | 26.0 | Invalid | 1,700 | Invalid | OFF |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4**

| Time (sec) | ABS Control Status | BOS Control Status | Brake Oil Pressure (Mpa) | Longitudinal Acceleration , VSC Sensor (m/s^2) | Yaw Rate (deg/s) | Steering Input (degrees) | Shift Position |
|---|---|---|---|---|---|---|---|
| -5.00 | OFF | OFF | 0.00 | 1.579 | -0.98 | -4.5 | D |
| -4.50 | OFF | OFF | 0.00 | 1.795 | 0.00 | -1.5 | D |
| -4.00 | OFF | OFF | 0.00 | 1.292 | 0.49 | 0.0 | D |
| -3.50 | OFF | OFF | 0.00 | 0.861 | 0.00 | -1.5 | D |
| -3.00 | OFF | OFF | 0.00 | 0.718 | 0.00 | -1.5 | D |
| -2.50 | OFF | OFF | 0.00 | -0.072 | 0.00 | -1.5 | D |
| -2.00 | OFF | OFF | 0.00 | 0.000 | 0.00 | -1.5 | D |
| -1.50 | OFF | OFF | 0.00 | 0.000 | -0.49 | -1.5 | D |
| -1.00 | OFF | OFF | 0.00 | 0.287 | 0.00 | -1.5 | D |
| -0.50 | OFF | OFF | 0.00 | -0.144 | 0.00 | -1.5 | D |
| TRG(0) | OFF | OFF | 0.48 | -1.507 | 61.00 | -19.5 | D |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4**

| Time (sec) | Sequential Shift Range | Cruise Control Status | VSC Control Status | READY Signal | Drive Mode, Power Train | Drive Mode, Snow | Drive Mode, EV |
|---|---|---|---|---|---|---|---|
| -5.00 | Undetermined | OFF | ON (enable) | Invalid | Normal | OFF | Invalid |
| -4.50 | Undetermined | OFF | ON (enable) | Invalid | Normal | OFF | Invalid |

# 17 EDR time data & ped events

**System Status at Pedestrian Event (1st Prior Event, TRG 1)**

| | |
|---|---|
| TRG Count (times) | 1 |
| TRG Count not for Pedestrian (times) | SNA |
| Ignition Cycle, Crash (times) | 353 |
| Odometer signal (miles [km]) | 1,195 [1,923] |
| Trip count (times) | 392 |
| Time count (msec) | 1,283,900 |
| Time count input system | Normal |
| Airbag Warning Lamp, On/Off | OFF |
| Pedestrian Warning Lamp, On/Off | OFF |
| Time from Previous TRG (msec) | 32767 or greater |

1. Odometer added
2. “Trip count” similar to key cycles but slightly higher,
3. Time Count (like Ford “Key on time”), Odometer
4. Pedestrian events if pedpro option bought

# NISSAN EDR's 21.4.1 (CDR 51%)

| Model | Location | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR cable number | | | | | | | | | | | | | | | | | |
| **INFINITI** | | | | | | | | | | | | | | | | | No info |
| G CONVERTIBLE | TUNNEL BETWEEN SEATS | | | | * | | Jul-12 | | 780 | | | | | | | | |
| G COUPE | TUNNEL BETWEEN SEATS | | * | Yellow = Proprietary Consult Tool Only | | | | | 780 | | | | | | | | |
| G SEDAN | TUNNEL BETWEEN SEATS | | * | No direct to module cables | | | Jul-12 | | 780 | | | | | | | | |
| M | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | | |
| Q30 | | | | | | | | | | | | | 780 | | | | |
| Q40 | TUNNEL BETWEEN SEATS | Yellow area not covered by CDR | | | | | | | | | 780 | | | | | | |
| Q50 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | | |
| Q60 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | 780 | | | | |
| Q70 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | | |
| EX | TUNNEL BETWEEN SEATS | | | Dealers do NOT have special software | | | | | 780 | | | | | | | | |
| FX | TUNNEL BETWEEN SEATS | | | required to read EDR | | | | | 780 | | | | | | | | |
| JX35 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | | |
| QX50 | | | | | | | | | | | | | 780 | | 828 | | |
| QX60 | | | | | | | | | | | | | 780 | | | | |
| QX80 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | 828 | |
| QX30 | TUNNEL BETWEEN SEATS | | | | | | | | | | | | 598 | | | | |
| **NISSAN** | | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 |
| 370Z | TUNNEL BETWEEN SEATS | | * | | | | Aug-12 | | 780 | | | | | | | | |
| ALTIMA COUPE | TUNNEL BETWEEN SEATS | | | * | | | | | 780 | INCLUDED IN SEDAN BELOW | | | | | | | |
| ALTIMA SEDAN | TUNNEL BETWEEN SEATS | | * | Yellow = Proprietary Consult Tool Only | | | | | 780 | | | | | | 828 | | |
| ARMADA | TUNNEL BETWEEN SEATS | * | | No direct to module cables | | | | | 780 | | | ? | 780 | | | | |
| CUBE | TUNNEL BETWEEN SEATS | | | | * | | | | 780 | | | | | | | | |
| FRONTIER | TUNNEL BETWEEN SEATS | * | | | | | Oct-12 | | 780 | | | | | | | | 780 |
| GT-R | TUNNEL BETWEEN SEATS | | | | * | | | | | 780 | | | | | | | |
| JUKE | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | | |
| KICKS | | | | | | | | | | | | | | 780 | 828 | | |
| LEAF | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | 828 | | |
| MAXIMA SEDAN | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | | | | | | 828 | | |
| MICRA | Check Manual | | | | | | | | | | 780 | | | | | | |
| MURANO | TUNNEL BETWEEN SEATS | | * | | * | | | | 780 | | | | | | 828 | | |
| MURANO CROSS C | TUNNEL BETWEEN SEATS | | | | | | | | included in above | | | | | | | | |
| NV200 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | | | |
| PATHFINDER | TUNNEL BETWEEN SEATS | * | | | | | | | 780 | | | | | | | | |
| QUEST | CTR CONSOLE NEAR FRONT | | * | | | | | | 780 | | | | | | | | |
| ROGUE | Check Manual | | | * | | | | | 780 | | | | | | | | 828 |
| ROGUE SPORT/QUASHAI | | | | | | | | | | | | | | 780 | | 828 | |
| ROOX (Japan) | | | | | | | | | | | | | | | | 828 | |
| SENTRA | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | | | | | | | 828 | |
| TITAN | TUNNEL BETWEEN SEATS | * | | | | | | | 780 | | | | | | | | |
| VERSA SEDAN | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | | | | | | | 828 | |
| VERSA NOTE HB | TUNNEL BETWEEN SEATS | | | | | | Sep-12 | | 780 | | | | | | | | 154 |
| XTERRA | TUNNEL BETWEEN SEATS | * | | | | | Oct-12 | | 780 | | | | | | | | |

# Nissan Pre-Crash Data

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)**

(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % full | Engine RPM | Motor RPM | Service Brake (On, Off) | Steering Input (deg) |
|---|---|---|---|---|---|---|
| -5.0 | 24 [ 39] | 16 | 2650 | 0 | Off (Brake Not Activated) | -32 |
| -4.5 | 23 [ 37] | 11.5 | 1750 | 0 | Off (Brake Not Activated) | -2 |
| -4.0 | 21 [ 34] | 9 | 1200 | 0 | On (Brake Activated) | 28 |
| -3.5 | 15 [ 24] | 9 | 1150 | 0 | On (Brake Activated) | 14 |
| -3.0 | 13 [ 21] | 4 | 950 | 0 | On (Brake Activated) | 4 |
| -2.5 | 11 [ 17] | 4 | 950 | 0 | On (Brake Activated) | 4 |
| -2.0 | 5 [ 8] | 4 | 900 | 0 | On (Brake Activated) | -48 |
| -1.5 | 3 [ 5] | 4 | 850 | 0 | On (Brake Activated) | -98 |
| -1.0 | 1 [ 2] | 4 | 850 | 0 | On (Brake Activated) | -160 |
| -0.5 | 3 [ 5] | 10.5 | 1100 | 0 | Off (Brake Not Activated) | -194 |
| 0.0 | 6 [ 9] | 16 | 1150 | 0 | Off (Brake Not Activated) | -218 |

Hybrid/Electric or Trans Input

Trans input tells you convertere slip

Table 2 – not rq'd

At AE

# Nissan Long ΔV Polarity Fixed?

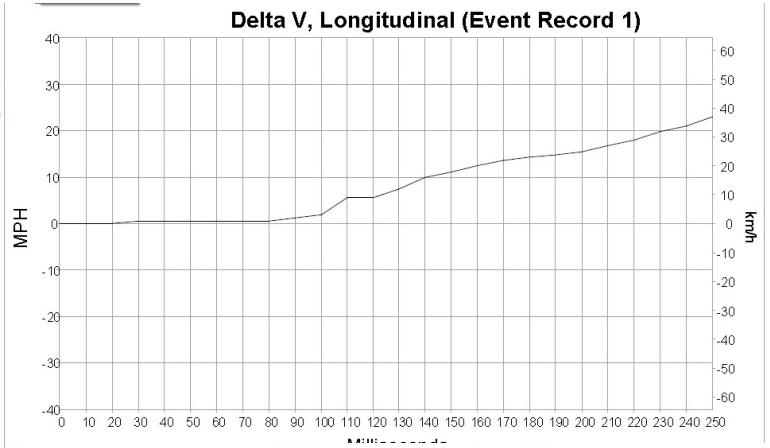
## Front hit *was* positive - ***Some exceptions still in 19.3!***

2013 Altima in CDR 12.2

**System Status at Event (Event Record 1)**

| | |
|---|---|
| Maximum Delta-V, Longitudinal (MPH [km/h]) | 29 [ 47] |
| Time, Maximum Delta-V, Longitudinal (msec) | 300 |
| Maximum Delta-V, Lateral (MPH [km/h]) | 5 [ 8] |
| Time, Maximum Delta-V, Lateral (msec) | 210 |
| Maximum Acceleration, Longitudinal (g) | 32.5 |
| Time, Maximum Acceleration, Longitudinal (msec) | 102.5 |
| Maximum Acceleration, Lateral (g) | 36 |
| Time, Maximum Acceleration, Lateral (msec) | 112.5 |

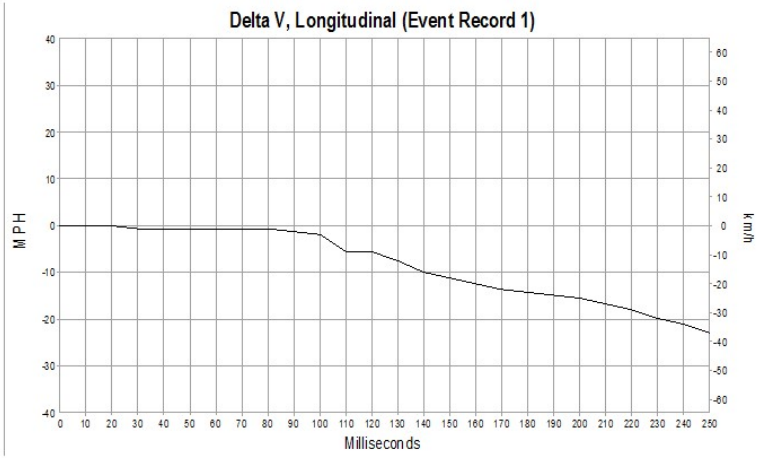
+29 at 300ms




2013 Altima in CDR 16.5

| | |
|---|---|
| Maximum Delta-V, Longitudinal (MPH [km/h]) | -29 [-47] |
| Time, Maximum Delta-V, Longitudinal (msec) | 300 |
| Maximum Delta-V, Lateral (MPH [km/h]) | 5 [ 8] |
| Time, Maximum Delta-V, Lateral (msec) | 210 |
| Maximum Acceleration, Longitudinal (g) | -32.5 |
| Time, Maximum Acceleration, Longitudinal (msec) | 102.5 |
| Maximum Acceleration, Lateral (g) | 36 |
| Time, Maximum Acceleration, Lateral (msec) | 112.5 |

-29 at 300ms



# Nissan Recording Threshold

1. Several files have been read with events under 5mph Delta V over 150ms.
2. From 2013-2019 we believed all Nissans had a wakeup threshold of AE
3. At 2020 EDR User Summit, it was reported that tests were done on a 2014 Pathfinder at low speeds. No events were created below 5mph Delta V, events were created over 5mph.
4. Threshold could be model dependent!

# Mazda CDR 21.2 Coverage 44%

| Model | Location | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR cable number shown | | | | | | | | | | | | | | | |
| **MAZDA** | | | | All Mazda use 387 adapter for direct to module | | | | | | | | | | | |
| CX-3 | TUNNEL BELOW IP | | | | | | | | 812 | | | | | | |
| Tribute/CX-5 | TUNNEL AFT SHIFTER | | | Read as Escape | | 779 | | | 811 | 812 | | | | | |
| CX-9 | TUNNEL BELOW IP | | | 778 | | | | | 811 | | | | | | |
| CX-30 | | | | | | | | | | | | | 843 | | |
| Mazda 2 | TUNNEL AFT SHIFTER | | | 792 | | | | not listed | 812 | not listed | | | | | |
| Mazda 3 | TUNNEL AFT SHIFTER | | | | 779 | | | | 811 | | | 843 | | | |
| Mazda 5 | TUNNEL BELOW IP | | | | 778 | | | | | not listed | | | | | |
| Mazda 6 | TUNNEL BELOW IP | | | 792 | | | 779 | | 811 | | 831 | | | | |
| MX-5 Miatta | TUNNEL BELOW IP | | | | | no coverage | | | 812 | | | | | | |
| MX-30 | | | | | | | | | | | | | | | 843 |

Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers.

Memo: The Mazda Tribute was sister to the Ford Escape thru 2011. A 2011 Tribute was read using a 2011 Escape VIN.

The source data for this chart is the Bosch CDR help file for Version 21.2
Always consult the latest help file for the most accurate coverage

# Volvo CDR 21.3 Coverage 32%

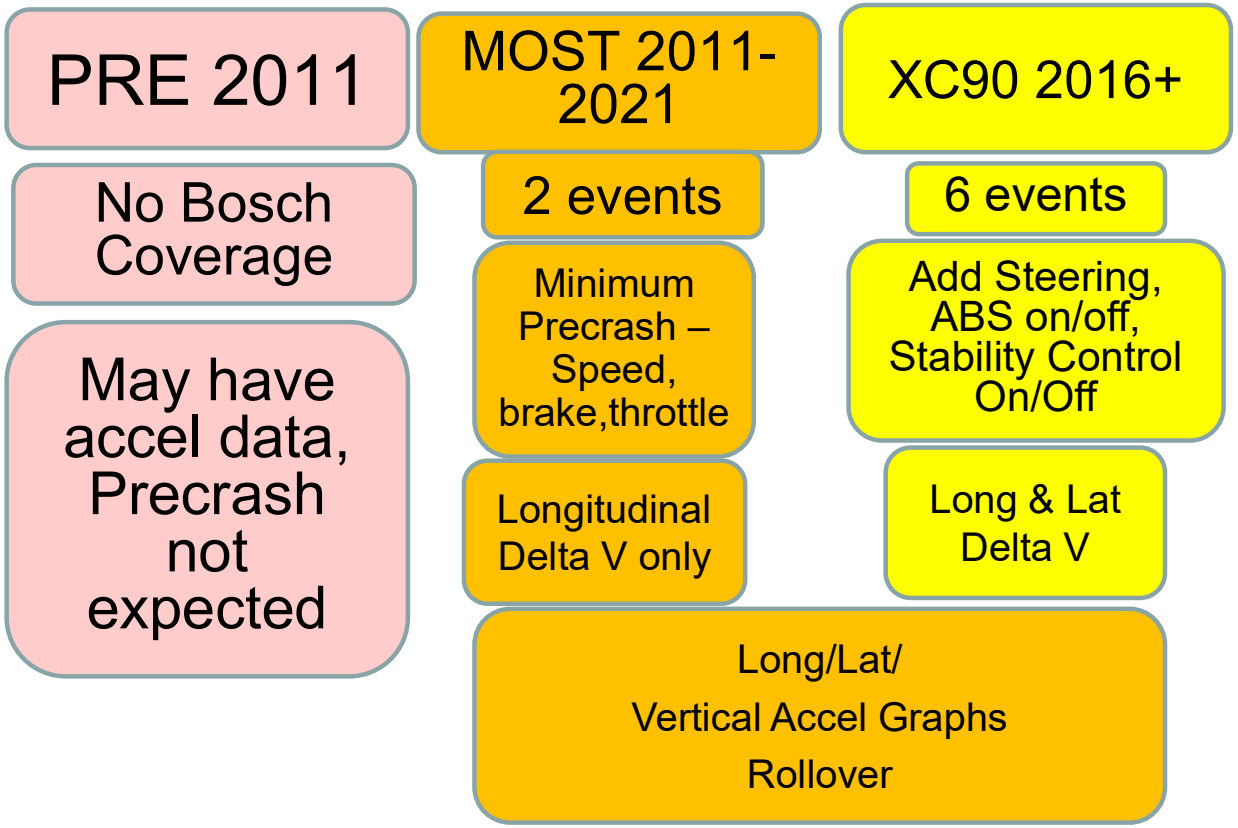
| Model | Location | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR cable number sh | | | | | | | | | | | | | |
| VOLVO | | All Volvo use 387 Adapter for DTM images | | | | | | | | | | | |
| S40 | | | | | | | | | | | | | |
| S60 | CTR TUNNEL UNDER CONSOLE | | 799 | | | | | | | 500/830 | | | |
| S80, S80L | CTR TUNNEL UNDER CONSOLE | | | | | | | | | | | | |
| S90 | CTR TUNNEL UNDER CONSOLE | | | | | | | | 816 | | | | |
| V40 | CTR TUNNEL UNDER CONSOLE | | | | | | | | 816 Euro | | | | |
| V60 | CTR TUNNEL UNDER CONSOLE | | 799 | | | | | | | 500/830 | | | |
| V70 | CTR TUNNEL UNDER CONSOLE | | | | 799 | | | | | | | | |
| V90 | CTR TUNNEL UNDER | | | | | | | | 816 | | | | |
| XC40 | CTR TUNNEL UNDER | | | | | | | | | 500/830 | | | |
| XC60 | CTR TUNNEL UNDER CONSOLE | | | | 799 | | | | | | 816 | | |
| XC70 | CTR TUNNEL UNDER CONSOLE | | | | 799 | | | | | | | | |
| XC90 | CTR TUNNEL UNDER CONSOLE | | | | | | | 816 | | | | | |

Old Volvo DV Law Enf only call Volvo

The source data for this chart is the Bosch CDR help file for Version 21.3
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# Volvo EDR Evolution



# 2016+ XC90 Sample Volvo File

## (vs. 2013 Volvo with minimum data)

**Pre-Crash Data -1 Sec (Event Record 1)**

| | |
|---|---|
| Ignition Cycle, Crash | 2,764 |
| Safety Belt Status, Driver | On, Belted |
| Safety Belt Status, Passenger | On, Belted |
| Frontal Airbag Warning Lamp | Off |
| Frontal Airbag Suppression Switch Status, Front Passenger | On |
| Seat Track Position Switch, Foremost, Status, Driver | Not Equipped |
| Seat Track Position Switch, Foremost, Status, Front Passenger | Not Equipped |
| Occupant Size Right Front Passenger Child | Not Equipped |

2020 still not added to other models

Added Data for XC90

**Pre-Crash -5 to 0 sec (Event Record 1)**

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full (%) | Service Brake (On, Off) | Steering input (%) | ABS Activity | Stability control status |
|---|---|---|---|---|---|---|
| -5.0 | 43.5 [70.0] | 100.0 | Off | 0.0 | Off | On |
| -4.5 | 47.8 [77.0] | 100.0 | Off | 0.0 | Off | On |
| -4.0 | 52.2 [84.0] | 100.0 | Off | 2.0 | Off | On |
| -3.5 | 55.9 [90.0] | 100.0 | Off | 3.0 | Off | On |
| -3.0 | 58.4 [94.0] | 100.0 | Off | 3.0 | Off | On |
| -2.5 | 61.5 [99.0] | 100.0 | Off | 11.0 | Off | Engaged |
| -2.0 | 62.8 [101.0] | 100.0 | Off | 6.0 | Off | Engaged |
| -1.5 | 65.2 [105.0] | 100.0 | Off | -9.0 | Off | Engaged |
| -1.0 | 67.1 [108.0] | 100.0 | Off | -8.0 | Off | Engaged |
| -0.5 | 69.6 [112.0] | 100.0 | Off | -9.0 | Off | Engaged |
| 0.0 | 70.2 [113.0] | 100.0 | Off | -13.0 | Off | Engaged |

# 2016+ XC90 Sample Volvo File

2021 still not added to other models

## System Status at Retrieval

| | |
|---|---|
| Vehicle Identification Number | YV1LF |
| Application Diagnostic Database Part Number | 31387255 AL |
| Ignition Cycle, Download | 2,765 |
| Lifetime Operating Timer (sec) | 2,154,311 |

## System Status at Event (Event Record 1)

| | |
|---|---|
| Data Area Status, Event Record 1 | Locked, Data Stored |
| Data Area Read Status, Event Record 1 | Data Not Read |
| Complete File Recorded (Yes/No) | Yes |
| Multi-Event, Number of Events (1,2) | Event Number 1 |
| Time from Preceding Event (sec) | Written but No Data Available |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | < -62 [-100] |
| Time, Maximum Delta-V, Longitudinal (msec) | 100 |
| Maximum Delta-V, Lateral (MPH [km/h]) | -19.9 [-32.0] |
| Time, Maximum Delta-V, Lateral (msec) | >300 |

Clipped Value (at Max)

Not present in 2013, Added for XC90

[redacted]

# BMW CDR US 21.5 Coverage

See help file for Rolls Royce

| Model | Location | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR adapter/cable number shown | | | | | | | | | | | | |
| **BMW** | **All BMW require 387 Adapter for DTM** | | | | | | | | | | | |
| **1 Series** | | | Not sold in US | | | | | | 500/822 | | | |
| **2 Series** | Driver side, behind IP, rt of strg | | | 796 | | | | | | | | 796 OR 500/807, 500/822 |
| **2 Series xdrive** | Center console between seats | | | | | | | | | | 500/822 | |
| **3 Series** | Passenger side, behind the glove box | | 796 | | | | | | | | 500/822 | |
| **Active Hybrid 3** | Driver side, behind IP, rt of strg | | | 796 | | | | | | | | |
| **4 Series** | Passenger side, behind the glove box | | 796 | | | | | 500/822 | | | | |
| **5 Series** | Passenger side, behind the glove box | | 796 | | | | | | | | 500/822 | |
| **6 Series** | See Service Manual | | | | | | | | 500/822 | | | |
| **6 Series GT** | Passenger side, behind the glove box | | 796 | | | 500/796 | 500/822 | | | | | |
| **7 Series** | Passenger side, behind the glove box | | | | | | | | 500/822 | | | |
| **8 Series** | Passenger side, behind the glove box | | 796 | | | | | | | | 500/822 | |
| **X3 xDrive** | Center Tunnel, Between Seats | | | 796 | | | | | 500/822 | | | |
| **X5** | Center Tunnel, Between Seats | | | | 798 | 500/798 | 500/807 | | also X2 | | | |
| **X1, X2** | See Help File - Various | | | | 796 | | | 500/822 | | | 500/807 | |
| **X3, X4** | Center Tunnel, Between Seats | | | | 796 | | | | | | 500/822 | |
| **X6** | See Service Manual | | | | | | | | 500/822 | | | |
| **X7** | Center Tunnel, Under Dash | | | | 798 | | | | 500/822 | | | |
| **Z4** | See Service Manual | | | | 798 | | | | 500/822 | | | |
| **iX** | | | | | | | | | | | | 500/822 |
| **i3, i8** | | | | 500/807 | | | | | | | | |

No “back model” coverage

500=flex ray

The source data for this chart is the Bosch CDR help file for Version 21.5

Always consult the latest help file for the most accurate coverage

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# BMW EDR Evolution

1. Basic EDR 2013
2. “Extension” 2014
3. Some driver assist mid-2015 - timing
4. Auto Steering (autonomous vehicle) Cameras all views 4 fps

   Not intended for CDR,

   only BMW can access
5. Clock and odometer in 2019 Mini Cooper

# Hardware/Software Glitch?

- Recently a 2020 BMW X5 was read DTM using CAN+/387/CDR500 and obtained a file with no events. The hex data shows memory section headings which all said “No Data Received”.
- Module was read using CDR900/CDR500 and retrieved a valid event.
- This is under investigation by Bosch, until resolved use CDR900 whenever possible.

# More data elements in 2020 X5 “ACSM5”

- Recent files show day/date/time of crash and second/third row seat belt use, more

| | |
|---|---|
| Multi-Event, Number of Events | 1. Event |
| Time From Previous Event to Current Event (msec) | Data Not Available |
| Time From Last Speed Data Sample (Precrash) to Time Zero (msec) | 57 |
| Time From Initial Event to Current Event (msec) | Data Not Available |
| Complete File Recorded, Public, Prio 1 Data | Completed Successfully |
| Ignition Cycle, Crash (Cycle) | 1,315 |
| Ignition Cycle, Download (Cycle) | 1,316 |
| Vehicle Clock, Date and Time at Event (YYYY-MM-DD, HH:MM:SS) | 2020-12-18, 09:03:54 |
| Vehicle Mileage (km) | 13,285 |
| Operating Time (min) | 20,794 |

| | |
|---|---|
| Safety Belt Status, 2nd Row, Left Side | Not Belted |
| Safety Belt Status, 2nd Row, Center | Not Belted |
| Safety Belt Status, 2nd Row, Right Side | Not Belted |
| Safety Belt Status, 3rd Row, Left Side | Data Not Available |
| Safety Belt Status, 3rd Row, Center | Data Not Available |
| Safety Belt Status, 3rd Row, Right Side | Data Not Available |

# Mercedes CDR Coverage 21.2 36%

| Model | Start of Sales | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR DTM cable number | | | | | | | | | | | |
| Mercedes | EDR Start | All Mercedes use 387 adapter for direct to module downloads | | | | | | | | | |
| A-Class Limo | | | | | | | | 832 | | | |
| AMG-GT | | | | | 800 | | | | | | |
| B-Class EV | after 4/1/2014 | | | 598 | | | | | | | |
| B-Class | after 6/2/2014 | | | 598 | | | | | 832 Limo | | |
| C Class Sedan | after 9/1/2014 | | | 801 | | | | | | | |
| C Class Coupe | tart of 2015 M.Y. | | | 800 | 801 | | | | | | |
| CLA Class | after 6/2/2014 | | | 598 | | | | | 832 | | |
| CLS-Class | | | | 800 | | | | | 801 | | |
| E Class (includes coupe) | after 4/1/2013 | | 800 | | | 801 w steering data | | | | | |
| EQA/B/C | | | | | | | | | 801 w strg | | |
| G Class | after 11/1/201 | | | 598 | | | | 801 | | | |
| GL Class | after 7/11/201 | | | 598 | ? | ? | | | | | |
| GLA Class | after 6/2/201 | | | 598 | | | | | | 832 | |
| GLC Class | | | | | 598 | | 801 | | | 832 | |
| GLE Class | | | | | 598 | | | | 832 | | |
| GLK | after 12/2013 | | 800 | | ? | ? | | | | | |
| GLS Class | | | | | 598 | | | | 832 | | |
| GT | | | | | | | | 801 | | | |
| G wagon | | | | | | | | | 801 | | |
| Maybach S600 | | | | | 801 | | | | | | |
| ML-Class | after 7/11/2014 | | | 598 | ? | | | | | | |
| S-class Sedan | after 9/2013 | | 801 | | | | | | | 500/841 | |
| S-Class Coupe | after 9/1/2014 | | | 801 | | | | | | | |
| SL | after 12/2013 | | 800 | | | | | | | | |
| SLC | | | | | | | 800 | | | | |
| SLK | after12/2013 | | 800 | | | | | | | | |
| Sprinter | | | | | | | | 832 | | | |
| Vito | | | | | 598 | | | | | 832 | |
| Smart & 2016 with "E" | after 9/1/2014 | | | 809 | | | | | | | |
| Smart with VIN 4th "F" | | | | | 820 | | | | | | |
| Smart for two, coupe, cabrio | | | | | | | 820 | | | | |
| Metris | | | | | | | 598 | | | | |

No "back model" coverage

# Mercedes Details

1. Mercedes phased in EDR in anticipation of the NPRM requiring EDR's for Sept. 2014
2. Precrash speed, brake, throttle (563 Table 1) & lat/longitudinal ΔV.
3. Can hold 5 events, D's lock, ND's FIFO
4. Recording threshold is 5 mph over 150ms
5. Speed is avg. of 4 wheel speed sensors
6. The new 2017 E Class has Steering, ABS, Stab Cont'l. This marks a departure from their philosophy to record only the minimum required by law.

# Mercedes Key Cycles

- Some Mercedes have unusually high key cycle counts for their age, and since event.
- Other brands have 2500 key cycles per yr or 20,000km.
- Working theory is that counter increments whenever key fob leaves range and comes back into range.

| | |
|---|---|
| Ignition Cycle, Crash (cycle) | 11,639 |
| Ignition Cycle, Download (cycle) | 12,113 |
| Vehicle Mileage (km) | 20,950 |
| Operating Time (min) | 29,712 |
| Vehicle Identification Number | 4JGDA5JB2HA****** |
| Event Counter (counts) | 1 |

# VW/Audi Brands EDR

1. EDR in 2014 EOS, most 2015+ VW/Audi.
2. EDR was installed in anticipation of the US NPRM requiring EDR's by 9/1/2014, which was never implemented.
3. **Bentley** has EDR 2016+.
4. **Lamborghini** has EDR in Adventadore/Hurracan
5. **Porsche no EDR** *until 2020 Taycan.*
6. US only 2015-17, Worldwide 2018+ in transverse engine platforms

# VW/Audi 21.3 Oct 2021 34+% (NC in 21_5)

| Model | Location | 2009-13 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR adapter/cable number | | | | See help file for Rolls Royce | | | | | | | |
| All VW/Audi Require the 387 Adapter to read DTM | | | Gold = US only (Canada not covered) | | | | Yellow=Worldwide | | | | |
| **VW** | | | | | | | | | | | |
| Arteon | UNDER CENTER STACK | | | | | | 805 | | | | |
| Atlas | UNDER CENTER STACK | | | | | | 805 | | | | |
| Beetle | UNDER CENTER STACK | | | 804 | | | Canada added | | | | |
| CC | UNDER CENTER STACK | | | 804 | | | | | | | |
| Crafter | UNDER CENTER STACK | | | | | | 805 | | | | |
| EOS | UNDER CENTER STACK | | 806 | | | | | | | | |
| Golf, eGolf | UNDER CENTER STACK | | | 805 | | | Worldwide | | | | |
| Jetta | UNDER CENTER STACK | | | 804 | | | 805 | | | | |
| Neo | | | | | | | | | | | |
| Passat | UNDER CENTER STACK | | | 804 | | | 805 | | | | |
| Polo | UNDER CENTER STACK | | | | | | 805 | | | | |
| Routan | UNDER CENTER STACK | | | | | | | | | | |
| Sportwagen + Alltrack | | | | Golf platform, not called out separtely in help file | | | | | | | |
| T-Roc | UNDER CENTER STACK | | | | | | 805 | 805 | | | |
| T-Cross | UNDER CENTER STACK | | | | | | | | | | |
| Tiguan | UNDER CENTER STACK | | | 804 | | 805 | | | | | |
| Touareg | UNDER GEAR SELECTOR | | | 804 | | | | 500/826 | | | |
| Touran | UNDER CENTER STACK | | | | | | 805 | | | | |
| **AUDI** | | | | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| A1 | Under Center Stack | | | | | | | 805 | | | |
| A3,S3, RS3 | Under Center Stack | | | 805 | | | | | 842 | | |
| A4, S4, A4 allroa | Tunnel between 1st/2nd row seats | | | 804 | | 500/813 | | | | | |
| A5, R5, S5, RS5 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/813 | | | | |
| A6, A6L, S6, Ava | Tunnel between 1st/2nd row seats | | | 804 | | | 500/826 | | | | |
| A7, S7, RS7 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/826 | | | | |
| A8, A8L, S8 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/826 | | | | |
| Q2 | Under Center Stack | | | | | | 805 | | | | |
| Q3 | Under Center Stack | | | 804 | | | | 805 | | | |
| Q4 | | | | | | | | | | | 842 |
| Q5, SQ5 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/813 | | | | |
| Q7 | Under Center Stack | | | | | 500/813 | | | 500/826 | | |
| Q8 | Under Center Stack | | | | | | 500/826 | | | | |
| R8 | Under Center Stack | | | | | 804 | | | | | |
| TT, TTS | | | | not covered at this time | | | | | | | |
| eTron | Under Center Stack | | | | | | | 500/826 | | | |
| The source data for this chart is the Bosch CDR help file for Version 19.6. | | Always consult the latest help file for the most accurate coverage | | | | | | | | | |
| Copyright Ruth Consulting LLC 2021- Authorized for distribution by UNF-IPTM & SAE | | | | | | | | | | | |

Chrysler Minivan Sister

No "back model" coverage

# VW/Audi Events

- Recorder holds up to 6 events in Chron order.
- Event types are Front, Rear, Side, Rollover, Pedestrian (if equipped), External Trigger, and HV (undefined). External triggers are Automatic Driver Assistance System (ADAS) events and are not crashes. If the driver is not braking hard enough, ADAS will override and brake harder.
- DL's say crash event threshold is 8 kph (5mph) lat or long Delta V over 150ms for ND's, D's will record at any magnitude. External Triggers may record with 0 ΔV.
- A small # of ND events have <5mph, this is the exception not the rule.

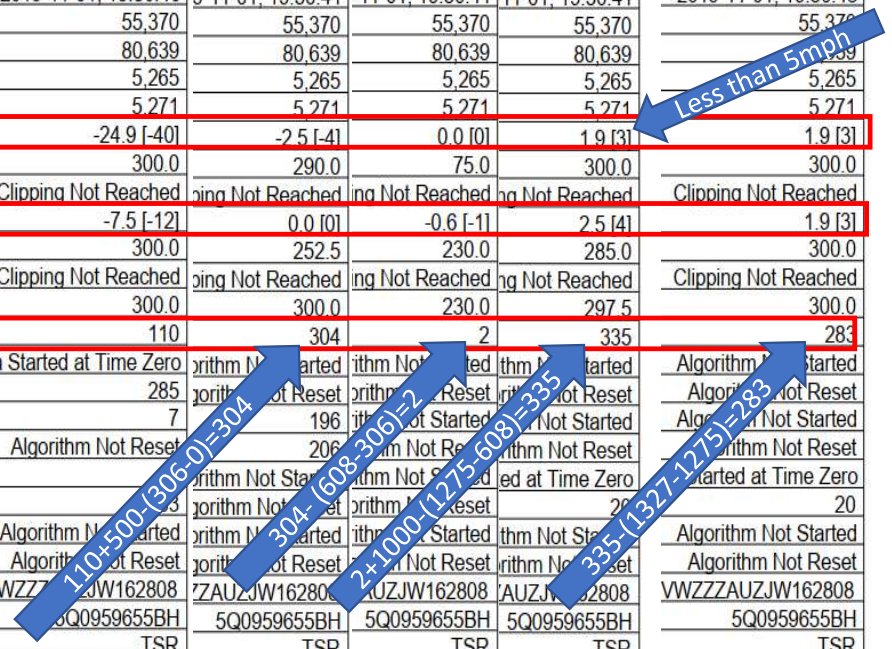
# VW/Audi Details

1. Same data set for all event types
2. Time from last precrash to AE noted
3. Some 2017+ modules use Flex Ray/500
4. Date/Time of Crash from clock (GPS if clock not set) in 2015+ Audi models, phased into VW models
5. Newer models require hood to be open to read through DLC, have gateway
6. Speed from speedometer

# Threshold & Time between events (Five Overlaid)

**System Status at Event (Record 5)**

| | | | | | |
|---|---|---|---|---|---|
| Event Counter at Event | 1 | 2 | 3 | 4 | 5 |
| Event Type | Frontal | Frontal | Frontal | ta Not Available | Data Not Available |
| Multi-Event, Number of Events | 1, Event | 2, Event | 3, Event | 4, Event | 5, Event |
| Time from Initial Event to Current Event (msec) | 0.0 | 306.0 | 608.0 | 1,275.0 | 1,327.0 |
| Vehicle Clock, Date and Time at Event (YYYY-MM-DD, HH:M | 2019-11-01, 19:50:40 | 9-11-01, 19:50:41 | -11-01, 19:50:41 | 11-01, 19:50:41 | 2019-11-01, 19:50:43 |
| Vehicle Mileage (km) | 55,370 | 55,370 | 55,370 | 55,370 | 55,37 |
| Operating Time (min) | 80,639 | 80,639 | 80,639 | 80,639 | 1,339 |
| Ignition Cycle at Event (Cycles) | 5,265 | 5,265 | 5,265 | 5,265 | 5,265 |
| Ignition Cycle at Download (Cycles) | 5,271 | 5,271 | 5,271 | 5,271 | 5,271 |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | -24.9 [-40] | -2.5 [-4] | 0.0 [0] | 1.9 [3] | 1.9 [3] |
| Time, Maximum Delta-V, Longitudinal (msec) | 300.0 | 290.0 | 75.0 | 300.0 | 300.0 |
| Clipping Time, Longitudinal Acceleration Sensor (msec) | Clipping Not Reached | ping Not Reached | ing Not Reached | ng Not Reached | Clipping Not Reached |
| Maximum Delta-V, Lateral (MPH [km/h]) | -7.5 [-12] | 0.0 [0] | -0.6 [-1] | 2.5 [4] | 1.9 [3] |
| Time, Maximum Delta-V, Lateral (msec) | 300.0 | 252.5 | 230.0 | 285.0 | 300.0 |
| Clipping Time, Lateral Acceleration Sensor (msec) | Clipping Not Reached | ping Not Reached | ing Not Reached | ng Not Reached | Clipping Not Reached |
| Time, Maximum Delta-V, Resultant (msec) | 300.0 | 300.0 | 230.0 | 297.5 | 300.0 |
| Time from Last Speed Data Sample (Precrash) to Time Zero | 110 | 304 | 2 | 335 | 283 |
| Time from Time Zero to Algorithm Start (Front) (msec) | ithm Started at Time Zero | orithm N arted | ithm No ted | thm N tarted | Algorithm tarted |
| Time from Time Zero to Algorithm Reset (Front) (msec) | 285 | gorith ot Reset | orithm Reset | rithm ot Reset | Algori Not Reset |
| Time from Time Zero to Algorithm Start (Side) (msec) | 7 | 196 | ith ot Started | Not Started | Alg Not Started |
| Time from Time Zero to Algorithm Reset (Side) (msec) | Algorithm Not Reset | 206 | hm Not Re | ithm Not Reset | rithm Not Reset |
| Time from Time Zero to Algorithm Start (Rear) (msec) | | orithm Not Star | hm Not S d | ed at Time Zero | tarted at Time Zero |
| Time from Time Zero to Algorithm Reset (Rear) (msec) | 3 | gorithm Not et | orithm eset | 20 | 20 |
| Time from Time Zero to Deployment (Rollover) (msec) | Algorithm N arted | orithm N arted | ith Started | thm Not Sta | Algorithm Not Started |
| Time from Time Zero to Algorithm Reset (Rollover) (msec) | Algorith ot Reset | gorith ot Reset | n Not Reset | rithm No set | Algorithm Not Reset |
| Vehicle Identification Number (VIN) | WVWZZ JW162808 | ZAUZJW16280 | UZJW162808 | AUZJ 2808 | VWZZZAUZJW162808 |
| Part Number, ACM | 5Q0959655BH | 5Q0959655BH | 5Q0959655BH | 5Q0959655BH | 5Q0959655BH |
| Supplier ID, ACM | TSR | TSR | TSR | TSR | TSR |
| Production Date, ACM | 171117 | 171117 | 171117 | 171117 | 171117 |
| Supply Voltage (Before Event) (V) | 14.7 | 12.6 | 12.6 | 12.6 | 12.6 |
| Complete File Recorded | Completed Successfully | eted Successfully | ted Successfully | ed Successfully | Completed Successfully |



# Subaru CDR EDR Coverage 21_5

| Model | Bosch CDR900 Interface Required for true Subaru models<br>Cables 614 and 616 were originally released for Toyota models but also service some Subaru's | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| **SUBARU US Models** | | | | | | | | | | | | | |
| Ascent | Not in production | | | | | | | | 835 | | | ? | |
| BRZ | n/a | n/a | Read as Toyota brand 86 model with Cable 614 (CAN+ or CDR900 interface) | | | | | | | | | 844 | |
| Crosstrek | n/a | n/a | 616 | | | | | | | | | | |
| Forester | No | No | 616* | * | | | | | 835 | | | | |
| Impreza | No | 614* | * | 614*/616 | | | | | | | Japan | | |
| Legacy | No | No | 614 | | 836 | | | | | 844 | | | |
| Outback | No | No | 614 | | 836 | | | | | 844 | | | |
| WRX (inlcudes STI) | No | No | No | No | 614 | | 616 | 614/616 | | | | 844 | 844 |
| Tribeca | No | No | No | No | Discontinued | | | | | | | | |
| | | | | | | | | | | | | | |
| **NON-US Models** | | | | | | | | | | | | | |
| Exiga (Australia, Japan) | | | 614 | | | | | | | | | | |
| XV (Europe) | | 614 | | | | | 614/616 | | | 616 | | | |
| Levorg (Japan) | | | | | 614 | | 614/616 | | | | 844 | 844 | |

# Subaru EDR Tools

1. Bosch CDR 900 interface and 21.3+ software reads all model and model years except for 2013-2020 BRZ, which you read as a Toyota brand Model 86. The 21.4 software is rejecting the BRZ, reload 19.6 software to read it until Bosch fixes that.
2. The Subaru Dealer tool previously used for EDR is no longer needed thanks to Bosch CDR coverage. It may still be useful for fault codes or freeze frame data. $5.8K initial cost, $2700/yr subscription .

# CDR 2015+ Outback Precrash Data

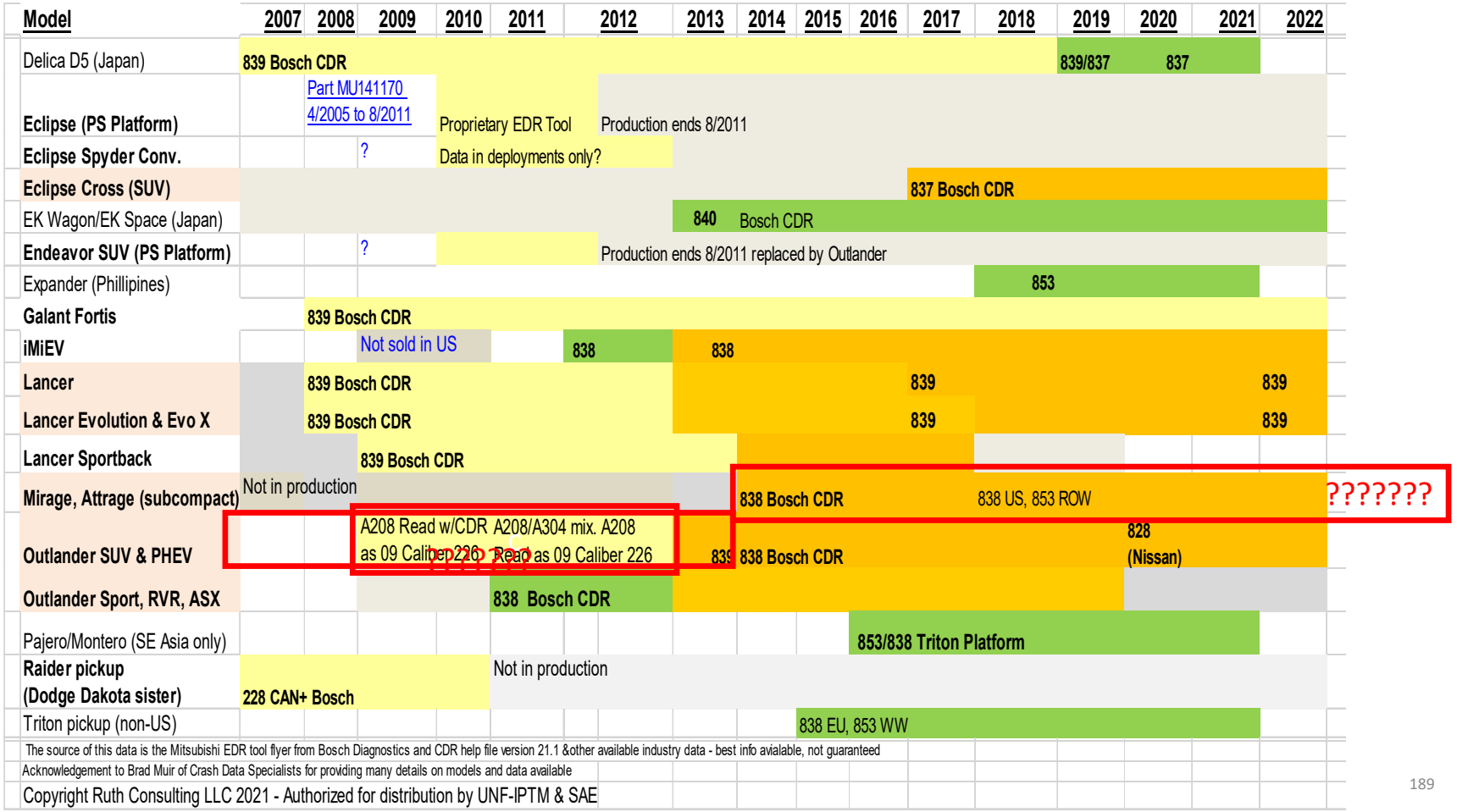
**Pre-Crash Data -5 to 0 Sec (Record 1)**

| Time (sec) | -5.0 | -4.5 | -4.0 | -3.5 | -3.0 | -2.5 | -2.0 | -1.5 | -1.0 | -0.5 | 0.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Speed, Vehicle Indicated (MPH [km/h]) | 21.1 [34] | 22.4 [36] | 23.6 [38] | 24.9 [40] | 24.9 [40] | 23.0 [37] | 19.3 [31] | 16.2 [26] | 10.6 [17] | 7.5 [12] | 6.8 [11] |
| Accelerator Pedal, % Full (%) | 23 | 28 | 24 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Service Brake, On/Off | OFF | OFF | OFF | OFF | ON | ON | ON | ON | ON | ON | ON |
| Engine RPM (RPM) | 1,900 | 2,100 | 2,300 | 1,800 | 1,300 | 1,100 | 1,200 | 1,100 | 1,000 | 1,000 | 1,000 |
| ABS Activity | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF | OFF |
| Stability Control (On, Off, Engaged) | ON | ON | ON | ON | ON | ON | ON | ON | ON | ON | ON |
| Steering Input (deg) | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | -2.5 | -2.5 | -5.0 | -5.0 | -2.5 | -2.5 |




1. Bosch CDR software 21.1 includes significantly expanded Mitsubishi coverage.
2. The only models not officially covered are pre-2013 and may be spoofable as Dodge Calibers.
3. The original Mitsubishi proprietary tool ($7,000) is not needed for any vehicles
4. Early Mitsubishi readouts had some incomplete files retrieved (no useful data retrieved). CDR coverage is not expected to affect that issue, it will still exist.

# Mitsubishi EDR's May 2021
# Most now supported by CDR, as of Dec '21 some not



# Help file cable links suggest some Mitsubishi listed in help file as supported are not (yet) – possibly in 21.4?

**MITSUBISHI ACM CABLE - P/N: 1699200783 / CABLE ID# 837**

This CDR cable is required for direct-to-module imaging of 2018 and newer Eclipse Cross vehicles sold into the various markets. This cable will be available end of June 2019.

**MITSUBISHI ACM CABLE - P/N: 1699200784 / CABLE ID# 838**

This CDR cable is required for direct-to-module imaging of 2011 and newer Mitsubishi vehicles (see below) sold into the various markets. This cable will be available end of June 201

2011 - 2019 ASX, Outlander Sport & RVR

2014 - 2019 Outlander

2018 - 2019 Outlander PHEV

*2014 and newer Mirage

*2017 and newer Attrage

*2012 and newer iMiEV

* not currently supported in CDR software, support planned for a future release

**MITSUBISHI ACM CABLE - P/N: 1699200785 / CABLE ID# 839**

This CDR cable is required for direct-to-module imaging of 2007 and newer Mitsubishi vehicles (see below) sold into the various markets. This cable will be available end of June 201

*2008 and newer Galant Fortis

*2007 and newer Delica-D5

*2009 and newer Lancer Evolution

*2007 - 2013 Outlander

* not currently supported in CDR software, support planned for a future release

# Typical Mitsubishi CDR Data

**Pre-Crash Data -5 to 0 Sec (Record 1)**

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full (%) | Service Brake, On/Off | Engine RPM (RPM) | Steering Input (deg) |
|---|---|---|---|---|---|
| -5.0 | 3.1 [5] | 0 | On | 797 | 4.5 |
| -4.5 | 2.5 [4] | 0 | On | 751 | 4.5 |
| -4.0 | 1.9 [3] | 0 | On | 648 | 4.0 |
| -3.5 | 1.2 [2] | 0 | On | 593 | 4.0 |
| -3.0 | 0.6 [1] | 0 | On | 641 | 4.0 |
| -2.5 | 0.6 [1] | 60 | Off | 719 | 4.5 |
| -2.0 | 1.2 [2] | 71 | Off | 1,297 | 14.0 |
| -1.5 | 4.3 [7] | 75 | Off | 1,840 | 46.5 |
| -1.0 | 7.5 [12] | 77 | Off | 2,010 | 90.0 |
| -0.5 | 9.9 [16] | 76 | Off | 2,180 | 124.5 |
| 0.0 | 11.2 [18] | 72 | Off | 2,206 | 146.0 |

- +Long & Lat Delta V and Acceleration
- +Roll Angle -1 to +5 sec @ 0.1 sec interval

# Typical Mitsubishi CDR Data

**System Status at Event (Record 1)**

| | |
|---|---|
| Ignition Cycle, Crash | 1,828 |
| Multi-event, Number of Events (1,2) | 1 |
| Time From Event 1 to 2 (ms) | 0.0 |
| Complete File Recorded (Yes, No) | Yes |
| Safety Belt Status, Driver | Belted |
| Safety Belt Status, Front Right Passenger | Unbelted |
| Frontal Air Bag Warning Lamp, On/Off | Off |
| Frontal Air Bag Suppression Switch Status | None |
| Seat Track Position Switch Status, Driver | Not Forward |
| Occupant Size Classification, Passenger, Child Size | OC 5TH UP |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | -25.5 [-41] |
| Time, Maximum Delta-V Longitudinal (ms) | 290 |
| Maximum Delta-V, Lateral (MPH [km/h]) | 29.2 [47] |
| Time, Maximum Delta-V Lateral (ms) | 110 |
| Time, Maximum Delta-V, Resultant (ms) | 290 |

# TESLA



1. Remember the definition of an EDR is something that records a time series of data **as a result of a crash**.
2. Tesla may record information as a result of turning the key on, but that is not an event data recorder. More on this later.
3. Tesla did not have any EDR in the ACM of its original 2008 Roadster models.
4. Tesla has installed EDR in The 2012+ Model S, the 2015+ Model X, and the 2017 Model 3.
5. Effective March 2018, Tesla announced they will sell a publicly available tool to download the Tesla EDR

# Tesla EDR

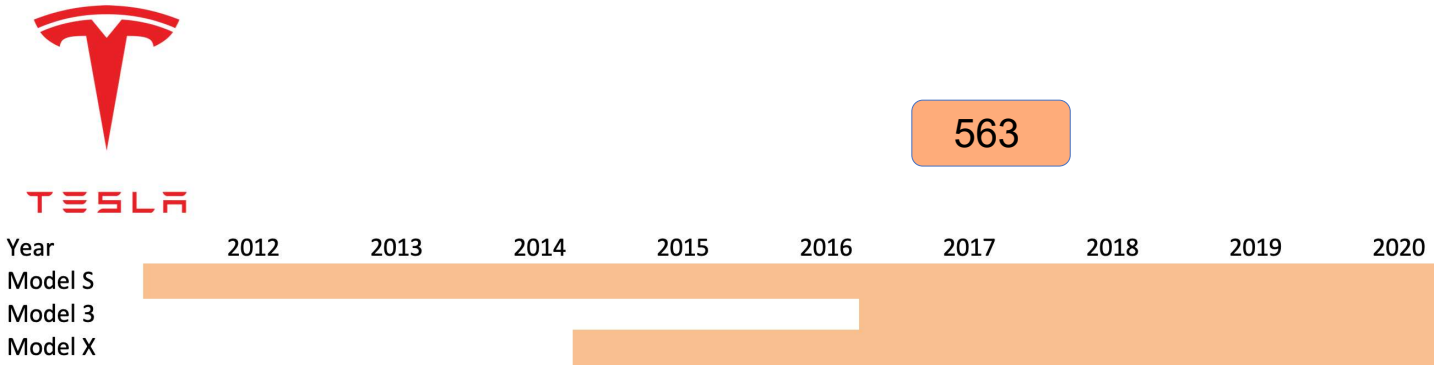
- Kit includes:
  - TIV-144:  In vehicle cable for Tesla Model S and X
  - TIV-145:  In vehicle cable for Tesla legacy Model S
  - TIV-996:  In vehicle cable for Tesla Model 3
  - TD2M-601:  Direct to Module cable for Tesla Model X and 3
  - TD2M-602:  Direct to Module cable for Tesla Model S
  - TD2M-139:  Direct to Module cable for Tesla Model S (NEW CABLE)
  - AC Power Adapter
  - PCAN-USB adapter
  - Hard Shell Case



COST:  $1200.00

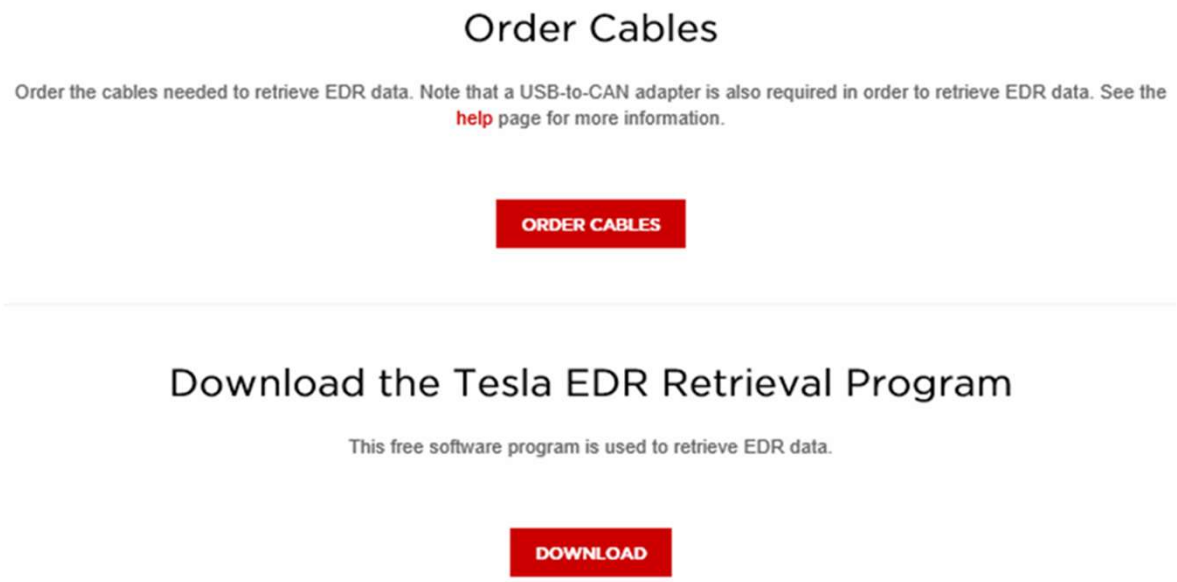
# Tesla EDR

## Coverage



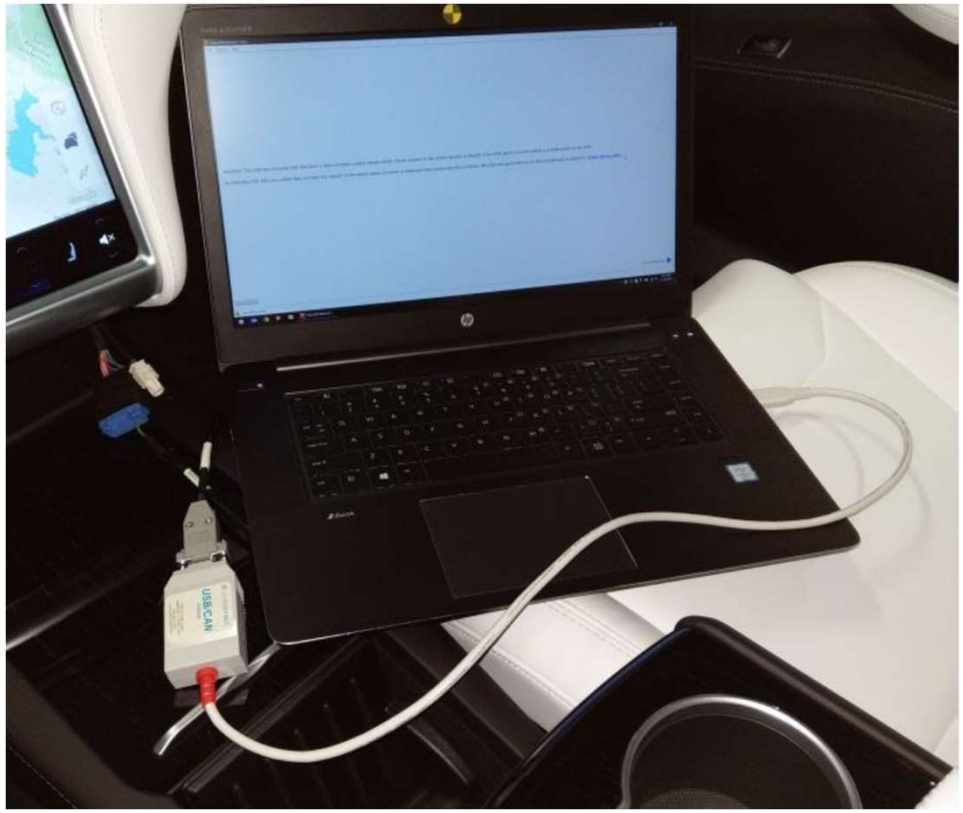
# TESLA

- More information is available on their website https://edr.tesla.com
- See “Other sources” later



# The Tesla EDR Tool

1. Tesla provides the software free of charge on their website that installs on your laptop. No special computer is needed.
2. The hardware kit sells for a list price of $1200.00.
3. The equivalent of the Bosch CDR interface manual is the “PCAN-USB” adapter. It is very small, and reminds you more of a USB to serial adapter, just a slight bit larger.

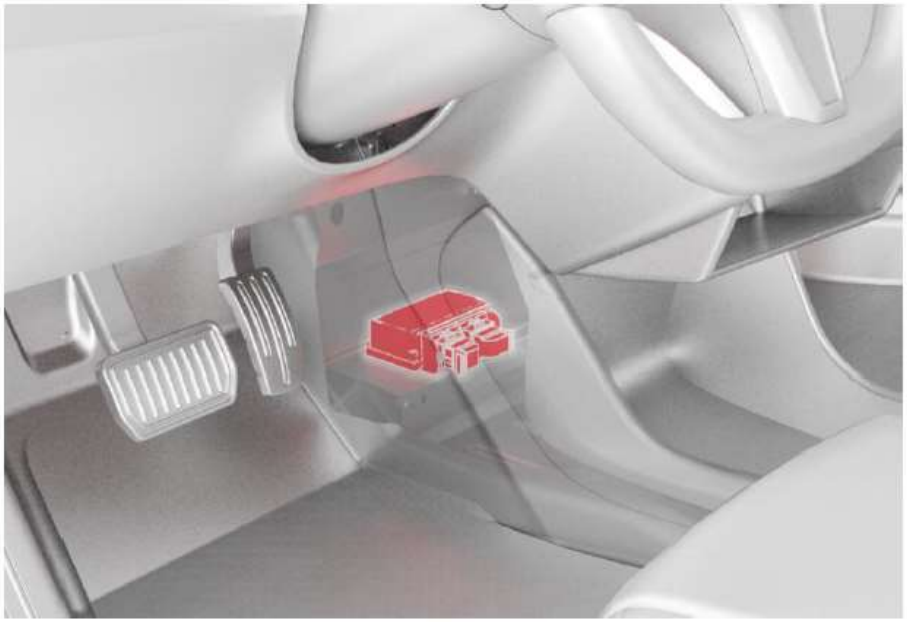


Plug in the PCAN adapter to the Body Harness. Model X and Model 3 are in other places & use a different connector described in Tesla's 27-page EDR guide at **edr.tesla.com/help**.



# TESLA also has 3 direct to module connectors

If the In-Vehicle Connection procedure is unsuccessful, you might be able to connect to the RCM directly, and establish a direct-to-module connection. The RCM is located forward of the center console, below the instrument panel and center screen.



Some 2018 Model 3 Cables must be replaced

# External Power must be used for DTM readouts

·. Connect 12V power to the Tesla Model S Direct-To-Module EDR Retrieval Harness.

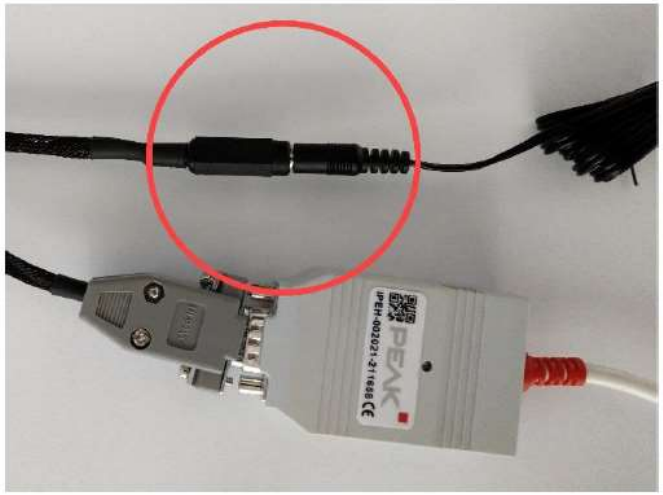



Figure 20. Connecting 12V power to the EDR retrieval harness

·. Proceed to “Retrieve Data” on page 26 to complete the data retrieval procedure.

# The Tesla 27-page Guide

- The Guide is VERY thorough and helpful
- It gives step by step instructions for the hardware assembly and to navigate the EDR program
- There are photographs of connector and module locations
- There are instructions for disassembly to get to the connectors and modules

# Tesla EDR

- Data varies by model

| Model | S | 3 | X |
|---|---|---|---|
| **Event Data** | | | |
| Max X Δv | X | X | X |
| Time to X Max ΔV | X | X | X |
| Max Y Δv | X | X | X |
| Time to Y Max ΔV | X | X | X |
| Time to Max Resultant Δv | X | X | X |
| Ignition cylcle Event | X | X | X |
| Ignition cycle runtime | X | X | X |
| Odomter at event | X | X | X |
| Driver Seat Track Posiition | X | X | X |
| Driver belt | X | X | X |
| Passenger Belt/Occupant | X | X | X |
| 2nd row Belt/Occupant | X | | X |
| 3rd row Belt/Occupant | | | X |
| File Complete | X | X | X |

# Tesla EDR

- Data varies by model

| | Model S | Model 3 | Model X |
|---|---|---|---|
| **PreCrash Data** | | | |
| Vehicle Speed | 2Hz | 5Hz | 2Hz |
| Accelerator Pedal | 2Hz | 5Hz | 2Hz |
| Rear Motor Speed | 2Hz | 5Hz | 2Hz |
| Service Brake | 2Hz | 2Hz | 2Hz |
| Stavility Control | 2Hz | 2Hz | 2Hz |
| ABS | 2Hz | 2Hz | 2Hz |
| Steering Wheel Angle | 2Hz | 10 Hz | 10 Hz |
| Lateral Acceleration | Not Reported | 10 Hz | 10 Hz |
| Longitudinal Acceleration | Not Reported | 10 Hz | 10 Hz |
| Roll Rate | Not Reported | 10 Hz | 10 Hz |
| Yaw Rate | Not Reported | 10 Hz | 10 Hz |

# Tesla EDR

- Data varies by model

| | Model S | Model 3 | Model X |
|---|---|---|---|
| **Crash Data** | | | |
| Longitudinal $\Delta v$ | 300ms | 300ms | 250ms |
| Lateral $\Delta v$ | 300ms | 300ms | 250ms |
| Longitudinal Acceleration | 500Hz (300ms) | 2000Hz (60ms) | 2000Hz (60ms) |
| Lateral Acceleration | 500Hz (300ms) | 2000Hz (60ms) | 2000Hz (60ms) |
| Normal Acceleration | Not Reported | Not Reported | 50Hz (1.4s) |
| Roll Angle | 100Hz (6s) | Not Reported | Not Reported |

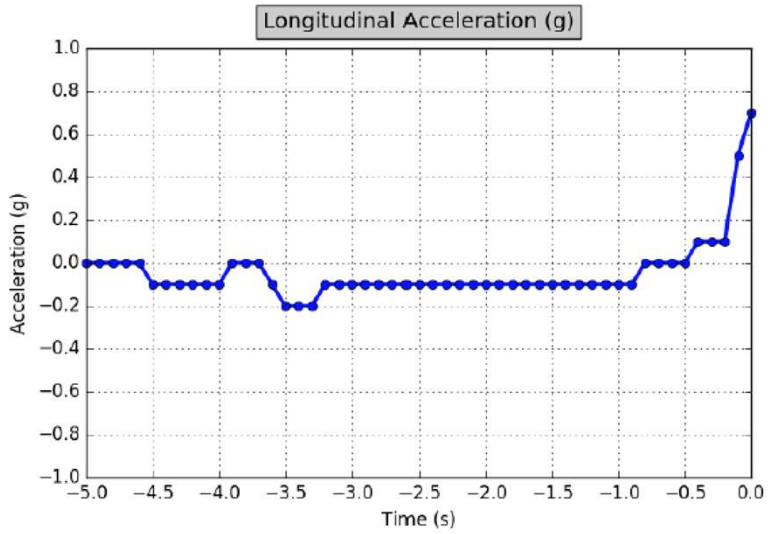
# Tesla EDR Sample File

- Model 3 Data

| Time (sec) | Service Brake | Stability Control | ABS Activity |
|---|---|---|---|
| -5.0 | Off | Off | Off |
| -4.5 | Off | Off | Off |
| -4.0 | Off | Off | Off |
| -3.5 | Off | Off | Off |
| -3.0 | Off | Off | Off |
| -2.5 | Off | Off | Off |
| -2.0 | Off | Off | Off |
| -1.5 | Off | Off | Off |
| -1.0 | Off | Off | Off |
| -0.5 | Off | Off | Off |
| 0.0 | On | Off | Off |
| -2.6 | 6.0 | 10.8 | 714 |
| -2.4 | 6.0 | 13.2 | 746 |
| -2.2 | 6.0 | 14.8 | 784 |
| -2.0 | 7.0 | 12.4 | 807 |
| -1.8 | 7.0 | 12.4 | 845 |
| -1.6 | 7.0 | 12.4 | 883 |
| -1.4 | 7.0 | 11.2 | 905 |
| -1.2 | 8.0 | 12.0 | 922 |
| -1.0 | 8.0 | 10.4 | 941 |
| -0.8 | 8.0 | 4.8 | 932 |
| -0.6 | 8.0 | 0.0 | 916 |
| -0.4 | 8.0 | 0.0 | 888 |
| -0.2 | 8.0 | 0.0 | 843 |
| 0.0 | 7.0 | 0.0 | 749 |

# Tesla EDR Sample File

- Model 3 Data




| Time (s) | Acceleration (g) | Time (s) | Acceleration (g) | Time (s) | Acceleration (g) |
|---|---|---|---|---|---|
| -5.0 | 0.0 | -3.2 | -0.1 | -1.4 | -0.1 |
| -4.9 | 0.0 | -3.1 | -0.1 | -1.3 | -0.1 |
| -4.8 | 0.0 | -3.0 | -0.1 | -1.2 | -0.1 |
| -4.7 | 0.0 | -2.9 | -0.1 | -1.1 | -0.1 |
| -4.6 | 0.0 | -2.8 | -0.1 | -1.0 | -0.1 |
| -4.5 | -0.1 | -2.7 | -0.1 | -0.9 | -0.1 |
| -4.4 | -0.1 | -2.6 | -0.1 | -0.8 | 0.0 |
| -4.3 | -0.1 | -2.5 | -0.1 | -0.7 | 0.0 |
| -4.2 | -0.1 | -2.4 | -0.1 | -0.6 | 0.0 |
| -4.1 | -0.1 | -2.3 | -0.1 | -0.5 | 0.0 |
| -4.0 | -0.1 | -2.2 | -0.1 | -0.4 | 0.1 |
| -3.9 | 0.0 | -2.1 | -0.1 | -0.3 | 0.1 |
| -3.8 | 0.0 | -2.0 | -0.1 | -0.2 | 0.1 |
| -3.7 | 0.0 | -1.9 | -0.1 | -0.1 | 0.5 |
| -3.6 | -0.1 | -1.8 | -0.1 | 0.0 | 0.7 |
| -3.5 | -0.2 | -1.7 | -0.1 | | |
| -3.4 | -0.2 | -1.6 | -0.1 | | |
| -3.3 | -0.2 | -1.5 | -0.1 | | |

# Troubleshooting

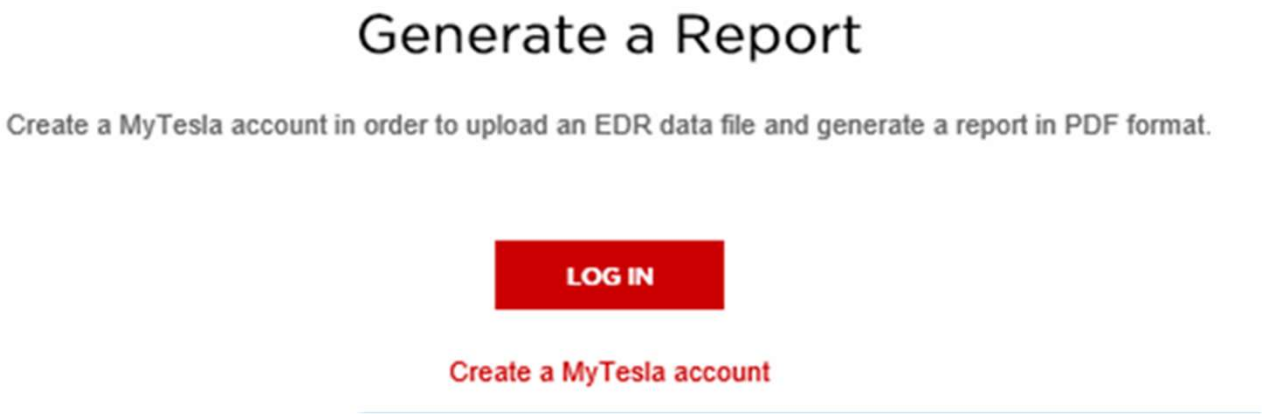
1. Some users report a “time out” error when trying to read data. The vehicle is going back to sleep. **Putting your foot on the brake pedal** and keeping it there for the whole readout keeps the vehicle alive.
2. Low battery power accentuates the time out problem. Tesla NEMA 5-15 adapter?



3. Or see the Help Guide for information how to backpower through the First-Responder loop.

# Converting Raw Data to a Report

The Tesla software extracts a raw data file.  To convert the raw data to a report, you must upload the raw data to the Tesla Server. A report is immediately returned in the latest version of software.



# Worried about report conversion being biased by Tesla?

1. Data is Encrypted.
2. The report is returned IMMEDIATELY. There is no time for Tesla to alter any data to Tesla's benefit.
3. However, some clients insist that no one has a copy of the data except them.
4. There is a "Save without VIN sequence" option to de-identify data
5. There is no alternative.

# What about OTHER data Tesla Data Records?

1. Most of the time you will NOT NEED to access the other data Tesla records, the EDR should be sufficient. But if you need a longer time history, Tesla records MANY data parameters every one second plus, when anything electronically monitored happens. It gets uploaded to the cloud.
2. It is NOT an "event" data recorder, it is a continuous recorder (does **NOT** meet Part 563 definition of a data recorder) (so **NO** PUBLICLY AVAILABLE tool to read it)
3. Law Enforcement contact Tesla and demonstrate proper legal authority
4. Tell them what you are interested in (they have so much you have to narrow it down)

# What about OTHER data Tesla Data Records?

1. Modern Tesla have recorded video that can be uploaded to Tesla
2. This data can be retrieved from Tesla
3. Contact legal@tesla.com



# Other sources for Tesla equipment

1. www.edrfinder.com has developed their own independent Tesla hardware kit. (They are affiliated with one of the Bosch CDR distributors for Europe, and are located in Germany
2. It utilizes the same interface PCAN-USB adapter
3. They have made an adapter from the PCAN adapter to a Bosch CDR DLC cable and the Bosch 783 direct-to-module cable
4. They still have to make a special cable for the Delphi DTM cable
5. Assuming you have a CDR system, this system is lower cost because it has fewer parts
6. Introductory pricing is $695

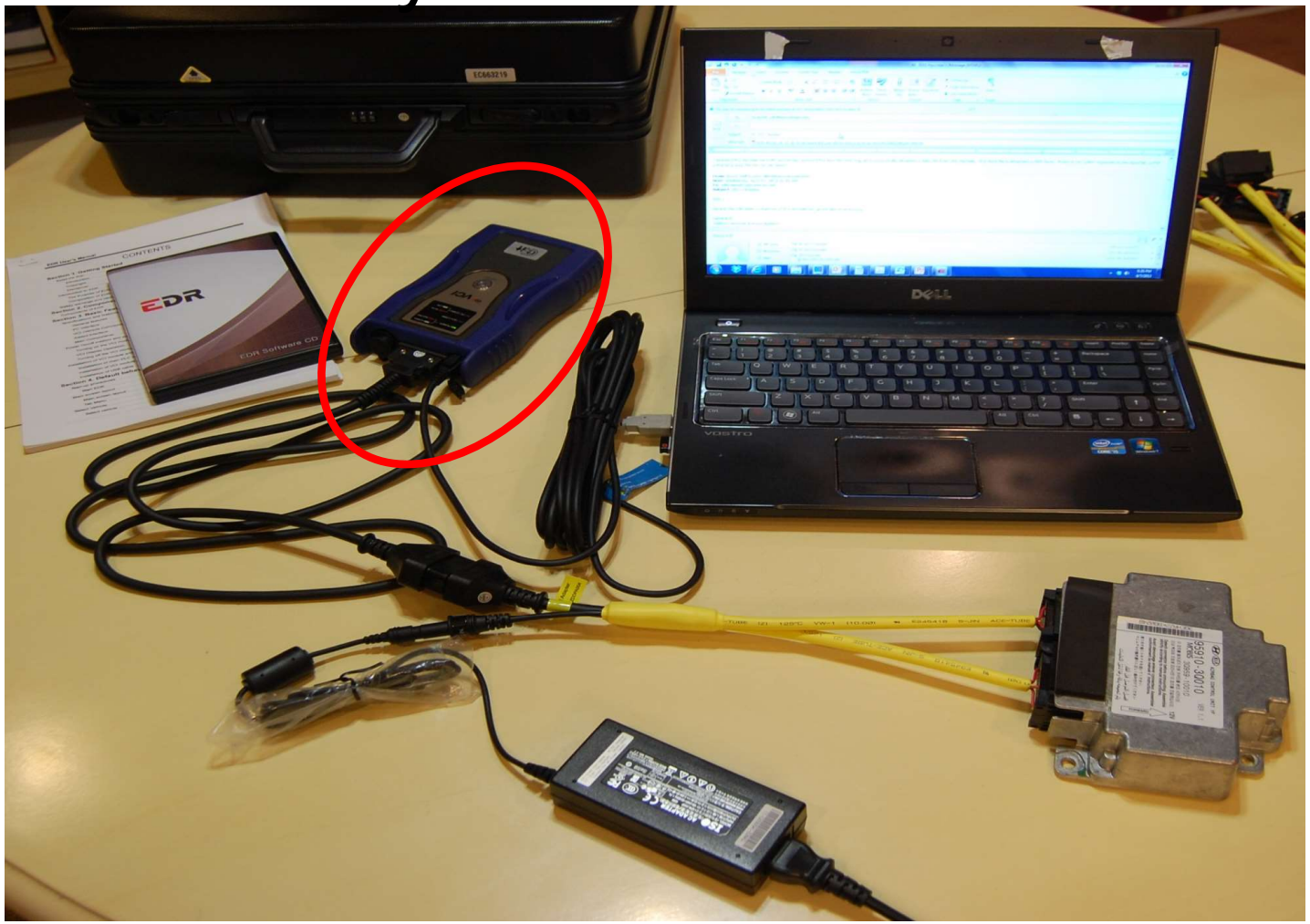
# Hyundai/Kia EDR Tools

- Starting Jan 2022, you may purchase U.S. DLC only kits from Crash Data Group $4950 per brand.
- Full Hyundai kit comes with 13 DTM cables thru #24, Kia with 15 DTM cables thru #23. 10 Cables are in both. New cables now cost $300 each . #25-30 for 2019/2020/2021/2022's not yet included in full kits.
- Hyundai full kit: Part G0Z**H**DMN001 ***$6450,*** Kia full kit: Part GIT0Z**K**DMN001 ***$6950 + tax+ship, +software $695/brand***
- Software updates **$695/yr/** brand from GIT Tool Company or Crash Data Group. Current V48 Hyundai/Genesis, V46 Kia.
- CAN-FD (Flexible Data Rate) DTM adapter needed for 2020 HV80 & other new models cost approx. $1,080
- Contacts: Patrick Dadson pdadson@gitamerica.com (888)549-4977
- Scott Baker **crash@crashdatagroup.com** (800) 280-7940

# Additional Info

1. K/H Interface modules are identical, but you must buy both kits to get software for both. Software updates won't download unless the correct brand interface module is attached to your laptop.
2. I read K/H modules for others for $500. More people are buying Hyundai  kits now that we have 10+ model years on the road.

# The Hyundai EDR Reader



# Officially Supported 2013+ Models V48H/45K

| Model | Model Years | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| KIA | 2010 | 2011 | 2012 | 2013* | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
| CADENZA (Azera sister) | | | | | 19(VG) | 19 | 19 | 13(YG) | 13 | 13 | | | |
| FORTE | 10 | 10 | 10 | 10 (TD) | 6 (YD) | 6 | 6 | 12(YD,YDM) | 12 | 21(BDm) | 21 | 21 | |
| K5 | | | | | | | | | | | | 26(D3LA) | |
| K900 | | | | | | 11 (KH) | 11 | 11 | 11 | 21(RJ) | 21 | | |
| NIRO (DE EV, DE HEV, DE PHEV) | | | | | | | | 21(DE) | 21 | 21 | 21 | 21 | |
| OPTIMA | No | 4 | 4 | 4 (TF/QF) | 4 | 4 | 14 (JF,JFA)* | 14 | 14 | 14(JFA) | 14 | ? | |
| OPTIMA HYBRID | No | 4 | 4 | 4 (TF HEV) | 4 | 4 | 4 | 23(JF HEV) | 23 | 23 | 23 | ? | |
| RIO | No | No | 8 | 8(UB) | 8 | 8 | 8 | 8 | 21 (SC) | 21 | 21 | 21 | |
| RONDO (Canada only 2011 MY+) | No | No | 9 | (not officially supported since non-US, read as a Kia Rio) | | | | | | | | | |
| SEDONA /CARNIVAL and EV6 | No | No | No | No | 10(VQ) | 12(YP) | 12 | 12 | 12 | 12 | 12 | 12 | 29+CFCI KA4 |
| SELTOS | | | | | | | | | | | | 28(SP2) | |
| SORENTO | No | 3 | 3 | 3(XM) | 3,7(XM) | 3,7 | 13 (UMA) | 13 | 13 | 13 | 13 | 29+CFCI | (MQ4) |
| SOUL | 1 see note 1 | 1 see note 1 | 1 | 1(AM) | 12(PS) | 12(PSEV) | 12 | 12 | 12 | 12,26(SK3 EV | 26 (SK3) | 26 | |
| SPORTAGE (SL) | No | Yes | 2 | 2(SL) | 2 | 2 | 2 | 13(QL) | 13 | 13 | 13 | 13 | |
| STINGER (CK) | | | | | | | | | 21 (CK) | 21 | 21 | 21 | 30+CFCI NQS |
| TELLURIDE (ON) | | | | | | | | | | | 25(ON) | 25 | |
| HYUNDAI | 2010 | 2011 | 2012 | 2013* | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | |
| ACCENT (RB) | No 10 | No? | 5 see note 2 | 5(RB) | 5 | 5 | 5 | 5 | 21(HC) | 21 | 21 | 21 | |
| AZERA (HG) | | likely | Yes | 7(HG) | 7 | 7 | 7 | 7 | | | | | |
| ELANTRA Coupe(JK), Sedan(UD/MD | No | Yes | 8 | 8(JK/UD/MD) | 8 | 8 | 8 | 21(AD, ADA) | 21 | 21 | 21 | Hyb 28 (CN7) | |
| ELANTRA GT (GD) (aka "i30") | old model | | new model | 6/12(GD) | 6,12 | 6,12 | 6,12 | 6,12 | 21(PD) | 21 | 21 | 21 | |
| EQUUS (VI)(stretch genesis) | ? | yes? | yes? | 9,11(VI) | 9,11(VI) | 9,11(VI) | 9,11(VI) | | | | | | |
| GENESIS (Coupe BK, Sedan BH/DH) | 9 shave keyway | 9 shave keyway | ? | 9(BK, BH) | 9 | 13(DH) | 13 | | | | | | |
| GENESIS G70 | | | | | | | | | | 21,25(IK) | 21,25 | 29+CFCI (JK1) | |
| GENESIS G80 | | | | | | | | 13(DH) | 13 | 13 | 13 | 29+CFCI | |
| GENESIS GV80 | | | | | | | | | | | 29+CFCI | 29+CFCI | |
| GENESIS G90 | | | | | | | | 13(HI) | 13 | 13 | 13 | 13 | |
| IONIQ EV and HEV | | | | | | | | 21(AE) | 21 | 21 | 21 | 21 | |
| KONA (i20, Rio sister)small SUV) | | | | | | | | | 24 (OS) | 24 | 24 | 24 | |
| KONA EV | | | | | | | | | | 27 (OS EV) | 27 | 27 | |
| NEXO | | | | | | | | | | 26 (FE) | 26 | 26 | |
| PALISADE | | | | | | | | | | | 25(LX2) | 25 | |
| SANTA CRUZ | | | | | | | | | | | | | 30+CFCI (NX4) |
| SANTE FE | 9 shave keyway note 3 | 9 shave keyway | 9 shave keyway | 7(NC,DMA) | 7 | 7 | 7 | 7 | 7 | 25 (TMA) | 25 | 25 | |
| SONATA | | 4 | ? | (YFA, YF HEV | 4 | 14 (LF) 4 (HEV) | 14 (LF HEV) | 14 | 14 | 14 | 25(DN8A) | 25 | |
| TUCSON | Yes | Yes | 2 | 2(LM) | 2 | 2 | 13(TL) | 13 | 13 | 13 | 13 | 30+CFCI (NX4) | |
| VELOSTER coupe | out of productio | | ? | 5(FS) | 5 | 5 | 5 | 5 | 5 | 14 (JS) | 14 | 14 | |
| VELOSTER N | | | | | | | | | | 28(JSN) | 28 | 28 | |
| VENUE | | | | | | | | | | | 30(QX) | 30 | |

# Smart Cruise Control info 21 Palisades

## Pre-Crash Information 2 (-5 ~ 0 sec)

| Time (sec) | Gear shift selector display | SCC information (Main switch status) | SCC information (Set Speed) | SCC information (SPEED UNIT) | SCC information (Mode status) | SCC information (Fail info) |
|---|---|---|---|---|---|---|
| -5.0 | D | ON | 85 | MPH | Engaged | System without Error |
| -4.5 | D | ON | 85 | MPH | Engaged | System without Error |

# Auto Braking info 21 Palisades

## Pre-Crash Information 3 (-5 ~ 0 sec)

Forward Collision Alert

| Time (sec) | FCA (Function status) | FCA (Warning Level) | FCA (Fail info) |
|---|---|---|---|
| -1.0 | ON | No Warning / Invalid or Not supported | Normal |
| -0.9 | ON | No Warning / Invalid or Not supported | Normal |
| -0.8 | ON | Level 1 | Normal |
| -0.7 | ON | Level 2 | Normal |
| -0.6 | ON | Level 2 | Normal |

Audible

0.2g Autobrake

Level 3 =Max autobrake

# Pre-2013 Kia Hyundai may have data

| Model | Vehicle Size, Descrip | Model Years 2010 | 2011 | 2012 | 2013* |
|---|---|---|---|---|---|
| **KIA** | | Cable # | Cable # | Cable # | Cable # |
| CADENZA (VG, 2014) (Azera sister) | 112.0" WB | YG requires cable 13 per software | | | |
| FORTE (TD 2013, YD 2014)) | 104.3"WB, compact | 10 | 10 | 10 | 10 (TD) |
| K900 (KH 2015) | 119.9"WB | | | | |
| NIRO | | | | | |
| OPTIMA (2013 TF QF, 2016 JF, JFA) | 110.0" WB sedan | No | 4 | 4 | 4 (TF/QF) |
| OPTIMA HYBRID | 110.0" WB sedan | No | 4 | 4 | (TF HEV) |
| RIO (UB) | 94.5"WB, compact | No | No | 8 | 8(UB) |
| RONDO (Canada only 2011 MY+) | | No | out of production | | |
| SEDONA (VQ '14, YP '15) | 113.8/118.9" WB miniv | No | No | No | No |
| SORENTO (XM to 2015)(UMA 2016) | 106.3" WB, crossover | No | 3 | 3 | 3(XM) |
| SOUL (AM 2013, PS 2014+) | 100.4" WB | 1 see note 1 | 1 see note 1 | 1 | 1(AM) |
| SPORTAGE (SL) | 103.9" WB, crossover | No | Yes | 2 | 2(SL) |
| **HYUNDAI** | | | | | |
| ACCENT (RB) | 94.5"WB, compact | No 10 | No? | 5 see note 2 | 5(RB) |
| AZERA (HG) | 109.4"WB, sedan | | likely | Yes | 7(HG) |
| ELANTRA Coupe(JK), Sedan(UD/MD) | 104.3"WB, compact | No | Yes | 8 | 8(JK/UD/MD) |
| ELANTRA GT (GD) (aka "i30") | former Elantra Touring | old model | | new model | 6(GD) |
| EQUUS (VI)(stretch genesis) | 119.9" WB, luxury seda | ? | yes? | yes? | 9(VI) |
| GENESIS (Coupe BK, Sedan BH/DH)) | 115.6" WB, luxury seda | 9 shave keyway | 9 shave keyway | ? | 9(BK, BH) |
| GENESIS G90 | | | | | |
| SANTE FE (Sport DMA, NC) | 106.3" WB, crossover | 9 shave keyway note 3 | 9 shave keyway | 9 shave keyway | 7(NC,DMA) |
| SONATA (YF sedan, YF HEV) | 110.0" WB sedan | | 4 | ? | 4(YF) |
| TUCSON (LM to 2015)(TL in 2016) | 103.9" WB, crossover | Yes | Yes | 2 | 2(LM) |
| VELOSTER coupe (FS) | 104.3"WB, compact | out of productior | | ? | 5(FS) |

# 2010-2012 EDR Access

1. The tool requires VIN input and only allows read of 2013+ VINS.
2. Input surrogate 2013 VIN to communicate with 2010-2012 models. Report will list surrogate VIN used.
3. Decoder changed slightly for 2013. Not all data decodes properly.
4. 2010-12 file admissibility must be presented skillfully (get help).
5. **Data successfully admitted in Virginia, Florida, and California.**

# K/H 2010-2012 data reliability?

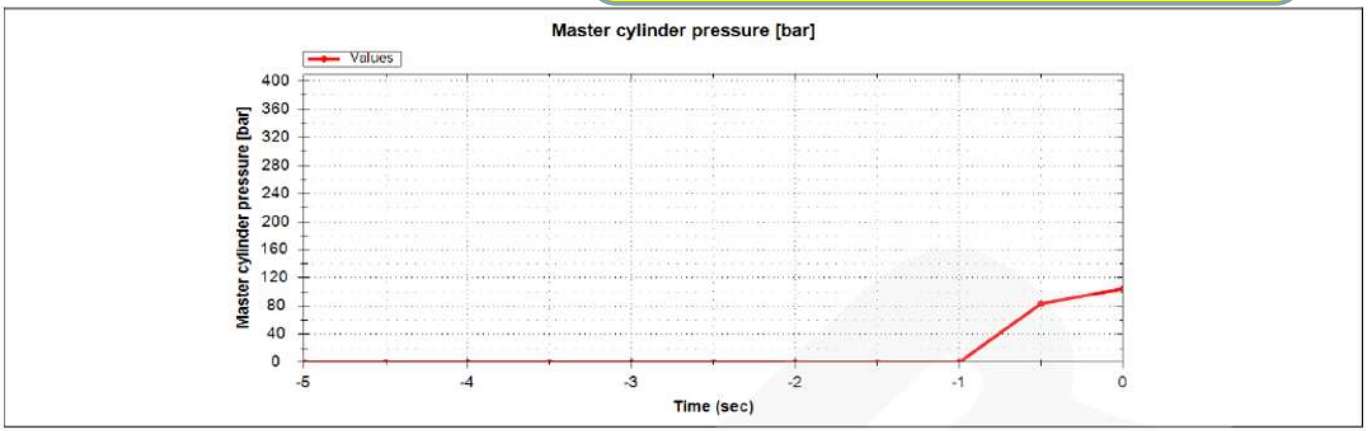
1. In 50+ NHTSA crashes, speed, delta V and belt use appear OK (one exception)
2. Steering angle,  and accel data appear implausible on some
3. Key cycles at download not in all models
4. SAE 2014-01-0502 has details.
5. Additional testing on 2012 Soul reported in SAE 2015-01-1445.
6. Admitted to court in VA, CA, one other

# Brake Pres. added to 2019 Forte

< Event 1 >

Master cylinder pressure

**Units in Bars. 1 Mpa=10 bars**
**Toyota max scale is 12 Mpa=120 bar**




| Num | Time (sec) | Master cylinder pressure [bar] |
|---|---|---|
| 1 | -5.0 | 0.2 |
| 2 | -4.5 | 0.2 |
| 3 | -4.0 | 0.2 |
| 4 | -3.5 | 0.2 |
| 5 | -3.0 | 0.2 |
| 6 | -2.5 | 0.2 |
| 7 | -2.0 | 0.2 |
| 8 | -1.5 | 0.2 |
| 9 | -1.0 | 0.2 |
| 10 | -0.5 | 83.7 |
| 11 | 0.0 | 104.6 |

# Newly published research

1. Every year at the SAE World Congress in April new research is presented.
2. A very brief summary follows, I encourage you to read these papers.
3. Other good sources of EDR info include Collision Magazine

https://www.collisionmagazine.com/article-index/

& Accident Reconstruction Journal.

# EDR SAE Papers 2020

**2020-01-1327 Performance of Event Data**

**Accuracy of Speed Change Measured by Event Data Recorders during Oblique Offset Frontal Impacts** by by Curtat, Wilkinson, Siegmund (MEA Forensic)

**2020-01-1328 Evaluation of General Motors Event Data Recorder Performance in Semi-Trailer Rear Underride Collisions** by Famiglietti, Fatzinger et al (Momentum Engineering)

**2020-01-TBD Analysis of Honda Accord EDR Data for Low- to High Speed Impact Severities** by Crosby/ Skiera/Bare (Exponent)

**2020-01-1329 Further Non-deployment and Deployment Laboratory Experiments Using a Toyota Auris 2007 Event Data Recorder** by Tabone (Univ. of Malta)

# EDR @ SAE 2021

1. **Accuracy of Gen 2 Toyota EDR in Low Speed Side Impacts**
2. **Analysis of Crash Data from a 2012 Hyundai Genesis Coupe Event Data Recorder**
3. **Validation of EEPROM Chip Removal and Reinstallation for Retrieval of Electronic Crash Data**

# Regulatory Update

1. NHTSA issued NPRM in Dec 2012 to **require** 563-EDR's by Sept 2014 (past!)
2. NPRM proposed EDR as FMVSS 405, OEM's would have to self certify EDR's. Strong OEM objections were filed.
3. Officially withdrawn Feb 2019 https://www.federalregister.gov/documents/2019/02/08/2019-01651/federal-motor-vehicle-safety-standards-event-data-recorders

4. Mfrs were NOT expected to remove EDR's – But one Land Rover model may have – now has Bosch freeze frame recorder.

# Regulatory Update

1. NHTSA/IIHS got 10 auto mfrs to agree on 9/11/2015 to making **A**uto **E**mergency **B**raking **(AEB)** standard (timeline 2022). Voluntary commitment is a relatively new strategy by NHTSA. Now 20 have agreed.
2. Next big issue is Autonomous Vehicles (self driving cars).
3. States are drafting conflicting regulations on type of certification required – could hinder development.

# Appeal for Files

- Manufacturers keep coming out with new generations of EDR's
- The way I keep up is asking my network of former students send me files from new vehicles
- PLEASE SEND ME FILES FROM 2019/2020/2021 MY VEHICLES
- If you don't de-identify them, I will
- Helpful if I know which way they turned to check polarity of steering, yaw, lat accel